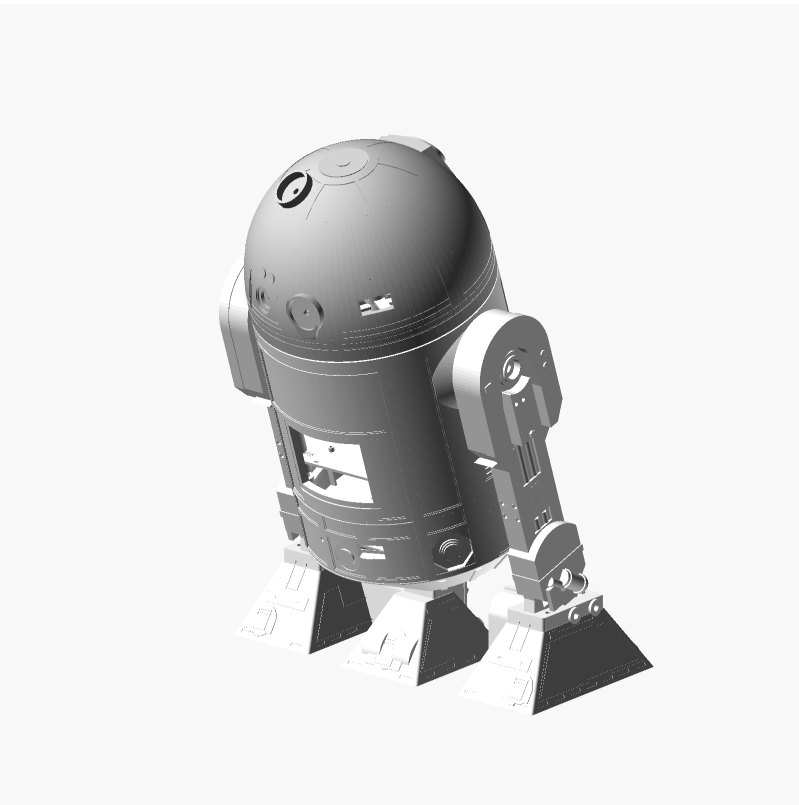


fable-r2d2

R2-D2 fabrication and assembly manual



Key number	Value
Body and dome diameter	317 mm (317 mm body ring, H2D 320 mm bed axis less 3 mm)
Overall height, three-leg driving stance	713 mm (dome crown above the floor)
Printed design files	9 STL files / 12 printed pieces
Estimated assembled mass	11.82 kg total, of which 6.16 kg is printed
Estimated printing	321 h and 10.1 kg of filament for the sliced parts
Static tip-back margin	81.0 mm (14.9 degrees)
Drawing set	25 MATLAB-style PNG drawings, generated 2026-09-12

Digital design; no physical build or test.

Nothing in this package has been printed, assembled, wired or driven. Every mass, current, runtime, torque and margin in this manual is arithmetic on CAD geometry and published component ratings. The acceptance tests in chapter 8 are the first real measurements, not a confirmation of these numbers.

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8. Ranked risks and the test that retires each

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Appendix A Package file list

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1 Overview and stance

A digital fabrication package for a 713 mm tall R2-D2 robot sized to a Bambu Lab H2D: the body and dome are 317 mm in diameter (the 320 mm bed axis minus the 3 mm buffer), every print stays under 315 mm tall (max height minus 5 mm), and the whole robot is nine STL files (twelve printed pieces). Three feet each carry two Adafruit 3777 TT motors and four Adafruit 3766 wheels; a seventh motor and wheel spin the dome through a spring-tensioned friction drive; a Raspberry Pi 4 runs the Wi-Fi control page, displays and sounds and an Adafruit KB2040 drives the motors and lights; a 12 V 7 Ah sealed lead-acid battery lives on a shelf in the lower body and charges through a panel port.

This is a design package. Nothing has been printed, assembled or driven. The physical tests in the manual are part of the build.

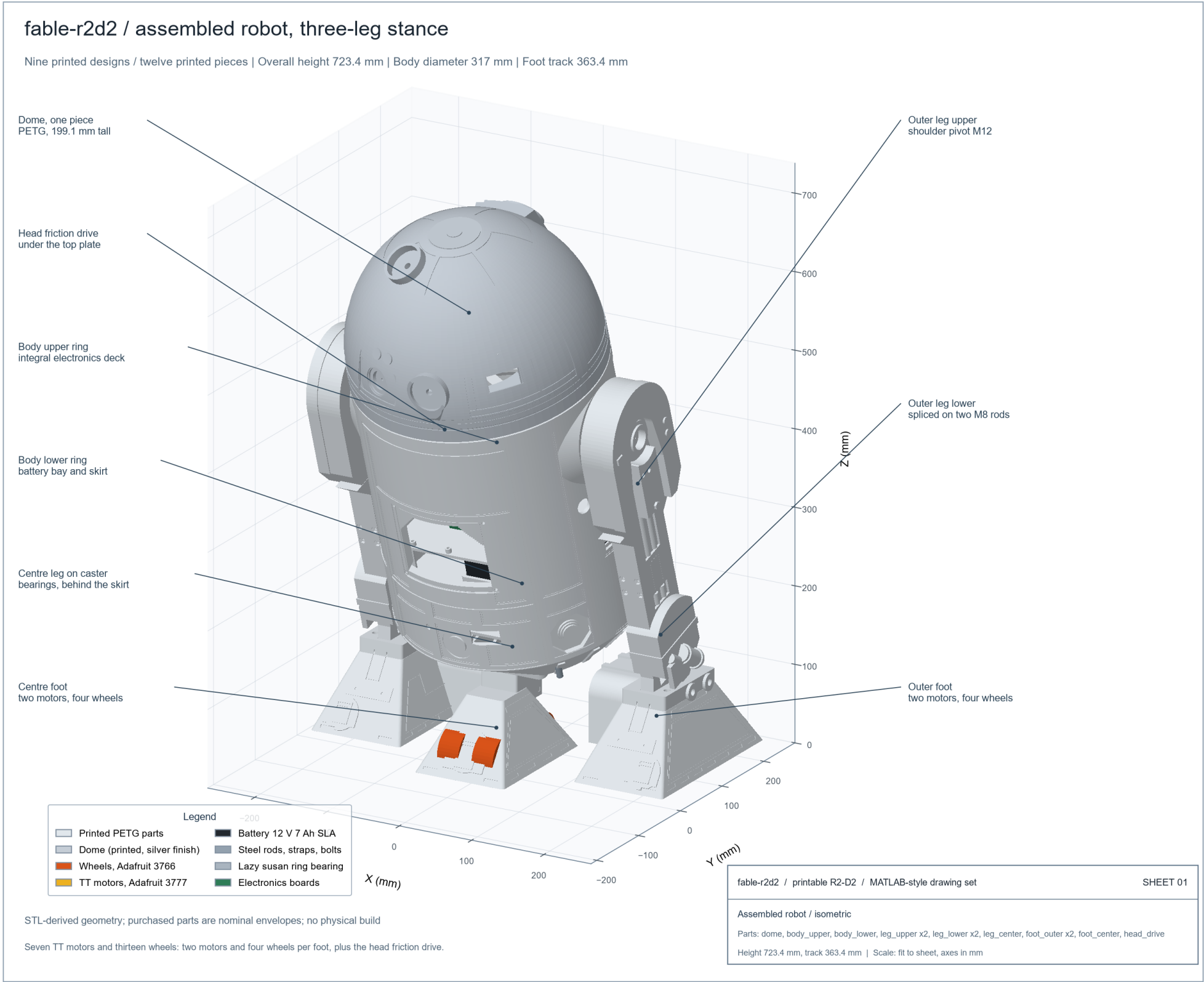
1.1 Stance and stability screening

These figures come from `docs/stability.json`, which sums the STL volumes at the modelled fill fraction and treats every purchased item as a point mass. It is a screening calculation, not a weighing. The tip-back margin is the number that governs how this robot may be handled: it is small, so never push the dome rearward and never lift the robot by the dome.

Measure	Value	Source
Centre of gravity, forward of the shoulder axis	55.2 mm	<code>docs/stability.json</code>
Centre of gravity, above the floor	303.6 mm	<code>docs/stability.json</code>
Support polygon, fore and aft	-25.8 to 161.2 mm forward of the shoulder axis	wheel contact patches
Tip-back margin (rearward)	81.0 mm / 14.9 degrees	the tightest margin
Tip-forward margin	105.9 mm / 19.2 degrees	<code>docs/stability.json</code>
Sideways tip angle	34.1 degrees	<code>docs/stability.json</code>
Centre-foot static load share	0.63 of the total mass (the rear foot carries more than its geometric share)	<code>docs/stability.json</code>
Print envelope check	pass	<code>cad/validation.json</code> , envelope [322, 317, 315] mm

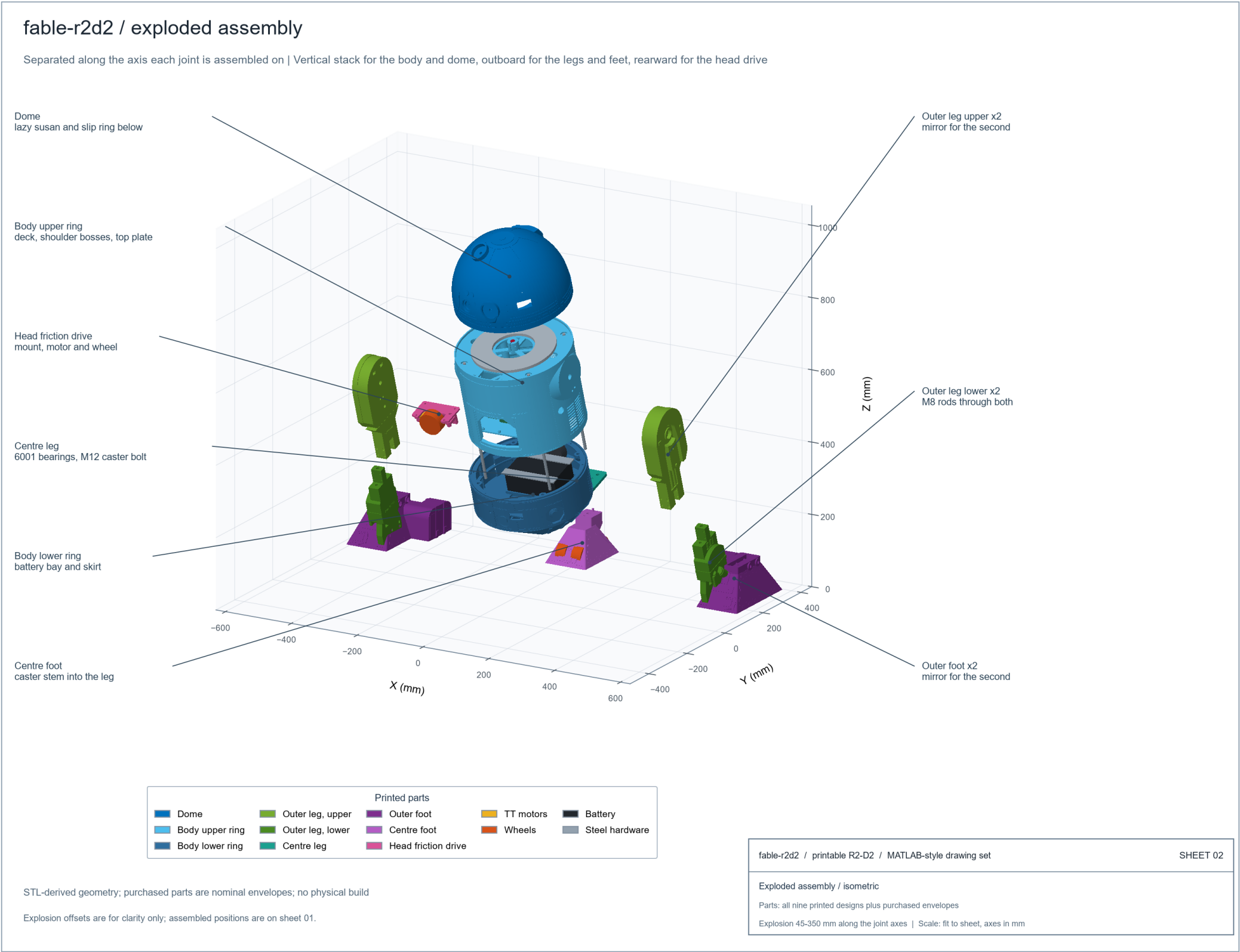
Method: STL volumes x 1.27 g/cm³ x fill fraction; purchased parts as point masses; screening only

Figure 1 - the assembled robot in its three-leg driving stance



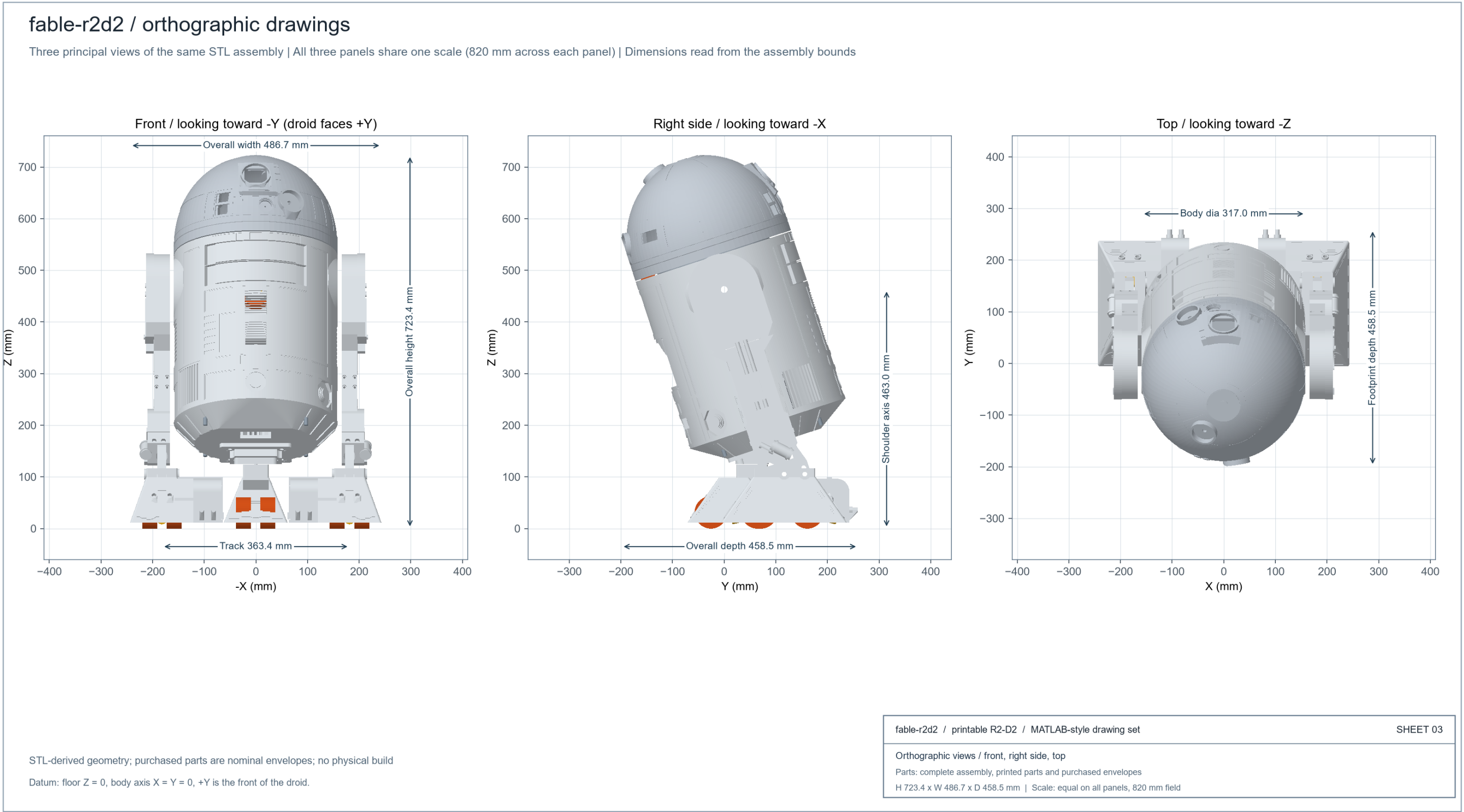
output/drawings/01_robot_assembled.png. Rendered from the nine package STL files with purchased parts drawn as nominal envelopes. Millimetre axes.

Figure 2 - the whole robot, separated along its assembly axes



output/drawings/02_robot_exploded.png. The offsets identify the parts; they are not assembly dimensions.

Figure 3 - orthographic views



output/drawings/03_robot_orthographic.png. Front, side, rear and top views with millimetre axes. Use the mechanical chapter for dimensions, not this drawing.

2 Printing

9 STL files make 12 printed pieces. Solid-material upper bound for the whole set is 13.69 kg before infill savings, supports and brim; the sliced prediction column is the number to buy filament against. Print the one outer foot first and assemble stage 1 on it before committing the rest.

2.1 Part manifest

Part	Qty	Material	X x Y x Z mm	Solid g	H2D slice prediction	Plate orientation
dome	1	PETG	316.8 x 316.8 x 199.1	1304	1243 g / 35.4 h each	Lip down on the plate; crown up
body_upper	1	PETG	317.0 x 317.0 x 264.0	3445	3115 g / 107.6 h each	Bottom (mating lip) down
body_lower	1	PETG	317.0 x 317.0 x 141.1	2144	1939 g / 59.2 h each	Skirt bottom down
leg_upper	2	PETG	289.7 x 139.4 x 50.8	1445	612 g / 20.0 h each	Shoulder face down; mirror for the second
leg_lower	2	PETG	232.0 x 100.0 x 58.2	666	315 g / 11.2 h each	Splice face down; mirror for the second
foot_outer	2	PETG	181.8 x 265.0 x 105.2	855	681 g / 19.6 h each	Sole down; mirror for the second
leg_center	1	PETG	140.0 x 100.0 x 34.3	247	128 g / 4.2 h each	Body flange down
foot_center	1	PETG	123.3 x 180.2 x 119.9	510	331 g / 9.5 h each	Sole down
head_drive	1	PETG	116.5 x 72.3 x 39.9	107	81 g / 3.0 h each	Base down

Source: bom/printed-parts.csv and cad/h2d-slice-check.json.

2.2 Slicer profile actually used

Machine: Bambu Lab H2D 0.4 nozzle, trimmed to single-nozzle mode (325 x 320 x 320). Process: 0.20mm Standard @BBL H2D. Plate: Textured PEI Plate.

Overrides: wall_loops = 5, top_shell_layers = 6, bottom_shell_layers = 6, sparse_infill_density = 30%, sparse_infill_pattern = gyroid, enable_support = 1, support_type = tree(auto), support_on_build_plate_only = 0, brim_type = no_brim, skirt_loops = 0.

Check run: Local Bambu Studio CLI slicing of package meshes; no printer connection or physical print on 2026-09-12T22:45:18 UTC. Totals for the parts that sliced: 10054 g of filament and 320.6 printer hours.

2.3 Orientation and support notes

Reproduced from docs/mechanical.md section 2.

2. The nine STL files

Nine files. Twelve printed pieces. leg_upper, leg_lower and foot_outer are each printed twice, the second one mirrored in the slicer. The supplied file is the right-hand part.

File	Qty	Printed size X x Y x Z mm	Orientation	Material	Supports
dome.stl	1	316.8 x 316.8 x 199.1	Lip down, crown up	PETG (PLA Basic option)	Tree, inside the crown only
body_upper.stl	1	317.0 x 317.0 x 264.0	Seam face down	PETG (PETG-CF option)	Tree, under the deck only
body_lower.stl	1	317.0 x 317.0 x 141.1	Skirt bottom down	PETG	Tree, under the battery shelf only
leg_upper.stl	2	289.7 x 139.4 x 50.8	Inboard face down	PETG (PETG-CF option)	None
leg_lower.stl	2	232.0 x 100.0 x 58.3	Inboard face down	PETG	6.5 mm block under the tongue
foot_outer.stl	2	181.8 x 265.0 x 105.2	Sole down	PETG	None, if the slicer bridges
leg_center.stl	1	140.0 x 100.0 x 47.5	Flange face down	PETG	None
foot_center.stl	1	123.3 x 180.2 x 119.9	Sole down	PETG	None, if the slicer bridges
head_drive.stl	1	116.5 x 72.3 x 39.9	Base mating face down	PETG	Tree, under the arm

Every size above is measured from the mesh by scripts/export_cad.py and recorded in cad/validation.json, which now reports **nine of nine parts passing**: watertight, one connected solid, consistent winding, inside the envelope.

body_upper measures 264.0 mm tall, not the 246.0 mm of the ring skin, because the shoulder fairing and its gussets stand below the seam plane in the print frame. Both rings still sit well inside the 315 mm Z limit.

2.1 Slicer settings

These are the settings scripts/slice_check.py used, recorded verbatim in cad/h2d-slice-check.json.

Setting	Value
Machine	Bambu Lab H2D 0.4 nozzle
Process	0.20mm Standard @BBL H2D
Wall loops	5
Top shell layers	6
Bottom shell layers	6
Sparse infill density	30 %
Sparse infill pattern	gyroid
Support	on, tree (auto), not build-plate only
Brim	none
Skirt loops	0
Bed	Textured PEI plate
Filament profile	Bambu PETG HF @BBL H2D 0.4 nozzle

p1an.md locks 0.20 mm layers, five walls and 30-40 percent gyroid. Use 30 percent for the dome and the feet, 40 percent for the two body rings, the four leg pieces and the centre leg. The Bambu PETG wiki guidance is to keep the wall count at or below 6 and the infill at or below 50 percent, so 5 walls and 40 percent stay inside it (research/components-mechanical.md, "Filament and print guidance").

2.2 Material choice

Material	Where	Numbers read from the TDS
Bambu PETG Basic	Default for every structural part	Tensile X-Y 51 +/- 1 MPa, Z 35 +/- 6 MPa; bending X-Y 75 +/- 3 MPa; HDT 68 C at 1.8 MPa; density 1.25 g/cm3
Bambu PETG HF	The profile the slice check used; discontinued	Tensile X-Y 34 +/- 4 MPa, Z 23 +/- 4 MPa; bending X-Y 64 +/- 3 MPa; HDT 62 C; density 1.28 g/cm3
Bambu PETG-CF	Option for body_upper (the shoulder bosses) and the leg cores	Tensile X-Y 59 +/- 4 MPa, Z 38 +/- 3 MPa; bending X-Y 83 +/- 4 MPa; HDT 68 C
Bambu PLA Basic	Option for the dome only	The dome carries only its own weight on the ring bearing

PETG-CF needs a hardened steel nozzle, is not compatible with AMS lite, runs 240-270 C with 100 percent cooling and prints below 200 mm/s. The H2D ships with a hardened steel nozzle. PETG Basic is the safe default and is the material the BOM buys (five 1 kg spools, bom/hardware.csv H31).

PETG softens at 62-68 C. Do not leave the robot in a hot car. This is a mechanical limit, not just an electrical one.

2.3 Support notes, part by part

dome - Lip down. The shell overhang passes 45 degrees above about z 151 and reaches about 75 degrees at the flat crown. Support goes **inside**, under the last 25 mm of the crown (above about z 168), under the pie-panel ring, the 66 mm flat top and the HP3 cup. Everything else is designed support-free: the eye housing underside is 48 degrees from horizontal, the holoprojector barrels have gussets and teardrop cups, the PSI bosses are teardrops, and every pocket has a 45 degree gable. Remaining bridges are the tops of the 44.1 mm holoprojector bores, the 52.6 mm bezel bore, the 27 and 34 mm PSI apertures, the 30 and 44 mm logic windows and two 2.3 mm inset ledges. The 14 degree wedge face at the eye's bottom-left corner (about 20 x 6 mm) is a downward face; paint a support blocker there or accept minor roughness. No brim: the first layer is a 57 to 158.5 mm annulus.

leg_upper - No supports. The horseshoe, hub, booster cover, strut rods and channel all rise from the plate. The horizontal bores print round (M12 13 mm, dowels 6.2 mm, wire 9 mm). The captive nut traps are 13.4 x 8.4 mm tunnels bridged at 23 mm. The finger slot is vertical.

leg_lower - One support block. The ankle tongue plate lies 6.5 mm above the bed (lg_tongue_x0), because it must stay centred on the foot slot. Put a 6.5 mm normal support block under the tongue area, about 100 x 60 mm. Everything else rises from the bed. The 22 mm rod counterbores are horizontal round holes.

leg_center - No supports. The column leans 18 degrees, so its rear face is an 18 degree overhang. The bearing seats, the swivel groove and the wire slot all open upward. The M12 head counterbore is a 28 mm hole from the bed with a 2 mm ledge ring at 13 mm.

foot_outer and foot_center - Sole down, as modelled, and in principle no supports. The sloped walls are 19.7 and 34.6 degrees from vertical on the outer foot, 22.3 and 35.0 on the centre foot. Every internal rib rises from the bed. The gearbox pocket ceilings are 12 mm wide tilted bridges over a 19.4 mm pocket. The can pockets are 21 mm round arches with 1 mm of sag allowance. The one real demand is the top plate: the two 20 mm strips beside the ankle slot bridge 51 mm, and the plate bridges the side walls over 58 mm (outer foot) and 45 mm (centre foot). Use bridge settings with 100 percent fan. **If the slicer will not bridge that, add tree supports under the top plate only.**

head_drive - Base mating face down, arm 4 degrees lowered. The base, its pads, the tension boss and the lugs need no support. The arm is a suspended body and tree supports under it are unavoidable: inside the open motor pocket, under the side beam, and under the boss-to-wall bridge. Every supported face is hidden inside the body. The base and the arm print as one solid, joined by two sacrificial webs 0.8 mm thick, 6 mm long and about 3.5 mm tall at x = +/-15 mm. **After printing, cut both webs with a knife and file the stubs flush.** The arm needs 3 mm of clearance to the base plate at the hinge ridge.

body_lower - Skirt down, 135.1 mm of ring plus the 6 mm seam lip. **Tree supports under the battery shelf only** (z 41 to 45), between the sixteen radial support ribs. Everything else is self-supporting: the skirt, the skirt ribs, the floor, the rod bosses, the ring ribs with their 45 degree undersides, the seam flange and the lip. The skirt flares 41.9 mm out over 41.3 mm of height, which is 45.4 degrees from vertical. That is right at the limit. It prints in PETG at 0.20 mm, but it is the one outside surface that benefits from a 45 degree speed reduction.

body_upper - Seam face down, 246 mm of ring skin. **Tree supports under the electronics deck only**, inside radius 131 mm; the sixteen 45 degree gussets carry the deck from radius 131 to 156 mm. Both support regions are reached through the open top of the ring after printing. Bridges: the front vent louvre slots bridge 45 mm, the side vent slots bridge 70 mm, and the rear access opening bridges 120 mm across its top edge.

Neither ring needs support on any outside surface. Every skin feature is a recess or a through cut. No brim on either ring; the first layer of each is an annulus.

2.4 Predicted mass and time

From `cad/h2d-slice-check.json`, which is a local Bambu Studio CLI slice. No printer was connected and nothing was printed.

Part	Qty	Predicted mass each g	Predicted time each h
foot_outer	2	679.9	19.63
foot_center	1	322.4	9.16
head_drive	1	81.3	3.00
leg_upper	2	615.0	20.15
leg_lower	2	314.1	11.14
leg_center	1	162.3	5.42
Sub-total, six parts, nine pieces		3,783.9 g	119.4 h

The dome is not in that total. Its row failed on the script's `--arrange` step with return code -50, not on the geometry. The same STL centred on the bed with the same support settings slices clean: **1,243.9 g, 35.37 h** (`cad/integration-round2.md`, headline). Adding it gives **5,027.8 g and 154.8 h for ten of the twelve pieces**.

The two body rings are modelled but not yet sliced: `cad/h2d-slice-check.json` was written before `cad/body.scad` existed. What is known from the meshes (`cad/validation.json`) is their solid volume: `body_upper` 2,618.6 cm³ and `body_lower` 1,537.2 cm³. At the 0.45 fill fraction `docs/stability.json` uses for screening, that is **1,496 g and 879 g**, or 2,375 g for the pair, against the 3,018 g `research/loads.md` budgets. Print time for the two rings is **estimated at 60 to 90 hours**.

Whole-robot printing is therefore **estimated at about 7.4 kg of filament and 215 to 245 hours**. Buy six 1 kg spools. Re-run `python scripts/slice_check.py` now that the body exists, and replace the estimates above with its numbers.

2.5 What is on the two body rings

From the header blocks of `cad/body.scad`. Skin features are quoted there as (s, y), where s is the arc distance from the front or rear centreline and y is the distance down from the body top. Positive front s lands on the droid's **left**.

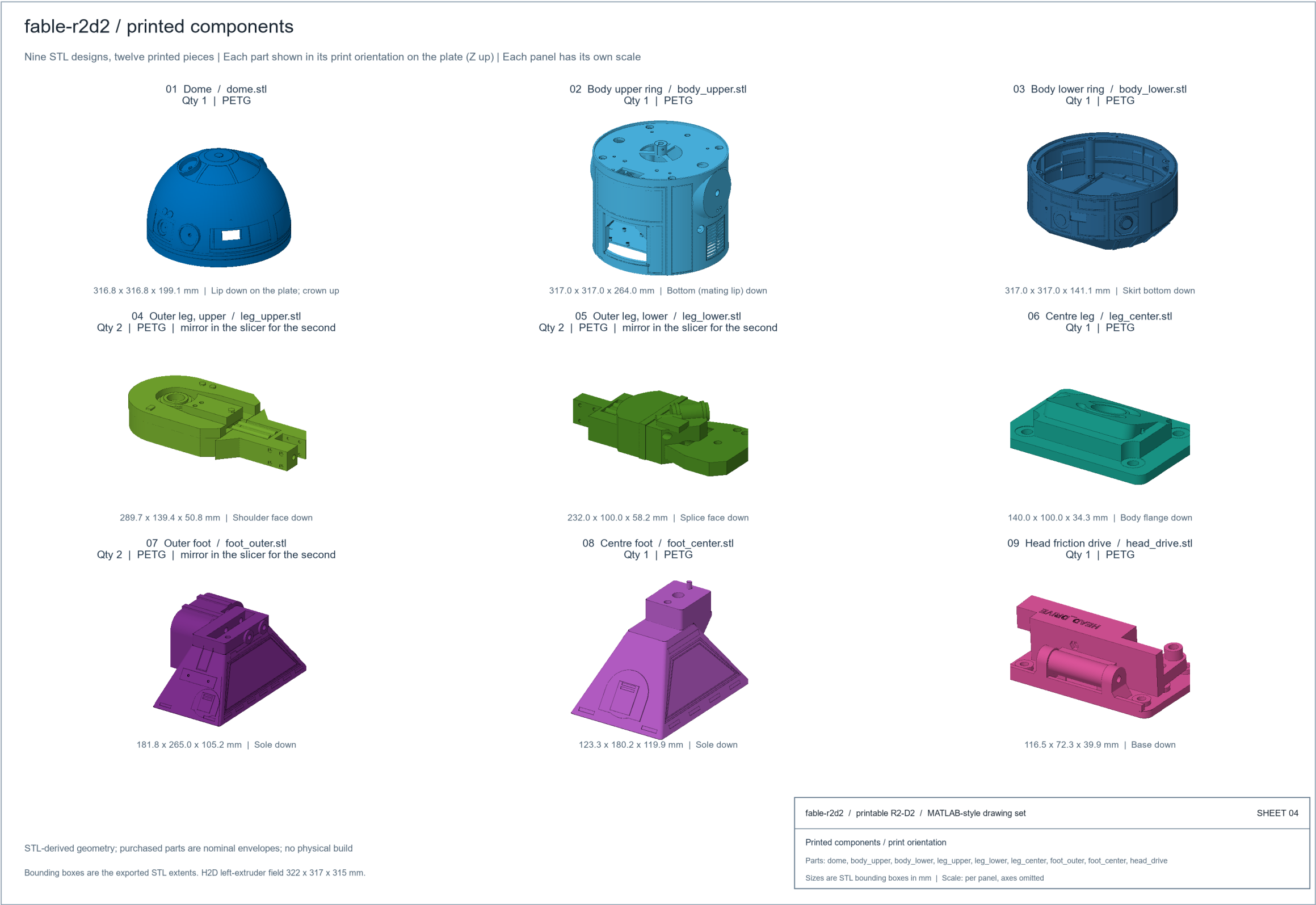
`body_lower`, **body-frame z 0 to 141.1**

Feature	Detail
Floor	6 mm plate on the skirt bottom. Four M8 clearance holes at center_leg_bolts with hex nut pockets on the top face , each in a 22 mm boss. One 60 x 40 mm cable opening centred at Y = -60. Recessed part label.
Battery shelf	4 mm plate at z 41 to 45, full ring width, on sixteen radial support ribs. An 8 mm curb runs round the SLA footprint, 11 mm outside the battery and 3 mm clear inside it. Four 26 mm socket windows sit over the four centre-leg nuts so a socket reaches them with the shelf in place. Four 3 mm strap slots. An 80 x 26 mm harness slot front and rear.
M8 rod columns	Four, on radius 138 mm. The nut is not under the floor . See 3.1.
Seam flange and lip	See 3.1.
Side vents	The notch is y 139 to 228, not 139 to 243.2: the lower 15 mm would have cut the seam flange. Both flanks carry one, formed by the front half plus the rear half of the skin.
Utility-arm relief	165 x 38 x 8 mm , not 165 x 43: the bay is only 41.3 mm tall.
Speaker grille	Nine 5 mm vertical louvres in the front pocket vent door, centred at (s, y) = (-158, 291). Vertical slots in a vertical wall need no bridging, and the reference photographs show vertical slats. Speaker bosses are behind it.
Charge port and switch	Pad on the rear lower band.

`body_upper`, **body-frame z 135.1 to 381.1**

Feature	Detail
Electronics deck	Integral, 4 mm thick, local z 36 to 40 (tray_z_upper is the top face). An annulus from radius 50 to 154.9 mm, cut off by a chord at y = -114.9, which leaves a 40 mm harness gap at the rear. Sixteen 45 degree gussets carry it from radius 131 to 156 mm.
Deck seats	Raspberry Pi 4: four M2.5 boss inserts on the pi4_holes pattern. KB2040: two M2 bosses. Four DRV8833: a 26 x 18 mm pad each with two M2 bosses. Three regulators: a 25.4 x 25.4 mm pad each with four 2.2 mm holes. ADS1115: two M2.5 bosses. Fuse holders: one 46 x 20 mm seat. Eight 6 x 2.5 mm cable slots through the deck, and six recessed labels: PI4, KB2040, DRV8833, 5V / 12V, ADC, FUSE.
Rear access opening	120 x 90 mm at y 126 to 216 , which is body-frame z 165.1 to 255.1. A 10 mm inner lip takes the cover, held by four M3 heat-set inserts . It sits well below the head-drive wheel sweep, so the lip cannot foul the tyre.
Shoulder pads and bosses	See 3.2. The fairing is clipped flat at the top plate.
Top plate	See 3.7 and 3.8.
Utility-arm bays	A backing pad behind the bay, radius 147.9 mm.

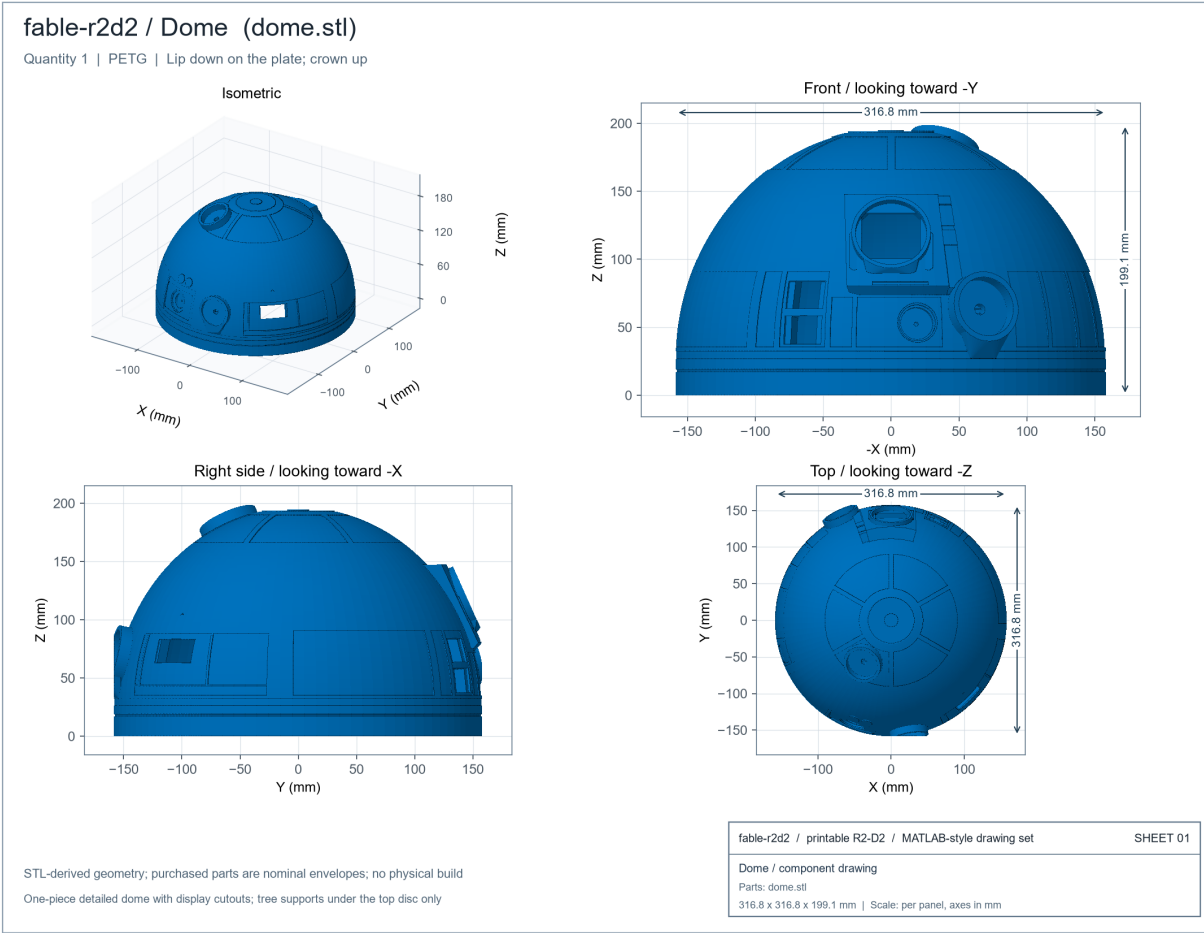
Figure 4 - every printed component in the package



2.4 Component sheets

One page per STL file: the part as it is modelled, with the numbers the slicer and the assembly stages need.

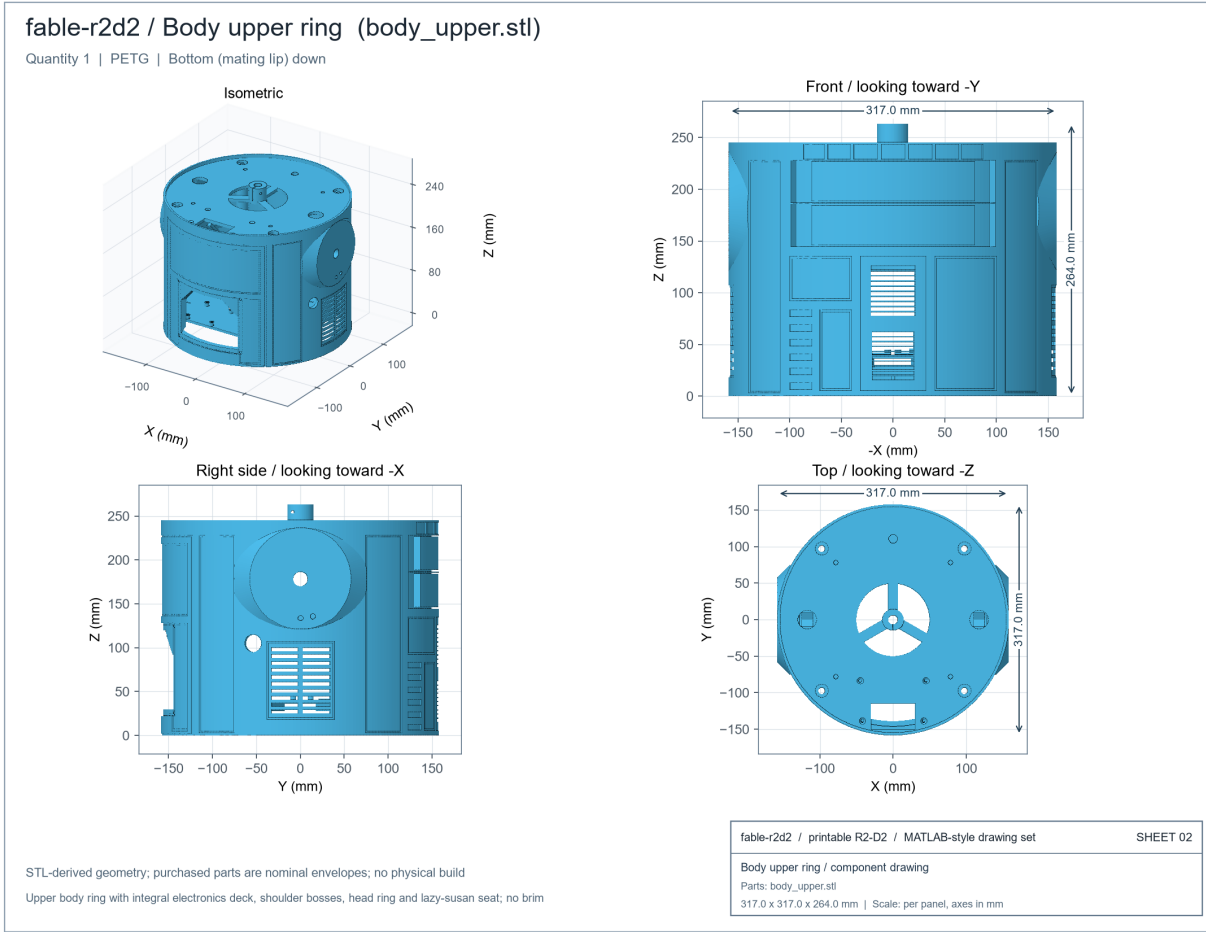
dome



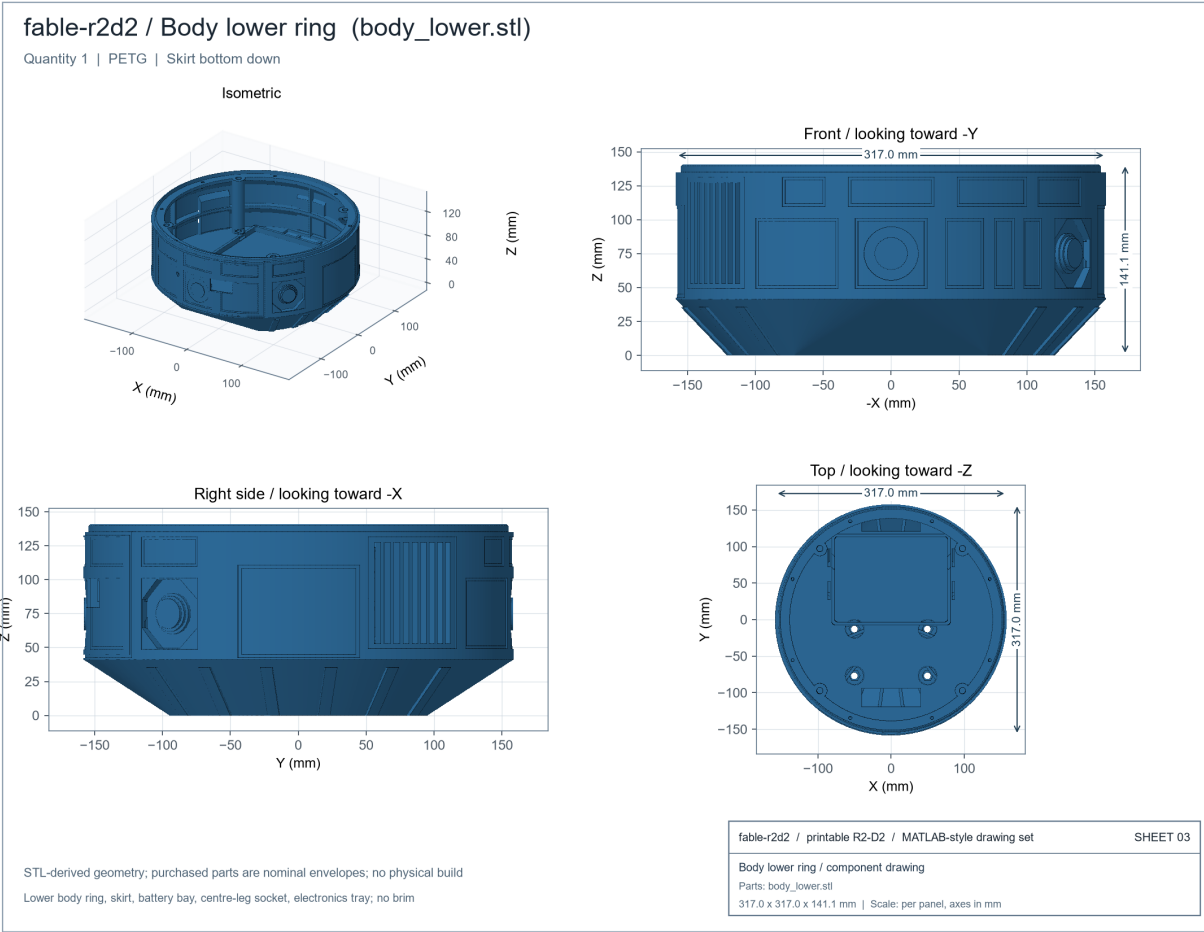
output/drawings/components/dome.png - One-piece detailed dome with display cutouts; tree supports under the top disc only

Field	Value	Field	Value
Quantity	1	Material	PETG
Envelope mm	316.83 x 316.83 x 199.14	Solid mass	1304.3 g
Mesh volume	1027.0 cm3	Triangles	49934
Watertight	True	Solids	1
Plate orientation	Lip down on the plate; crown up	Mirror for the pair	False
Sliced mass	1243.4 g	Sliced time	35.36 h

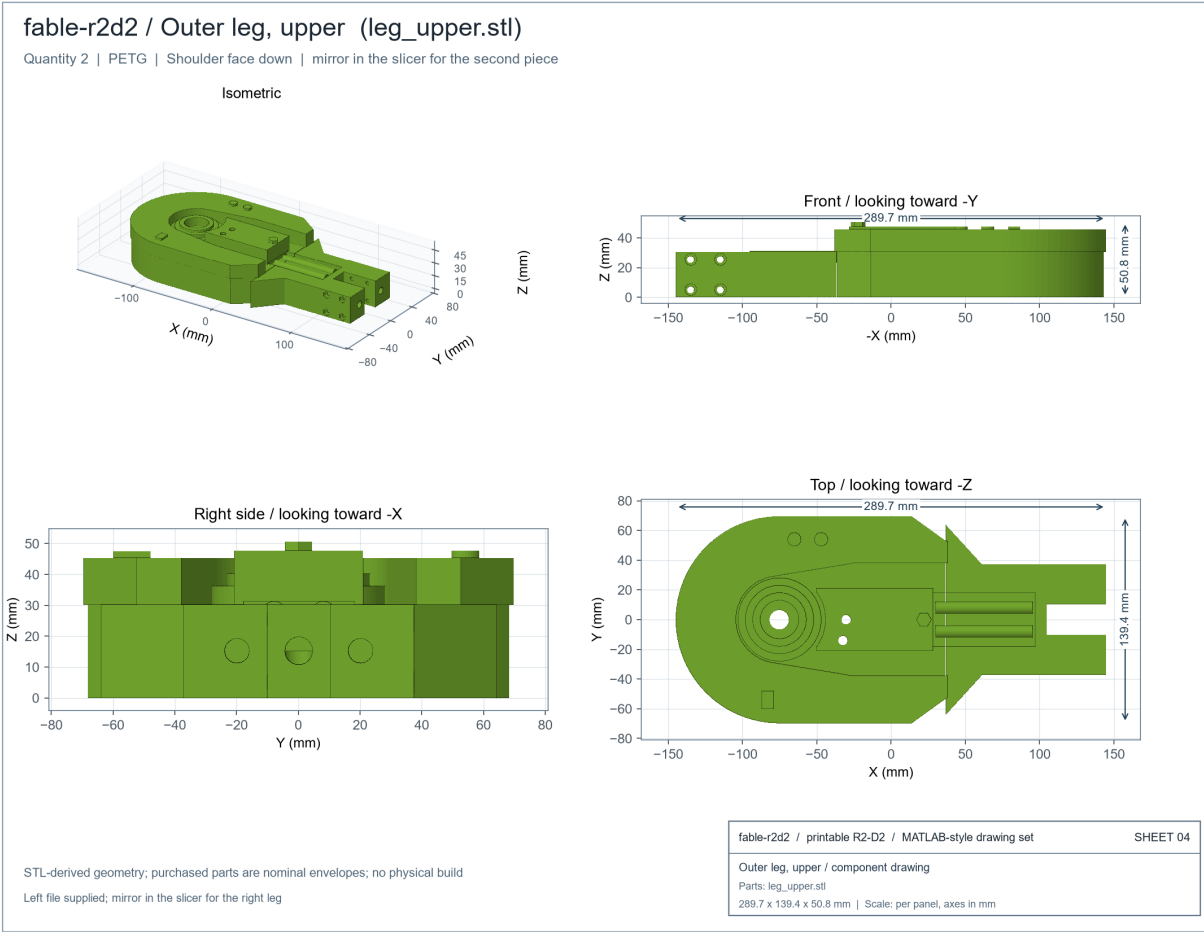
body_upper



body_lower



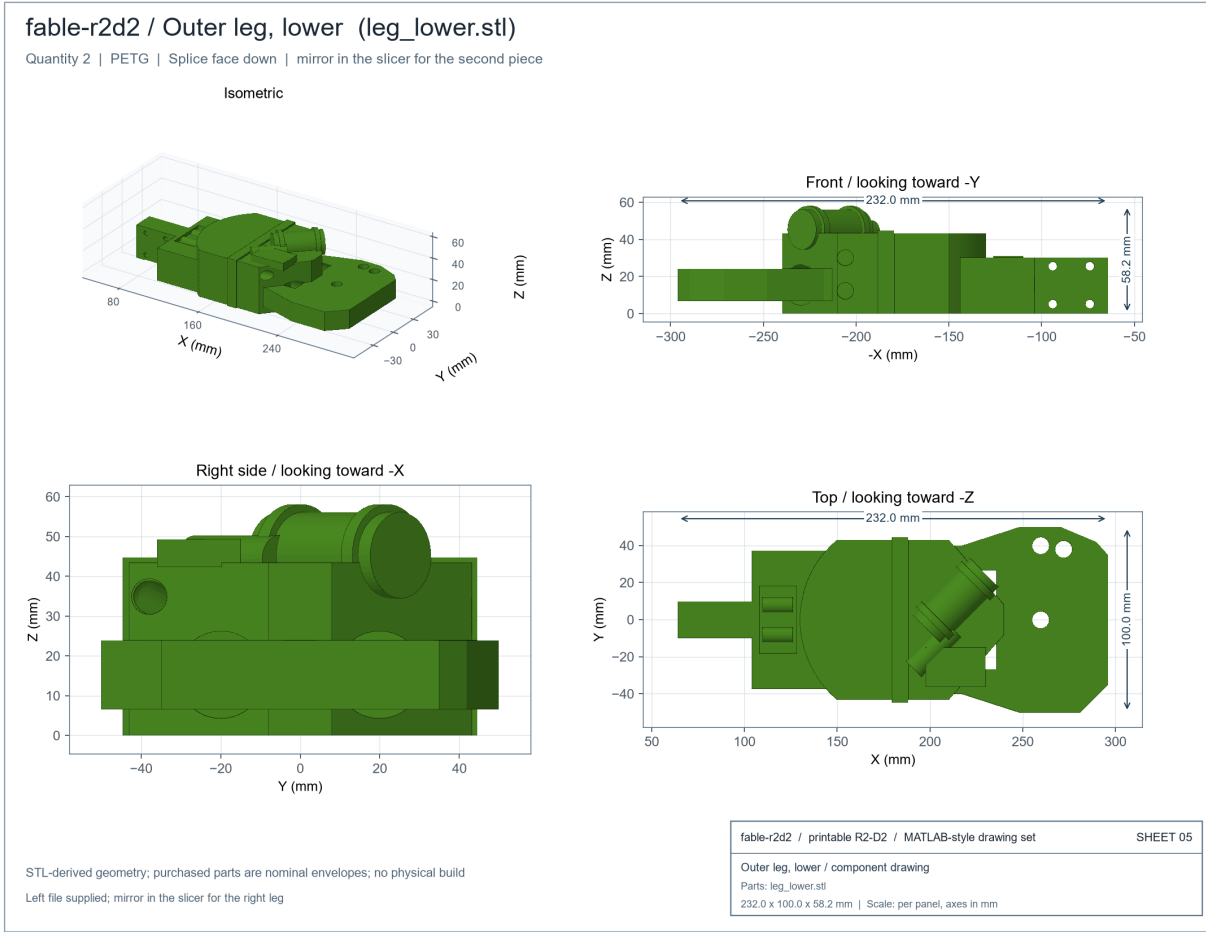
leg_upper



output/drawings/components/leg_upper.png - Left file supplied; mirror in the slicer for the right leg

Field	Value	Field	Value
Quantity	2	Material	PETG
Envelope mm	289.70 x 139.40 x 50.79	Solid mass	1445.2 g
Mesh volume	1138.0 cm3	Triangles	10820
Watertight	True	Solids	1
Plate orientation	Shoulder face down	Mirror for the pair	True
Sliced mass	612.4 g	Sliced time	20.04 h

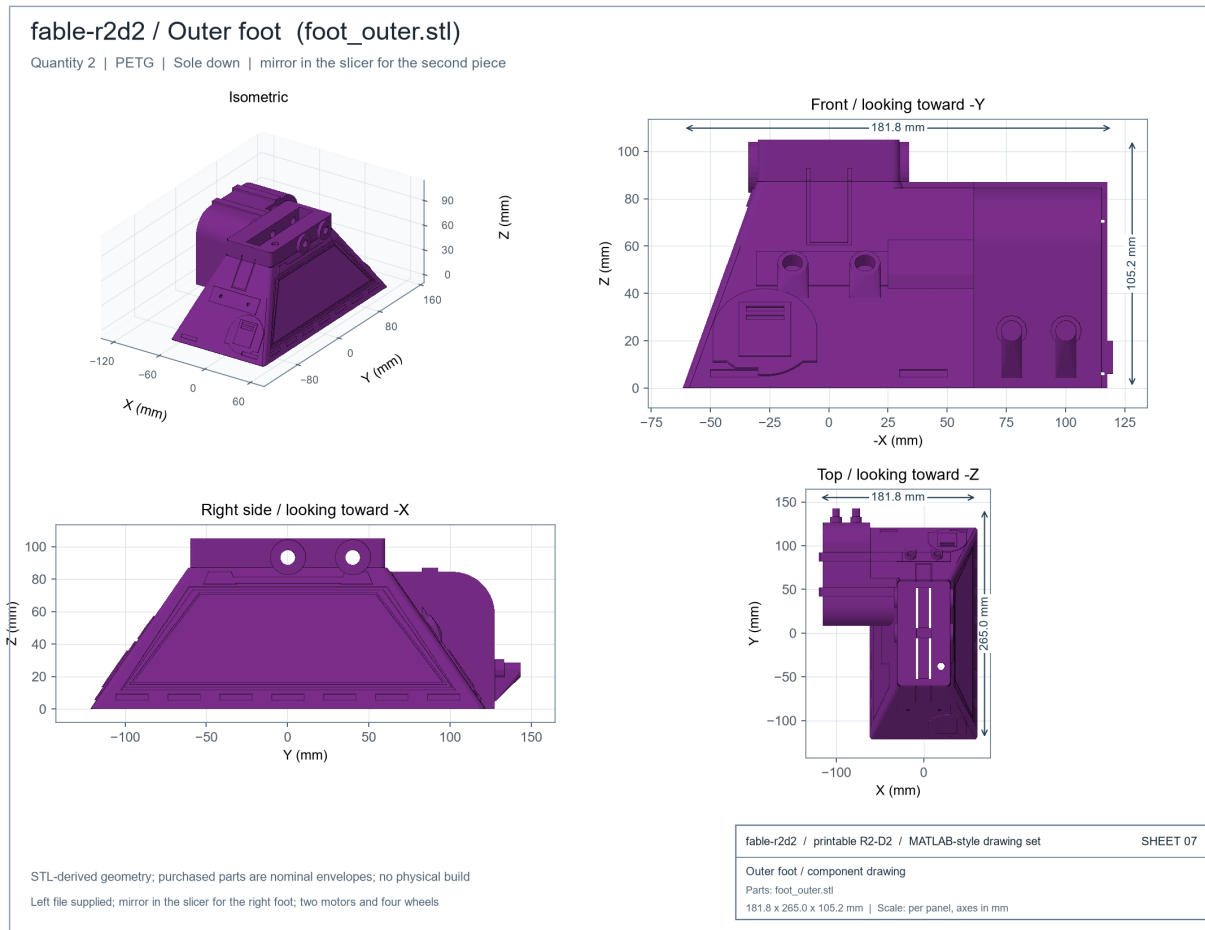
leg_lower



output/drawings/components/leg_lower.png - Left file supplied; mirror in the slicer for the right leg

Field	Value	Field	Value
Quantity	2	Material	PETG
Envelope mm	232.00 x 100.00 x 58.25	Solid mass	666.0 g
Mesh volume	524.4 cm3	Triangles	7314
Watertight	True	Solids	1
Plate orientation	Splice face down	Mirror for the pair	True
Sliced mass	314.5 g	Sliced time	11.20 h

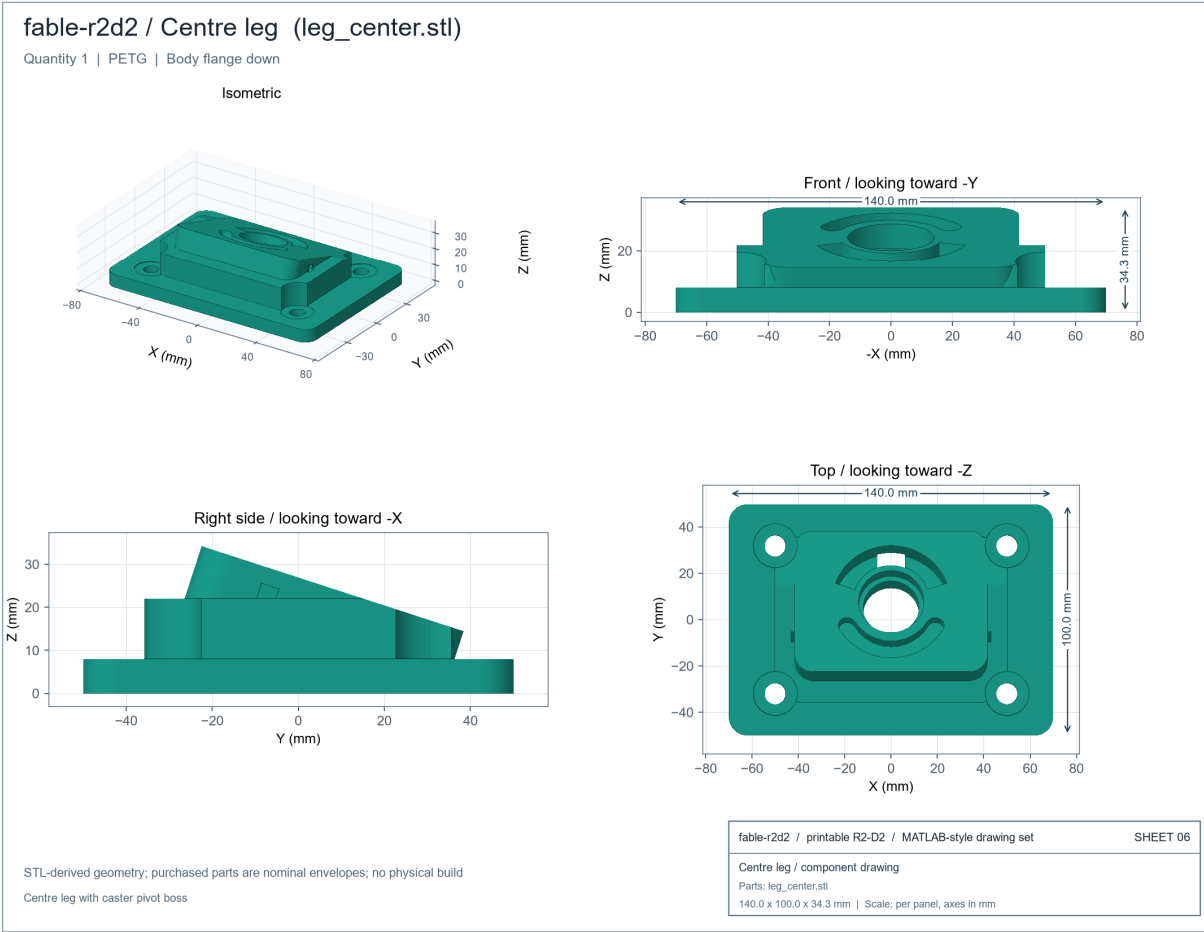
foot_outer



output/drawings/components/foot_outer.png - Left file supplied; mirror in the slicer for the right foot; two motors and four wheels

Field	Value	Field	Value
Quantity	2	Material	PETG
Envelope mm	181.80 x 265.00 x 105.20	Solid mass	855.0 g
Mesh volume	673.2 cm3	Triangles	15452
Watertight	True	Solids	1
Plate orientation	Sole down	Mirror for the pair	True
Sliced mass	680.9 g	Sliced time	19.59 h

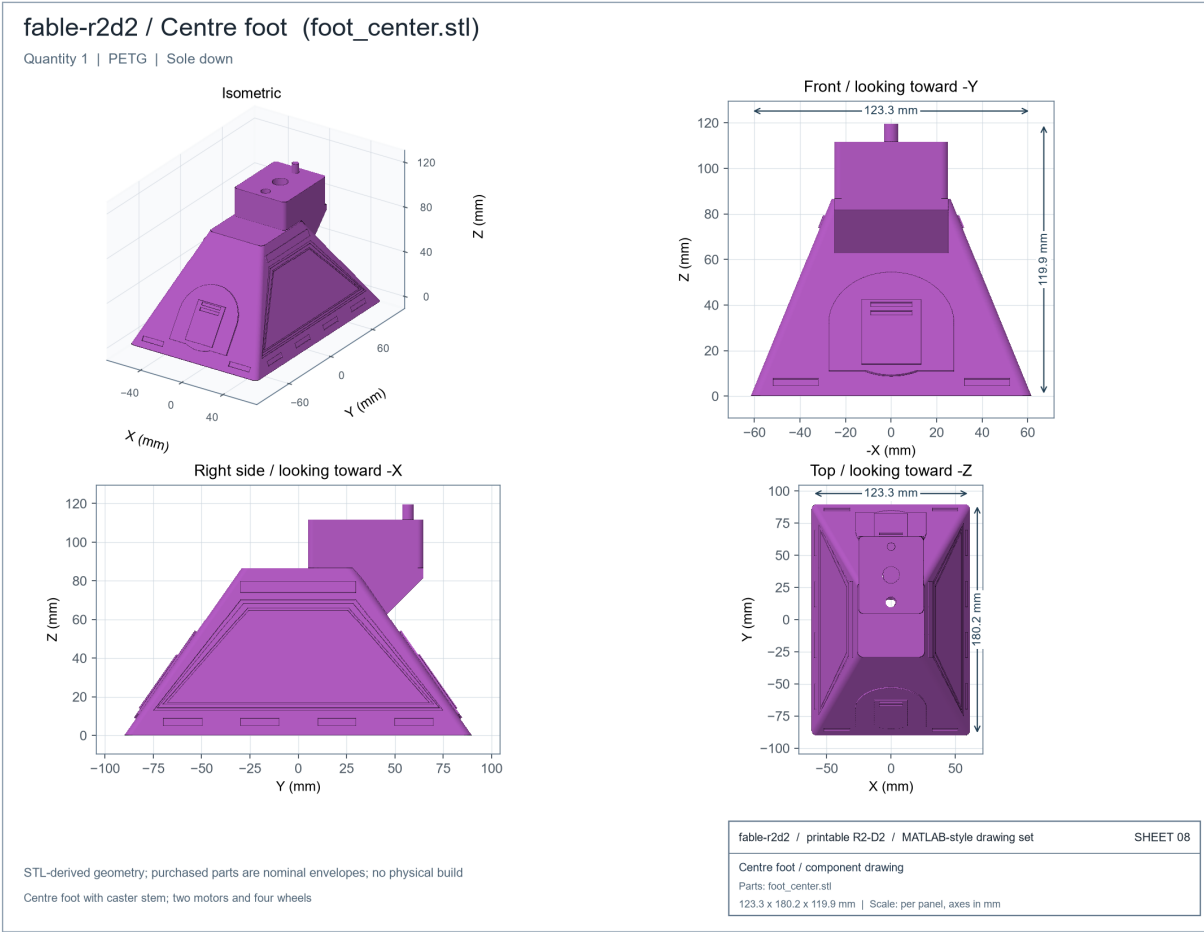
leg_center



output/drawings/components/leg_center.png - Centre leg with caster pivot boss

Field	Value	Field	Value
Quantity	1	Material	PETG
Envelope mm	140.00 x 100.00 x 34.29	Solid mass	246.8 g
Mesh volume	194.3 cm3	Triangles	7198
Watertight	True	Solids	1
Plate orientation	Body flange down	Mirror for the pair	False
Sliced mass	128.3 g	Sliced time	4.25 h

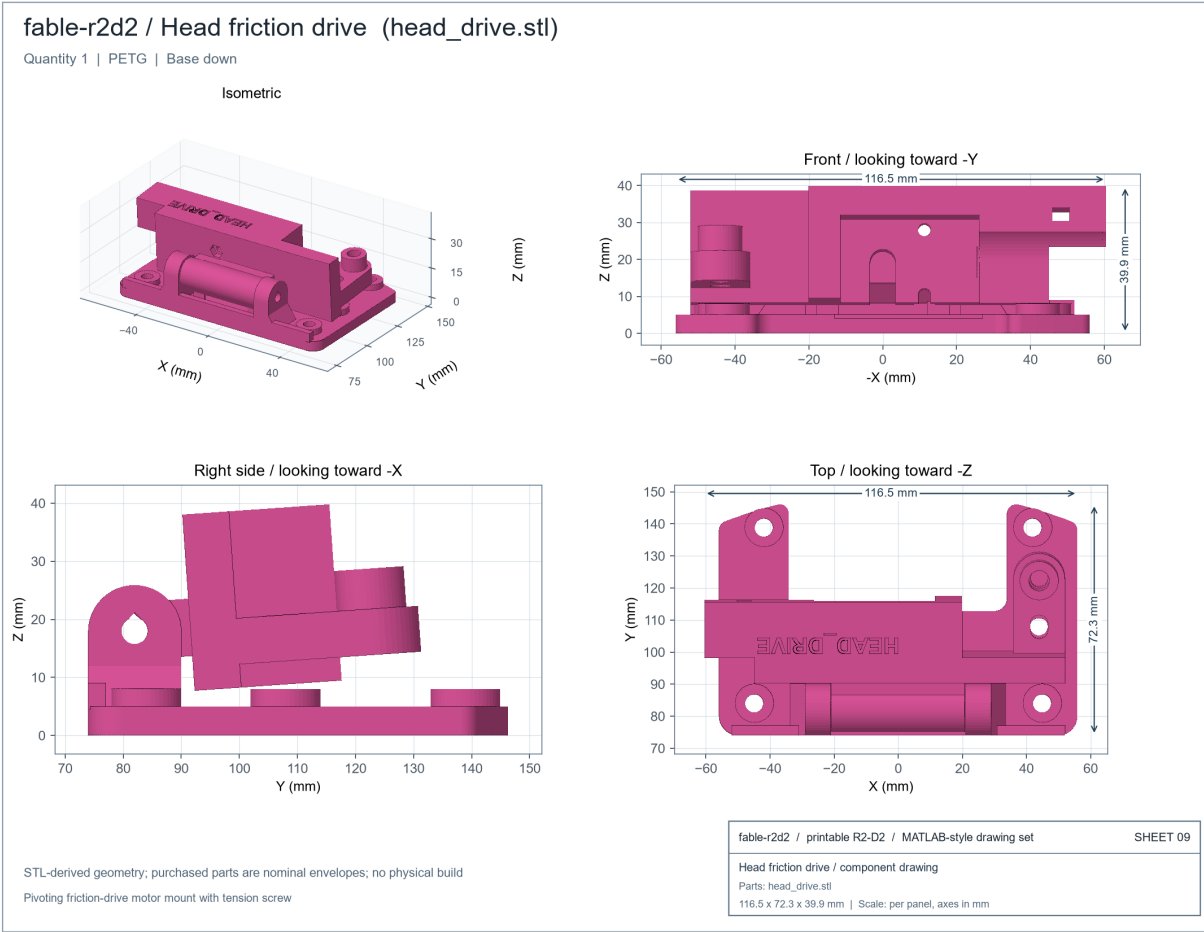
foot_center



output/drawings/components/foot_center.png - Centre foot with caster stem; two motors and four wheels

Field	Value	Field	Value
Quantity	1	Material	PETG
Envelope mm	123.30 x 180.20 x 119.90	Solid mass	510.1 g
Mesh volume	401.6 cm3	Triangles	8636
Watertight	True	Solids	1
Plate orientation	Sole down	Mirror for the pair	False
Sliced mass	330.7 g	Sliced time	9.48 h

head_drive



output/drawings/components/head_drive.png - Pivoting friction-drive motor mount with tension screw

Field	Value	Field	Value
Quantity	1	Material	PETG
Envelope mm	116.50 x 72.28 x 39.91	Solid mass	106.8 g
Mesh volume	84.1 cm3	Triangles	7752
Watertight	True	Solids	1
Plate orientation	Base down	Mirror for the pair	False
Sliced mass	81.3 g	Sliced time	3.00 h

3 Purchased parts

Every item that is not printed. Prices were researched on 2026-09-12 and are listed per item or per stated pack; they are not quotations. Nothing has been ordered. Check availability and the exact rating notes before buying, because a substitute with a different envelope will not fit the printed pockets.

3.1 Electronics, motors and power

Reference designators match the four circuit sheets in chapter 5 and electronics/wiring.csv.

Ref	Qty	Item, part number and supplier	Unit USD	Line USD	Fit, rating and source
A1	1	Raspberry Pi 4 Model B, 4 GB - control server, Wi-Fi, head displays, audio <i>Adafruit 4296 - Adafruit</i>	120.00	120.00	Powered from the 5 V regulator through a USB-C plug breakout (J12), not from a wall adapter. Needs the 32 GB microSD SD1. source
A2	1	Adafruit KB2040 (RP2040) - motor and NeoPixel controller <i>Adafruit 5302 - Adafruit</i>	8.95	8.95	Powered and commanded from a Pi USB 2.0 port over cable J11. Pin map is fixed by firmware/kb2040/code.py. source
U1	1	DRV8833 dual H-bridge - driver 1, left foot (lf-front, lf-rear) <i>Adafruit 3297 - Adafruit</i>	5.95	5.95	VMOTOR from the 6 V regulator U6. Keep both 0.2 ohm sense resistors: 1 A per motor. source
U2	1	DRV8833 dual H-bridge - driver 2, right foot (rf-front, rf-rear) <i>Adafruit 3297 - Adafruit</i>	5.95	5.95	VMOTOR from the 6 V regulator U7. source
U3	1	DRV8833 dual H-bridge - driver 3, centre foot (cf-front, cf-rear) <i>Adafruit 3297 - Adafruit</i>	5.95	5.95	VMOTOR from the 6 V regulator U6. source
U4	1	DRV8833 dual H-bridge - driver 4, head friction drive (channel A only) <i>Adafruit 3297 - Adafruit</i>	5.95	5.95	VMOTOR from the 6 V regulator U7. BIN1/BIN2 tied to GND; BOUT1/BOUT2 unused. source
U5	1	Pololu D36V50F5 step-down regulator, 12 V to 5 V 5.5 A - Pi, displays, NeoPixels, amplifier <i>Pololu 4091 - Pololu</i>	39.95	39.95	Fed from the 12 V bus through fuse F3 (3 A). Pololu stock is rationed; order early. source
U6	1	Pololu D36V50F6 step-down regulator, 12 V to 6 V 5.5 A - motor rail A (drivers U1 and U3) <i>Pololu 4092 - Pololu</i>	39.95	39.95	Fed from the 12 V bus through fuse F4 (5 A). Four motors x 1 A limit = 4 A worst case. source
U7	1	Pololu D36V50F6 step-down regulator, 12 V to 6 V 5.5 A - motor rail B (drivers U2 and U4) <i>Pololu 4092 - Pololu</i>	39.95	39.95	Fed from the 12 V bus through fuse F5 (5 A). Three motors x 1 A limit = 3 A worst case. source
U8	1	PAM8302 mono class D amplifier, 2.5 W into 4 ohm - dome and body sound <i>Adafruit 2130 - Adafruit</i>	3.95	3.95	Runs on the 5 V rail. Input is the Pi 3.5 mm jack, left and right summed through two 10 k resistors (R4). source
U9	1	ADS1115 16 bit 4 channel ADC, STEMMA QT - battery voltage sense <i>Adafruit 1085 - Adafruit</i>	14.95	14.95	Address 0x48 on the Pi I2C-1 bus, single-ended input A0, read by firmware/pi/battery.py. Divider R2 (100 k top) over R3 (15 k bottom). source
SR1	1	Slip ring, 12 wire, 2 A per wire - dome power and signals <i>Adafruit 1195 (Prosper SRC012C-12) - Adafruit</i>	24.95	24.95	Body 12.4 mm dia x 19.5 mm, no flange; clamp in the printed post on the body top plate. 28 AWG leads about 150 mm each side. source
DS1	1	Mini 0.8 inch 8x8 LED matrix with I2C backpack, yellow-green - front logic, upper window <i>Adafruit 872 - Adafruit</i>	9.95	9.95	HT16K33 address 0x70: no address jumper bridged. source
DS2	1	Mini 0.8 inch 8x8 LED matrix with I2C backpack, yellow-green - front logic, lower window <i>Adafruit 872 - Adafruit</i>	9.95	9.95	HT16K33 address 0x71: bridge jumper A0. source
DS3	1	Mini 0.8 inch 8x8 LED matrix with I2C backpack, red - rear logic, left half <i>Adafruit 870 - Adafruit</i>	9.95	9.95	HT16K33 address 0x72: bridge jumper A1. source

Ref	Qty	Item, part number and supplier	Unit USD	Line USD	Fit, rating and source
DS4	1	Mini 0.8 inch 8x8 LED matrix with I2C backpack, red - rear logic, right half <i>Adafruit 870 - Adafruit</i>	9.95	9.95	HT16K33 address 0x73: bridge jumpers A0 and A1. source
DS5	1	1.28 inch 240x240 round TFT, GC9A01A, EYESPI - radar eye <i>Adafruit 6178 - Adafruit</i>	17.50	17.50	Pi SPI0 through the slip ring. Backlight pin tied to 5 V through R1 (100 ohm) inside the dome. source
LED1	1	NeoPixel Jewel, 7 x 5050 RGB - front PSI <i>Adafruit 2226 - Adafruit</i>	5.95	5.95	Chain positions 0-6. Data in from slip ring wire 5, data out to LED2. source
LED2	1	NeoPixel Jewel, 7 x 5050 RGB - rear PSI <i>Adafruit 2226 - Adafruit</i>	5.95	5.95	Chain positions 7-13. Data out to LED3. source
LED3-LED5	1	Diffused 8 mm through-hole NeoPixels, pack of 5 - holoprojectors HP1, HP2, HP3 <i>Adafruit 1734 - Adafruit</i>	4.95	4.95	Three used, two spare; chain positions 14-16. Page reads Out of stock on 2026-09-12; alternate is Adafruit 1655 (5050 with driver, 10 pack, \$4.50) on a hand-made carrier. source
LS1	1	Speaker, 3 inch, 4 ohm 3 W - behind the front vent <i>Adafruit 1314 - Adafruit</i>	1.95	1.95	Page reads Out of stock on 2026-09-12. Noted alternate: Adafruit 1313, 3 inch 8 ohm 1 W, \$1.95, same 77.8 mm footprint and 60 mm tab spacing, about half the output. source
M1	1	TT gearmotor 1:48, 3-6 V - left foot, front <i>Adafruit 3777 - Adafruit</i>	2.95	2.95	Driver U1 channel A. 22 AWG extension to a JST-XH pair at the driver. source
M2	1	TT gearmotor 1:48, 3-6 V - left foot, rear <i>Adafruit 3777 - Adafruit</i>	2.95	2.95	Driver U1 channel B. source
M3	1	TT gearmotor 1:48, 3-6 V - right foot, front <i>Adafruit 3777 - Adafruit</i>	2.95	2.95	Driver U2 channel A. source
M4	1	TT gearmotor 1:48, 3-6 V - right foot, rear <i>Adafruit 3777 - Adafruit</i>	2.95	2.95	Driver U2 channel B. source
M5	1	TT gearmotor 1:48, 3-6 V - centre foot, front <i>Adafruit 3777 - Adafruit</i>	2.95	2.95	Driver U3 channel A. source
M6	1	TT gearmotor 1:48, 3-6 V - centre foot, rear <i>Adafruit 3777 - Adafruit</i>	2.95	2.95	Driver U3 channel B. source
M7	1	TT gearmotor 1:48, 3-6 V - head friction drive <i>Adafruit 3777 - Adafruit</i>	2.95	2.95	Driver U4 channel A, under the body top plate on the tensioned mount. source
W1-W14	14	Orange wheel for TT motor, 63 x 29 mm <i>Adafruit 3766 - Adafruit</i>	1.50	21.00	Thirteen used (four per foot, one on the head drive) and one spare. Page reads Out of stock on 2026-09-12. source
BT1	1	Battery, 12 V 7 Ah sealed lead acid, F2 terminals <i>Power-Sonic PS-1270 F2 - TRC Electronics</i>	19.00	19.00	151 x 65 x 94 mm, 100 mm over the terminals, about 2 kg. Sits on the shelf in body_lower, strapped. Never charge it inverted. source
G1	1	SLA charger, 3 stage, 12 V 1.5 A, 5.5 x 2.1 mm centre-positive barrel plug <i>PowerStream PST-P2012-A12B - PowerStream</i>	22.00	22.00	Bulk 1.5 A, absorb 14.75 V, float 13.75 V. External; it plugs into J1 with the robot switched off. source
J1	1	Panel charge jack, 5.5 mm OD barrel, threaded bushing <i>Switchcraft L722A (DigiKey SC1151-ND) - DigiKey</i>	6.65	6.65	Rated 5 A 48 V. Mounts in the rear lower band of body_lower. Wired to the battery side of SW1 through fuse F2, so it charges with the robot off. source
SW1	1	Main power switch, SPST rocker, 20 A 12 V DC, snap-in <i>Carling V1D2S00B-GZCXX-1XX-XFWD1 - DigiKey</i>	8.63	8.63	Panel cutout 36.83 x 21.08 mm in the rear lower band, beside J1. 0.250 inch quick-connect terminals. source
FH1-FH6	6	In-line ATO/ATC blade fuse holder, 20 A 32 V, 16 AWG leads <i>Littelfuse FHAC0001ZXJ (DigiKey F3209-ND) - DigiKey</i>	7.37	44.22	Six: FH1 main 15 A, FH2 charge 3 A, FH3 Pi rail 3 A, FH4 motor rail A 5 A, FH5 motor rail B 5 A, FH6 dome 5 V feed 2 A. The design brief listed five and left the dome holder out; the dome fuse needs its own holder. source
F1	1	Fuse, ATO blade, 15 A 32 V - main, at the battery positive lead <i>Littelfuse 0287015.PXCN - DigiKey</i>	0.44	0.44	Within 100 mm of the battery terminal. Buy spares; they cost \$0.44 each. source
F2	1	Fuse, ATO blade, 3 A 32 V - charge port branch <i>Littelfuse 0287003.PXCN - DigiKey</i>	0.44	0.44	Charger delivers 1.5 A maximum. source
F3	1	Fuse, ATO blade, 3 A 32 V - 5 V regulator input <i>Littelfuse 0287003.PXCN - DigiKey</i>	0.44	0.44	Measured 12 V draw of the 5 V rail is 0.65 A idle, 1.4 A worst case. source

Ref	Qty	Item, part number and supplier	Unit USD	Line USD	Fit, rating and source
F4	1	Fuse, ATO blade, 5 A 32 V - 6 V motor rail A input <i>Littelfuse 0287005.PXCN - DigiKey</i>	0.44	0.44	Four motors at the DRV8833 1 A limit is 24 W, 2.2 A from 12 V. source
F5	1	Fuse, ATO blade, 5 A 32 V - 6 V motor rail B input <i>Littelfuse 0287005.PXCN - DigiKey</i>	0.44	0.44	Three motors at the 1 A limit is 18 W, 1.7 A from 12 V. source
F6	1	Fuse, ATO blade, 2 A 32 V - dome 5 V feed through the slip ring <i>Littelfuse 0287002.PXCN - DigiKey</i>	0.44	0.44	Dome worst case is 1.6 A (17 NeoPixels at 60 mA, four matrices, one TFT). Two slip-ring wires carry the 5 V, two carry the return. source
TB1	6	F2 female spade terminal, 0.250 inch, insulated crimp, 14-16 AWG <i>TE 3-520408-2 - DigiKey</i>	0.59	3.54	Two on the battery, two on SW1, two spare. source
WR1	1	Silicone wire, 16 AWG, red and black, 25 ft each colour <i>BNTECHGO 16 AWG 50 ft kit - BNTECHGO</i>	19.68	19.68	Battery, switch, fuse and regulator input runs. 22 A chassis rating. source
WR2	1	Silicone wire, 22 AWG, six colours, 30 ft each <i>BNTECHGO 22 AWG 6 colour kit - BNTECHGO</i>	19.68	19.68	Motor extensions, 5 V and 6 V branches, display and NeoPixel signals. 7 A chassis rating. source
J2-J9	8	JST-XH 2 pin cable matching pair, 2.5 mm pitch, 200 mm <i>Adafruit 4872 - Adafruit</i>	0.95	7.60	Seven motor connections plus one spare. Male half at the driver, female half on the motor lead. source
J10	4	Anderson Powerpole 15-45 A housing (two red, two black) <i>Anderson 1327-BK - DigiKey</i>	0.64	2.56	One stacked pair on the battery lead and one on the harness side. source
J10C	4	Anderson Powerpole contact, 15 A, 16-20 AWG, silver <i>Anderson 1332-BK - DigiKey</i>	0.43	1.72	Crimped onto 16 AWG; one per housing. source
J11	1	USB A to USB C cable, 1 m - Pi to KB2040 power and serial <i>Adafruit 4474 - Adafruit</i>	4.95	4.95	Carries the USB CDC link the Pi sees as /dev/ttyACM1 and powers the KB2040. source
J12	1	USB Type C plug breakout - 5 V feed into the Pi USB-C power input <i>Adafruit 5978 - Adafruit</i>	2.95	2.95	Solder 18 AWG from U5 VOUT to VBUS and GND. The board's 5.1 k CC1 resistor is what makes the Pi accept the supply. Keep the run under 250 mm. source
R1	1	Through-hole resistors, 100 ohm 5% 1/4 W, pack of 25 <i>Adafruit 4293 - Adafruit</i>	0.75	0.75	One used: TFT backlight pin to 5 V inside the dome. Page reads Out of stock on 2026-09-12; any 100 ohm 1/4 W part works. source
R2	1	Through-hole resistors, 100 k ohm 5% 1/4 W, pack of 25 <i>Adafruit 2787 - Adafruit</i>	0.95	0.95	One used: top leg of the battery divider, 12 V bus to the ADS1115 A0 node. source
R3	1	Through-hole resistor, 15 k ohm 5% 1/4 W <i>Stackpole CF14JT15K0 - DigiKey</i>	0.50	0.50	One used: bottom leg of the battery divider, ADS1115 A0 node to ground. Price not read from the page; DigiKey singles are a few cents, minimum order applies. source
R4	1	Through-hole resistors, 10 k ohm 5% 1/4 W, pack of 25 <i>Adafruit 2784 - Adafruit</i>	0.75	0.75	Three used: two sum the Pi left and right audio into the PAM8302 A+, one pulls up the dome index sensor. Page reads Out of stock on 2026-09-12. source
SD1	1	microSD card, 32 GB, A2 class - Raspberry Pi OS <i>Official Raspberry Pi A2 32 GB (Adafruit 6010) - Adafruit</i>	13.69	13.69	Allowance: the Adafruit listing reads No longer stocked on 2026-09-12. Any A2 class 32 GB card works; the price is the Adafruit list price. source
SQ1	1	Hall effect switch US5881LUA - optional dome index (home) sensor <i>Adafruit 158 - Adafruit</i>	2.00	2.00	Optional. Mounts on the body top plate facing a magnet on the dome plate; open drain to KB2040 A3, which has its internal pull-up enabled in firmware. Fit the R4 10 k pull-up as well if the run is long. source
Subtotal, 54 line items				618.66	

Source: bom/electronics.csv. Subtotal USD 618.66, quantity times unit price, tax, shipping, tools and filament excluded.

3.2 Mechanical hardware, bearings and fasteners

Steel carries every load path that matters: the shoulder bolts, the threaded rods, the caster bearings and the lazy susan.

Ref	Qty	Item, part number and supplier	Unit USD	Line USD	Fit, rating and source
H01	1	Lazy susan ring bearing, 9 inch round steel, 228.6 mm OD, 114.3 mm centre hole <i>Triangle 9C (WW Hardware TR09C) - WW Hardware</i>	6.79	6.79	Dome bearing between the body top plate and the dome plate. 750 lb rating; four M5 screws per race through the access holes. source
H02	1	Ball bearings 6001-2RS, 12 x 28 x 8 mm, pack of 10 <i>PGN 6001-2RS 10 pack - Amazon</i>	11.45	11.45	Six used: two per shoulder pivot (four) and two in the centre-foot caster (params.scad caster_bearing_od 28, caster_bearing_t 8). Four spare. source
H03	2	Hex bolt M12-1.75 x 130 mm, class 8.8 zinc - shoulder pivot <i>Bolt Depot 6272 - Bolt Depot</i>	2.83	5.66	One per shoulder, through the leg hub, the bushings and the 30 mm body boss. Use a steel crush sleeve so the printed boss is never clamped directly. source
H04	1	Nylon-insert lock nuts M12-1.75, DIN 985, stainless, pack of 20 <i>BNUOK M12 nyloc 20 pack - Amazon</i>	8.99	8.99	Two used on the shoulder bolts; the rest are spares and centre-leg hardware. source
H05	8	Fender washer M12, 12 x 37 mm, zinc <i>Bolt Depot 17834 - Bolt Depot</i>	0.56	4.48	Large washers spread the shoulder clamp load over the printed faces; loads.md asks for about 40 mm diameter, so 37 mm is the closest stock size. source
H06	8	Flat washer M12, zinc <i>Bolt Depot 4531 - Bolt Depot</i>	0.08	0.64	Under the nyloc and against the bearing inner races. source
H07	2	Flanged bronze bushing, 12 mm bore x 16 mm OD x 20 mm, pack of 2 <i>HARFINGTON flanged bushing - Amazon</i>	12.99	25.98	Four bushings, two per shoulder, as the low-friction sleeve between the leg hub and the pivot bolt. source
H08	1	Dowel pins 6 x 30 mm, steel, pack of 50 <i>Fabory 6x30 dowel pin 50 pack - Amazon</i>	34.90	34.90	Four used: two index pins per shoulder at radius 45 mm, at the two-leg (0 deg) and three-leg (18 deg) positions. source
H09	4	Threaded rod M8 x 1.25, 1 m, zinc steel <i>Bolt Depot 23774 - Bolt Depot</i>	11.25	45.00	Cut plan: two rods give the four body rods at 385 mm each (2 x 385 = 770 mm, 230 mm offcut); two rods give the four leg rods at 425 mm each (2 x 425 = 850 mm, 150 mm offcut). Deburr and chamfer every cut end. source
H10	40	Hex nut M8-1.25, class 8.8 zinc <i>Bolt Depot 4788 - Bolt Depot</i>	0.06	2.40	Two per rod end (jam pairs) on eight rods plus the centre-leg flange bolts. \$4.84 per 100 at Bolt Depot. source
H11	1	Nylon-insert lock nuts M8-1.25, pack of 50 <i>Yinpecly M8 nyloc 50 pack - Amazon</i>	6.99	6.99	Final nut on every M8 rod end, the four centre-leg flange bolts and the two ankle bolts per outer foot. source
H12	40	Flat washer M8, zinc <i>Bolt Depot 4529 - Bolt Depot</i>	0.05	2.00	Under every M8 nut that does not get a fender washer. \$2.33 per 100. source
H13	16	Fender washer M8, 8 x 24 mm, zinc <i>Bolt Depot 17832 - Bolt Depot</i>	0.28	4.48	One at each end of the eight rods, so the 1 kN preload in loads.md sits on about 5 MPa of printed face. source
H14	12	Socket cap screw M5 x 0.8 x 12 mm, stainless 18-8 <i>Bolt Depot metric socket cap 5 mm x 0.8 mm x 12 mm - Bolt Depot</i>	0.30	3.60	Eight used: four into M5 heat-set inserts in the body top plate and four into the dome plate, through the lazy-susan access holes. Bolt Depot prices load by script and were not readable on 2026-09-12. source 5mm_x_0.8mm)
H15	1	Heat-set inserts M5 x 9.5 mm, brass, pack of 50 (ruthex RX-M5x9.5) <i>ruthex RX-M5x9.5 - Amazon</i>	9.99	9.99	Eight for the lazy susan, two for the head-drive tensioner. CAD pocket 6.63 mm diameter x 10.0 mm deep. source
H16	4	Socket cap screw M5 x 0.8 x 40 mm, stainless - head-drive tensioner <i>uxcell M5 x 40 socket cap, 10 pack - Amazon</i>	0.40	1.60	Two used on the tensioner, two spare. Listed at \$3.99 per 10 in a search result; the page itself was not fetched, so the price is an allowance. source
H17	1	Knurled thumb nuts M5-0.8, 304 stainless, pack of 10 <i>uxcell M5 thumb nut 10 pack - Amazon</i>	9.39	9.39	Two used: tool-free tension stop on the head-drive adjuster screws. source
H18	1	Compression spring 10 mm OD x 25 mm free length, 0.6 mm wire, 304 stainless, pack of 5 <i>Compression spring 10 x 25 x 0.6 - Amazon</i>	2.24	2.24	Two used on the head-drive tensioner; stack two per screw if the friction wheel slips on the dome ring. source
H19	1	Socket cap screws M4 x 0.7 x 12 mm, 12.9 alloy, pack of 50 <i>M4 x 12 socket cap 50 pack - Amazon</i>	7.29	7.29	Twenty-four used: eight at the body ring seam, eight at the two leg splices, four on the head-drive mount, four holding the electronics tray. source

Ref	Qty	Item, part number and supplier	Unit USD	Line USD	Fit, rating and source
H20	40	Hex nut M4-0.7, class 8.8 zinc <i>Bolt Depot 4784 - Bolt Depot</i>	0.05	2.00	Behind every M4 that does not land in a heat-set insert. \$2.54 per 100. source
H21	50	Flat washer M4, zinc <i>Bolt Depot 4525 - Bolt Depot</i>	0.05	2.50	One under every M4 head and nut. \$1.28 per 100. source
H22	1	Heat-set inserts M4 x 8.1 mm, brass, pack of 50 (ruthex RX-M4x8.1) <i>ruthex RX-M4x8.1 - Amazon</i>	9.99	9.99	Head-drive mount, electronics tray bosses and the leg splice plates. CAD pocket 5.83 mm diameter x 8.6 mm deep. source
H23	1	Socket and button head screw, nut and washer kit, M2 / M3 / M4, 1080 pieces <i>M2 M3 M4 screw kit 1080 pcs - Amazon</i>	12.99	12.99	Source of the M3 x 10 motor-saddle screws (14), the M2 x 6 screws for the four matrix backpacks and the KB2040, and general M3 and M4 spares. source
H24	2	Heat-set inserts M3 x 4 mm, brass, pack of 50 <i>Adafruit 4255 - Adafruit</i>	5.95	11.90	Motor saddles in the three feet (14), dome display mounts, slip-ring clamp and the tray standoff bosses. CAD pocket 4.24 mm diameter x 4.5 mm deep. source
H25	1	Nylon machine screw and stand-off set, M2.5 thread, 380 pieces <i>Adafruit 3299 - Adafruit</i>	16.95	16.95	M2.5 x 8 mm screws and standoffs for the Raspberry Pi (four holes, 58 x 49 mm pattern) and the round TFT (two 2.5 mm holes 22.8 mm apart). Page reads Out of stock on 2026-09-12; Adafruit 3658 is the same set in white. source
H26	1	Nylon standoff kit M3, 260 pieces, 6-20 mm <i>HVAZI M3 standoff kit - Amazon</i>	9.99	9.99	Stands the four DRV8833 boards, both 6 V regulators, the 5 V regulator, the ADS1115 and the amplifier off the electronics tray. source
H27	1	Hook-and-loop cinch straps, 25 mm x 400 mm, pack of 6 <i>MECCANIXITY cinch straps - Amazon</i>	7.59	7.59	Two across the battery on the body_lower shelf, anchored in the printed strap slots. One strap does not reach round the battery on its own. source
H28	1	Cable ties, 8 inch, pack of 100 <i>Skalon 8 inch cable ties - Amazon</i>	3.99	3.99	Harness bundling, strain relief at the slip ring and every leg pass-through. source
H29	1	Adhesive cable-tie mounts, 3/4 inch, pack of 100 <i>GTSE tie mounts - Amazon</i>	14.99	14.99	Route the harness along the inside of the body rings and the legs; each mount also takes an M3 screw if the adhesive lets go. source
H30	1	Closed-cell foam tape, 3/4 x 3/8 inch x 17 ft <i>Simply Conserve foam tape - Amazon</i>	4.99	4.99	Under the battery, behind the speaker and around the dome plate. The buy-box price was not extractable on 2026-09-12; \$4.99 was the first listed price. source
H31	5	Filament, Bambu PETG Basic, 1.75 mm, 1 kg - structural prints <i>Bambu PETG Basic - Bambu Lab US store</i>	17.99	89.95	Body rings, legs, feet, tray and head drive at 0.20 mm layers, five walls, 30-40 percent gyroid. loads.md estimates 4-5 kg for the whole robot. source
H32	1	Filament, Bambu PETG-CF, 1.75 mm, 1 kg - shoulder bosses (optional) <i>Bambu PETG-CF - Bambu Lab US store</i>	31.99	31.99	Optional: highest X-Y bending strength (83 MPa) for the shoulder bosses and leg cores. Needs a hardened steel nozzle and is not AMS lite compatible. source
H33	1	Filament, Bambu PLA Basic, 1.75 mm, 1 kg - dome (optional finish) <i>Bambu PLA Basic - Bambu Lab US store</i>	19.99	19.99	Optional: the dome carries only its own weight on the ring bearing, and PLA holds the panel detail better. Keep the robot out of a hot car either way. source
		Subtotal, 33 line items		433.68	

Source: bom/hardware.csv. Subtotal USD 433.68, quantity times unit price, tax, shipping, tools and filament excluded.

3.3 Purchased-parts total

USD 1,052.34 for both lists together, computed from the CSV files at build time. Filament, tools, the printer, paint, tax and shipping are not in that number. The consumables table in chapter 6 lists what else you must have on hand.

4 Mechanical design

The design envelope and stance, every load-carrying joint, the fits that make the printed parts go together, and the load screening. Reproduced from docs/mechanical.md sections 1, 3, 4 and 5.

1. Design envelope and stance

1.1 Print envelope

Axis	Machine	Rule	Limit used
X	H2D left extruder 325 mm	minus the 3 mm buffer the operator asked for	env_x = 322 mm
Y	H2D left extruder 320 mm	minus the 3 mm buffer	env_y = 317 mm
Z	H2D left extruder 320 mm	"print at max height minus 5 mm"	env_z = 315 mm

Two rules follow from this and they drive the whole design.

- **Maximum height rule.** Every part is made as tall as its geometry allows, up to 315 mm. A taller part is a part that does not need a joint.
- **Minimal part rule.** The package is nine STL files and twelve printed pieces. The electronics tray was folded into body_upper as an integral deck for this reason. The body is two rings because 381.1 mm of body cannot be one 315 mm print. The outer leg is two prints because it is 480 mm long and no bed axis or diagonal reaches 480 mm.

The body diameter is set by the envelope: $\text{body_od} = 317.0 \text{ mm} = \text{env_y}$. A cylinder is limited by the smaller bed axis. The whole model is then scaled 0.68386 from the club-standard 463.55 mm R2-D2 (scale_factor, research/proportions.md section 6).

1.2 Principal sizes

Item	Parameter	Value mm
Body outside diameter	body_od	317.0
Body height, skirt bottom to top edge	body_height	381.1
Ring seam, measured down from the body top	body_seam_from_top	246.0
Upper ring height	body_upper_h	246.0
Lower ring height, skirt included	body_lower_h	135.1
Skirt height	skirt_h	41.3
Body skin wall	body_wall	3.6
Dome outside diameter	dome_od	317.0
Dome height, band bottom to crown	dome_height	196.8
Dome wall	dome_wall (shell 3.2 in cad/dome.scad)	3.0
Dome gap above the body top edge	dome_gap	1.9
Outer leg, shoulder axis to ankle pivot	leg_len	375.9
Leg strut section (fore-aft x thickness)	leg_strut_w x leg_strut_t	74.6 x 30.4
Foot track, centre to centre	leg_track	363.4
Shell bottom edge above the floor	foot_clear	12
Ankle pivot above the floor, legs vertical	ankle_z	105.5

1.3 Three-leg operating stance

This is the stance the robot drives in. The body tilts back 18 degrees (body_tilt) and the two outer legs lean 18 degrees (leg_lean) so the outer feet stand forward of the shoulders. The centre leg trails at the rear and carries the castering third foot.

Measure	Value	Source
Shoulder axis above the floor	463.0 mm	shoulder_z_three_leg
Skirt bottom above the floor	164.5 mm	r2d2.scad skirt_bottom_z_three_leg

Measure	Value	Source
Skirt bottom forward of the shoulder axis	97.0 mm	r2d2.scad skirt_bottom_y_three_leg
Outer foot centre forward of the shoulder axis	116.2 mm	r2d2.scad ankle_y_three_leg
Centre foot centre forward of the shoulder axis	77.0 mm	skirt_bottom_y_three_leg minus caster_trail
Dome crown above the floor	715.9 mm (computed)	$463.0 + 265.9 \cos 18$

research/proportions.md section 6 gives 713.9 mm for this height as an estimate from the club drawings. The 715.9 mm figure is the arithmetic on params.scad and the two agree within 2 mm.

Note on the wheelbase. In this stance the centre foot sits 39.2 mm *behind* the outer foot line, not ahead of it. research/loads.md section 7 assumed the centre foot 290 mm *ahead* of the outer feet with the outer legs vertical. The two geometries are not the same, so every centre-of-gravity and foot-share number in that section must be re-run against params.scad. Section 5 below flags what this changes. The wheel contact patches still overlap: the outer foot wheels sit at 116.2 +/- 45 mm and the centre foot wheels at 77.0 +/- 45 mm (foot_axle_y = 45), so the support polygon runs from 32.0 mm to 161.2 mm forward of the shoulder axis.

1.4 Two-leg display stance

The display stance is the tall, straight, two-legged pose. The centre leg is unbolted and lifted out as one assembly with its foot.

Measure	Value	Source
Shoulder axis above the floor	481.4 mm	ankle_z + leg_len
Dome crown above the floor	747.3 mm (computed)	$481.4 + 67.2 + 1.9 + 196.8$

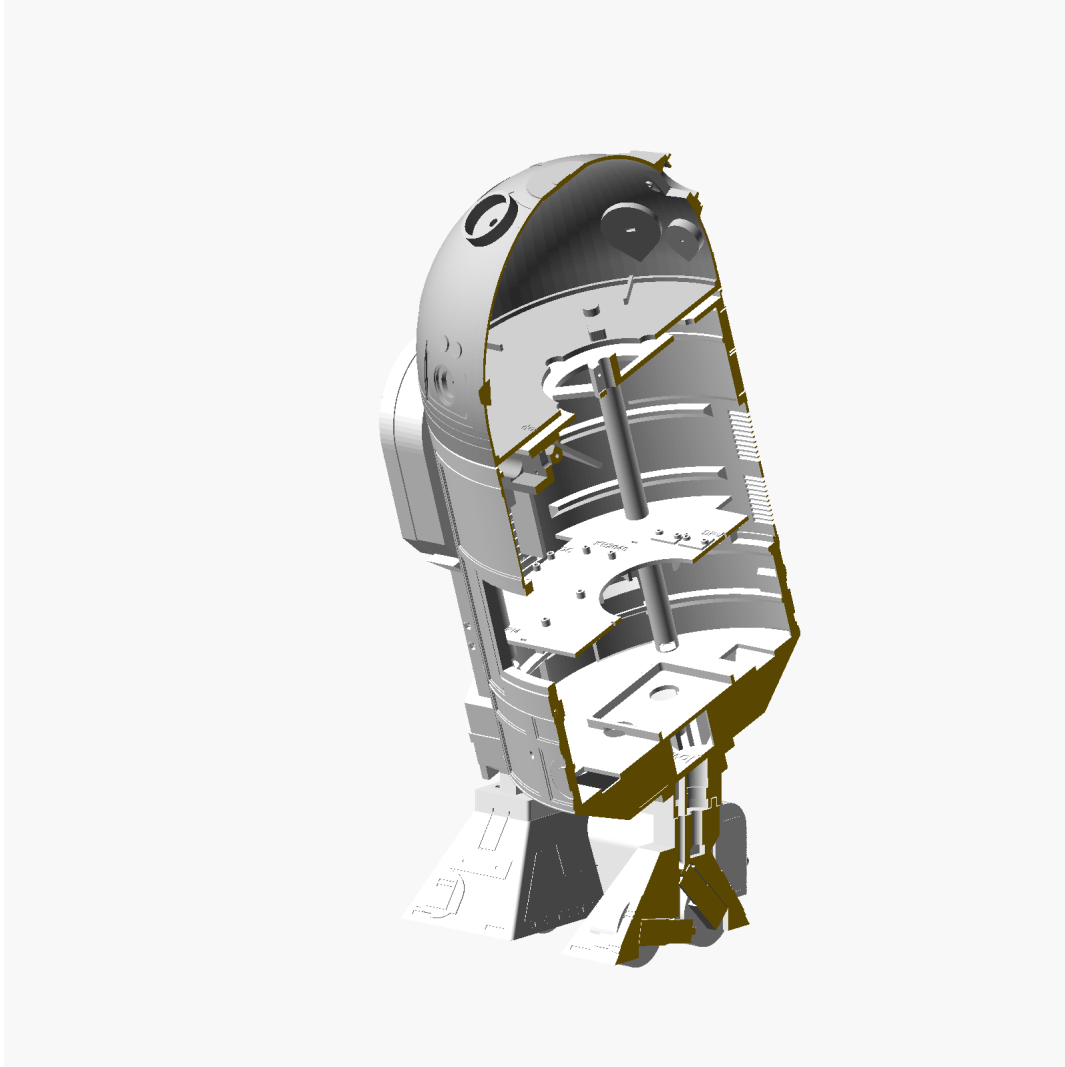
research/proportions.md section 6 gives 745.4 mm scaled from the club drawing. The two agree within 2 mm.

Changing stance needs three things per side and nothing else:

1. the 6 mm index dowel moved from the leg's 18 degree hole to its 0 degree hole;
2. the M8 ankle lock bolt moved from tongue hole B to tongue hole A;
3. the four M8 centre-leg flange bolts removed and the centre leg taken out.

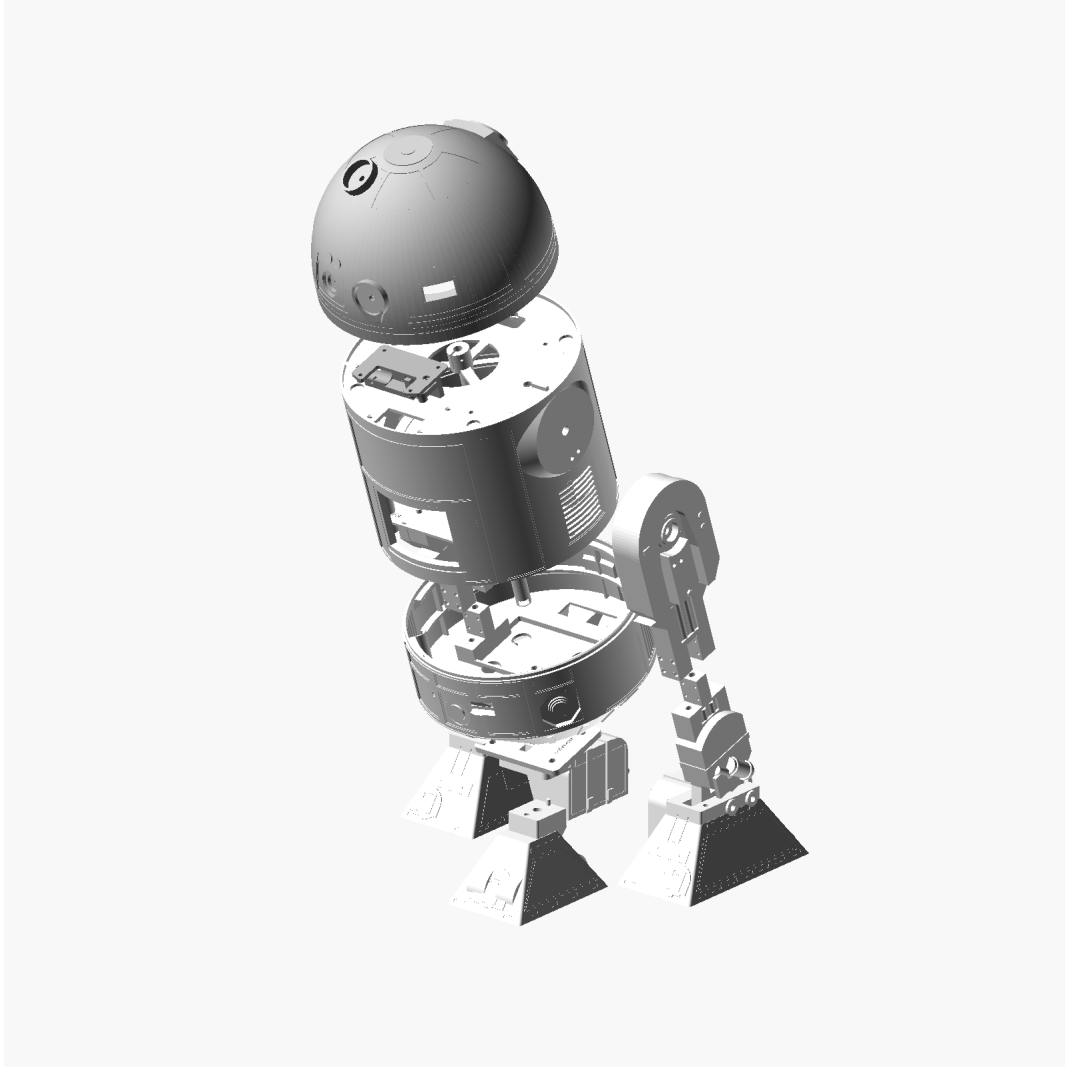
The robot **cannot drive in the two-leg stance**. It has no balance system. The display stance is for standing still, with the robot supported or watched.

Section through the assembly



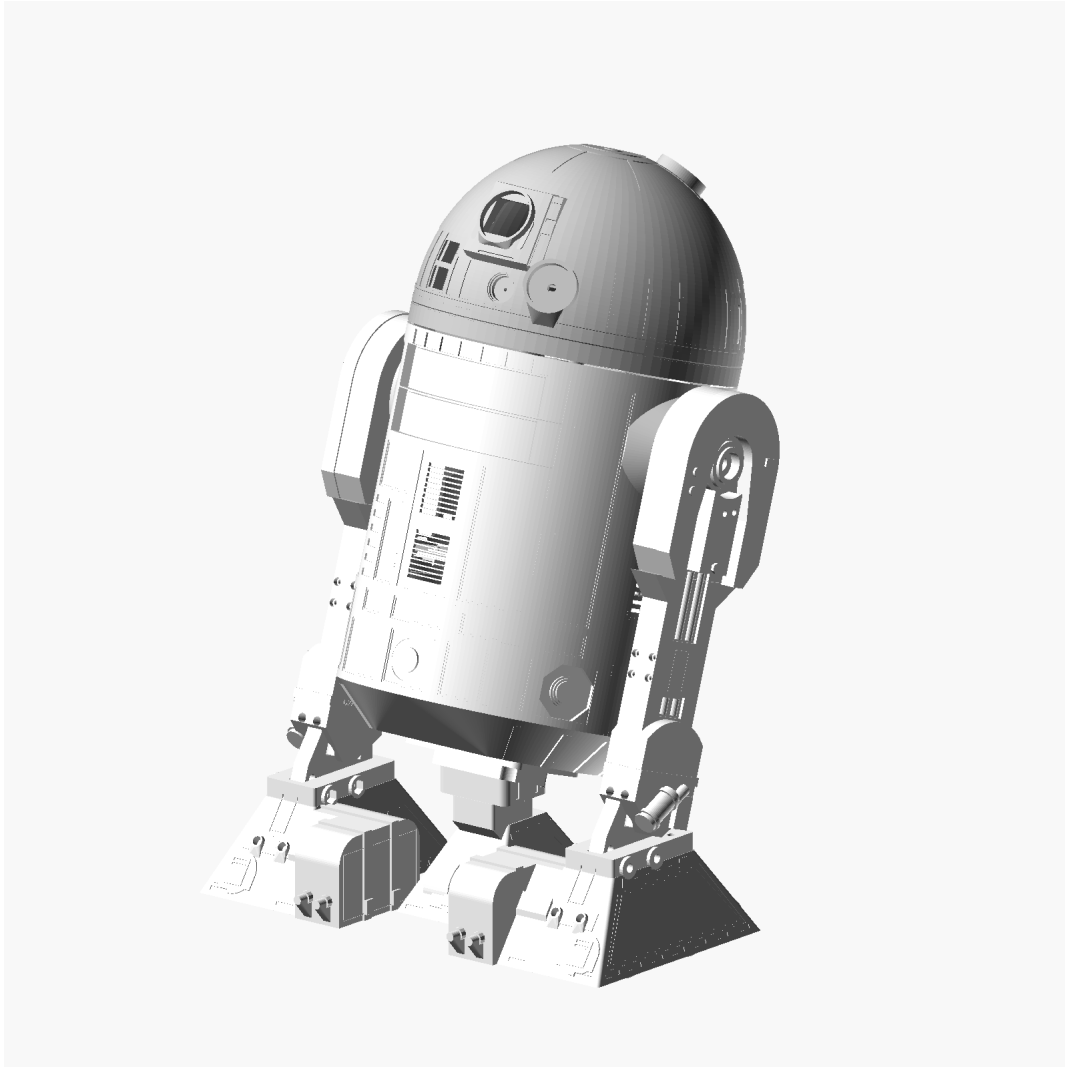
cad/section.png - the native OpenSCAD model cut on the centre plane. Battery bay, electronics deck, ring seam and the dome bearing seat.

Assembly order



cad/exploded.png - the offsets identify the parts and are not assembly dimensions.

Rear access



cad/rear.png - keep the charge port, the switch and the service fasteners reachable with the robot standing.

3. Every joint and how it carries load

3.1 Ring seam - locating lip, 8 x M4, 4 x M8 rods

The two body rings meet 246.0 mm below the body top edge. Three features share the work.

- **Locating lip.** `body_lower` carries a lip at radius 151.6 to 154.6 mm, z 135.1 to 141.1 (`seam_lip_h` tall, `seam_lip_t` thick), with a 15 degree lead-in chamfer on its top outer edge. `body_upper` has the matching socket, radius 151.3 to 154.9, z 135.1 to 141.5. That is **0.3 mm of clearance each side**. The lip takes the shear and sets the concentricity. It is not a clamping feature.
- **Eight M4 bolts.** An internal flange 12 mm wide and 5 mm thick (`seam_flange_w`, `seam_flange_t`) runs round both rings: radius 142.9 to 154.9, z 130.1 to 135.1 on the lower ring, and radius 142.9 to 151.3, z 135.1 to 140.1 on the upper. The bolt circle is radius **148.9 mm at 22.5, 67.5, 112.5 ... degrees** (`seam_bolt_n` = 8). Fit **M4 x 20 socket cap screws from above**; the hex nut pockets open downward on the lower flange.
- **Four M8 threaded rods.** Four rods on a 138 mm radius at 45, 135, 225 and 315 degrees (`rod_n`, `rod_r`, `rod_angle0`) run up to the top plate inside 20 mm bosses (`rod_boss_d`). They put the whole body in compression and carry the dome, the head drive and the shoulder reaction down the ring. The M4 bolts hold the seam closed; **the rods are what makes the body a single column**.

The lower rod nut is not under the floor. Radius 138 mm lies outside the skirt-bottom plate, whose maximum radius is `skirt_bottom_r` = 116.6 mm, so there is no floor under the rod to bolt to. Each rod column therefore starts **on the battery shelf**, and its nut is a **captive hex trap at z 46**, opened downward with a slot facing the body axis. Slide the nut in, feed the rod down into it, and do all the tightening at the top. The top nut sits in an 18 mm diameter x 4 mm counterbore in the top plate, reachable from above until the lazy susan goes on; the nut then stands 2.8 mm proud and clears the dome plate by 5.1 mm.

Preload each rod to about 1 kN. `research/loads.md` section 7 gives 1.6 N.m of torque for that on an M8 with a 16 mm washer, and 1 kN is 12 percent of the 8.2 kN proof load of an M8 class 4.6 rod. Use a 24 mm fender washer under each nut: 1 kN over 402 mm² of washer face is 2.5 MPa, which PETG holds without creeping.

3.2 Shoulder - M12 bolt, bushings, index pins

Each shoulder is a bolted steel pivot. It is not motorised.

- The body side is a flat pad 116 mm across (`shoulder_pad_d`) whose outer face is exactly the tangent plane at $|X| = \text{body}_r = 158.5$ mm, with an internal boss 120 mm across and 30 mm deep behind it (`shoulder_boss_d`, `shoulder_boss_t`), 313.9 mm up the body frame (`shoulder_z`). A hex pocket 13 mm deep opens to the inside of the ring for the M12 nyloc. The fairing that blends the pad into the skin is clipped flat at the top plate, so nothing of it stands above z 375.1.
- The leg side is a U-shaped plate 30.4 mm thick with the horseshoe raised on its **outboard** face. The inboard face is flat, and it sits at $|X| = 166.5$ mm, which is the 8 mm `shoulder_spacer` stack. `cad/body.scad` carries **no horseshoe**: the pad is a plain disc, exactly as `cad/legs.scad` requires.
- One **M12 x 130 class 8.8 hex bolt** (`shoulder_bolt_m`, `bom/hardware.csv` H03) passes through the leg hub, the bushing stack and the 30 mm boss. Its head sits in a 26 mm counterbore 10 mm deep inside the leg hub. A nyloc goes on inside the body.
- Two **flanged bronze bushings**, 12 mm bore x 16 mm OD x 20 mm long with a 20 mm flange 2 mm thick, plus washers, make up the 8 mm spacer stack between the pad and the leg (`shoulder_spacer` = 8).
- **Do not clamp printed faces with a torqued M12.** Fit a steel crush sleeve through the boss so the bolt clamps steel to steel, and use M12 fender washers, 37 mm OD, the closest stock size to the 40 mm `research/loads.md` asks for. At 5 N.m the preload is about 2.1 kN and the face pressure about 1.9 MPa. A 24 mm washer would give 6.6 MPa, which creeps.

Index pins carry the moment, not the bolt. The leg plate carries two 6.2 mm holes (`shoulder_index_d`) at a 45 mm radius (`shoulder_index_r`), at 0 and 18 degrees (`shoulder_index_angles`) from straight below the axis, placed toward the **rear**. The body pad carries **two** holes at the same radius and angles, 20 mm deep, placed toward the **front**: (y, z) = (0, -45) and (13.90, -42.80).

That pairing has a useful consequence. Rotating the leg forward by `leg_lean` puts leg hole 18 on body hole 0 **and** leg hole 0 on body hole 18 at the same time. So the **three-leg operating stance takes two dowels per shoulder** and the two-leg display stance takes one. Use both in the operating stance: `research/loads.md` section 7 asks for two pins rather than relying on one. Both leg holes run through the plate and the booster cover, so each 6 x 30 mm steel dowel is pushed in from outboard.

Load screening (research/loads.md section 7): bolt shear 1.6 MPa against a 48.9 kN proof load; bearing on the printed boss 0.49 MPa against about 8 MPa allowable, a 16x margin; index pin bearing 1.2 MPa, a 7x margin. Steel is never the limit. Creep under the clamp is the risk, and the crush sleeve retires it.

3.3 Leg splice - comb lap joint, 4 x M4, two full-length M8 rods

The outer leg splits at 220 mm below the shoulder axis (leg_split_z). The joint is a comb, not a butt.

- leg_lower carries a centre tongue 20 mm wide, full strut thickness, 40 mm tall (leg_splice_len). leg_upper has the matching slot between two fingers, and the fingers carry the rod bores. Fit clearance is 0.2 mm each side (lg_finger_clear). Both halves print flat with no floating lap piece.
- **Four M4 x 80 bolts** pass through the whole strut width in Y, at x = 5 and 25.4 mm and z = -210 and -190 mm. Heads sit in 8 mm counterbores in the front face; hex nut pockets are in the rear face.
- **Two M8 threaded rods** run the full length of the leg at y = +/-20 mm (leg_rod_offset) on the strut centre plane. The top nut of each rod sits in a captive hex trap that opens on the inboard face at z 40 to 48.4 mm. The rod is fed in from the ankle end, threaded through that captive nut, and the bottom nut and washer are tightened with a 13 mm socket in a 22 mm counterbore in the ankle underside.

The rods are the joint. research/loads.md section 7: the moment at mid-leg under a 35 N side push at the foot is 7.6 N.m, across a layer-line face whose interlayer strength is 18-23 MPa. Two rods preloaded to 1 kN with a 25 mm lever to the face edge hold the joint closed to 50 N.m, a 6.6x margin. Preload each to **1.6 N.m**.

Rod length: the bore runs from lg_rod_bottom_z = -336 mm up to leg_rod_top_z = +55 mm, 391 mm. cad/legs.scad names M8 x 400 stock. Cut each leg rod to **390 mm**. bom/hardware.csv H09 currently plans 425 mm cuts; 425 mm does not fit the bore and the cut plan needs correcting.

3.4 Ankle - tongue in a slot, M8 pivot, M8 lock, two stance holes

The leg ends in a tongue 17.4 mm thick and 100 mm wide (leg_tongue_t, leg_tongue_w) that drops 30 mm into a slot in the foot (foot_slot_depth). The slot is 17.6 mm wide (foot_slot_w) and is widened for the 18 degree swing.

- The foot carries a raised ankle block 60 x 120 x 18 mm on its top plate. The **pivot bore** runs along X through both block cheeks at 6.3 mm above the shell top (ft_pivot_up), with 21.7 mm bosses 4 mm proud (ankle_cyl_d) and an M8 nut pocket in the inboard boss.
- The **lock bore** is the same size, on the pivot plane, 40 mm ahead of the pivot (ft_lock_r = ankle_bolt_spacing = 40), also through both cheeks with bosses and an inboard nut pocket. It is a **single** bore in the foot.
- The tongue carries **two 9 mm lock holes on the same 40 mm arc** about the pivot (lg_lock_r, lg_lock_holes): **hole A** at (y +40, z 0) for the two-leg display stance, and **hole B** at (y +38.04, z -12.36) for the 18 degree three-leg lean. They are 12.5 mm apart with a 3.5 mm web between them. Swinging the leg brings each in turn onto the foot's single lock bore.
- Fasteners per outer foot: the **M8 x 80 pivot bolt** and the **M8 x 90 lock bolt**, heads outboard, nuts in the inboard boss pockets.

This interface is now closed on both sides. cad/legs.scad cuts exactly the two holes cad/feet.scad expects; its old single bore 25 mm below the pivot is gone; params.scad sets ankle_bolt_spacing = 40; and an assert() in each file ties its own radius to ankle_bolt_spacing, so a change to params cannot move one part without the other.

The pivot bolt takes the shear. The lock bolt takes the moment and sets the stance. Neither is torqued against the printed cheeks; the nyloc is run down until the play is gone and the tongue still turns by hand with the lock bolt out.

3.5 Centre leg - four M8 flange bolts

The centre leg bolts to the underside of the skirt floor. The flange is 140 x 100 x 8 mm (center_leg_flange) and lies in the tilted body plane, so the leg and the 18 degree body tilt are built into one part.

- **Four M8 through-bolts** at (+/-50, +/-32) mm (center_leg_bolts, center_leg_bolt_m). Heads go below, in 19 mm socket notches cut through the collar corners. The nuts sit in **hex pockets moulded into the top face of the floor**, each in a 22 mm boss, so they cannot turn.
- cad/legs.scad calls for **4 x M8 x 30 and nuts**.

- The battery shelf above carries **four 26 mm socket windows** over these four nuts, so a socket reaches them after the shelf and the battery are in place.
- Below the flange the collar is center_leg_section (100.3 x 71.7 mm) by 14 mm, then the column 83.3 x 64 mm down to the bearing housing.

Torque these to about **2 N.m (estimated)** with a 24 mm fender washer under each nut. That gives about 3 MPa on the printed floor, below the 6.6 MPa at which research/loads.md warns of PETG creep. Four bolts are the only thing holding the third leg on, so check them at every service.

3.6 Centre foot caster - M12 bolt, two 6001 bearings, swivel stop pin

The centre ankle is a vertical caster pivot. This is why six fixed-axle drive motors can turn the robot without scrubbing the centre foot.

- Two **6001-2RS bearings, 12 x 28 x 8 mm** (caster_bearing_od, caster_bearing_t), sit in seats in the centre leg: the lower one open to the bottom face at z 26 to 34.2 mm, the upper one open to the 28 mm head counterbore. Centres are **18 mm apart** (caster_bearing_gap = lg_bearing_cc = 18). A 20 mm spacing does not fit: the stack would be 72 mm against the 62.2 mm available under the tilted plane. params.scad now carries 18 and an assert() in cad/legs.scad holds the two in step.
- A **12 mm ID x 9.8 mm steel spacer tube** goes between the inner races, so the M12 preload never squeezes the outer races.
- One **M12 x 70 hex bolt** comes down from the leg, head in the counterbore. **It is tightened from above before the body is fitted.** Its nut sits in a nut slot under the centre foot's top plate (19.4 mm across flats, 8.4 mm deep), which is loaded from +X through the open sole before the motors go in.
- The **swivel stop** is an arc groove in the leg's bottom face, 7.6 mm wide and 8 mm deep, at a pin radius of 22 mm (lg_stop_r), spanning +/-60 degrees about forward (caster_stop_deg). The foot's **6 x 8 mm pin stands on the stem-block top** at (0, caster_trail + 22) and runs in that groove. The pin moved in from radius 35 to 22 so the leg groove could reach it; radius 22 also puts it on solid stem-block material, so the 16 mm ledge fin the old radius needed is gone. ft_stop_r and lg_stop_r are tied by an assert() in each file.
- Trail is 20 mm (caster_trail): the pivot axis stands ahead of the foot's axle midpoint so the caster follows.

Load screening: the caster carries roughly 117 N with a 3x dynamic factor, against a 6001 static rating of 2,390 N. The bearings are sized by the 12 mm bolt, not by load.

Wire route. The centre-foot wire hole is at (X 0, Y caster_trail - 21) = (0, -1), which is leg-frame radius 21, on the leg's wire arc slot (radius 16.5 to 22.5 mm) at every swivel angle. It moved there from (-18.5, caster_trail), which was only on the slot near zero swivel. ft_wire_r and lg_wire_r are tied by an assert() in each file. The skirt floor carries the matching 60 x 40 mm cable opening at Y = -60.

3.7 Dome bearing - Triangle 9C lazy susan, 8 x M5

The dome turns on a purchased steel ring bearing, not on a printed race.

- **Triangle 9C round lazy susan:** 228.6 mm OD (susan_od), 114.3 mm centre hole (susan_id), 7.9 mm thick (susan_t), 22 gauge steel, 750 lb rating. Four mounting holes per race on a 156.9 mm square, which is a radius of 110.9 mm (susan_hole_r) at 45, 135, 225 and 315 degrees (susan_hole_angles).
- **Bottom race:** four M5 x 12 screws down into four M5 heat-set inserts in the body top plate. The pockets are pressed from **above**, each in a 14 mm boss that hangs 8 mm under the plate.
- **Top race:** four M5 x 12 screws up into four M5 heat-set inserts in the dome plate. The dome plate carries 12 mm tall bosses for these.
- **Access holes:** 12 mm holes at 0, 90 and 270 degrees in the body top plate (susan_access_angles, susan_access_d). You rotate the dome to bring each top-race screw over an access hole and drive it from below. cad/body.scad **omits the 180 degree hole altogether**, because cad/head_drive.scad shows the drive base covers (0, -110.9) and makes it unusable. Three holes reach all four screws.

The dome plate is an annulus from 57 mm to 153.5 mm radius, 5 mm thick (dome_plate_r_in, dome_plate_r_out, dome_plate_t), printed as the first layers on the bed. Its outer edge clears the body lip inner face by 1.5 mm.

3.8 Head drive - hinge bolt, spring tension, thumb nut, lift stop

One printed part does the whole job: a base bolted under the body top plate and an arm that swings on it. One Adafruit 3777 motor turns one Adafruit 3766 wheel about a radial axis. The wheel top comes up through the slot in the top plate and presses on the flat underside of the dome plate, 7.9 mm above the plate's top face. Rolling the wheel drags the dome round.

- **Mount:** four M4 x 12 countersunk screws (DIN 7991) come down through plain, 90 degree x 9 mm countersunk holes in the body top plate into four M4 heat-set inserts in the base. The holes are at (+/-45, -84) and (+/-42, -139) in the native frame (hd_mount). The plate underside must be **flat** over x +/-56, y -148 to -74: no bosses, no ribs.
- **Hinge:** one M4 x 70 socket head, two M4 washers and an M4 nyloc through both base lugs and the arm boss. The boss half-width is 20.8 mm, leaving 0.2 mm each side. **Snug the nyloc. Do not torque it.** The arm must swing freely.
- **Motor:** two M3 x 30 button heads and two M3 hex nuts hold the gearbox to the arm wall, through the two 3.0 mm holes 17.6 mm apart. Heads go on the wheel side and must be fitted before the wheel is pressed on. The lower nut goes into a blind hex pocket on the inboard face; the upper nut drops into a hex slot from the wall top. One 3.6 mm cable tie goes round the can through the tunnel.
- **Tension:** one M5 x 60 hex head (DIN 933, full thread; a socket cap head would spin in the pocket) hangs head-up in a pocket on the base's mating face, so the top plate captures the head. Below it: the arm's ear slot, a 15 mm OD washer, a compression spring 10 mm OD x 25 mm free x 1.0 mm wire (about 2 N/mm), a plain washer, and an M5 knurled thumb nut. Turning the nut up compresses the spring against the ear, lifts the arm, and presses the tyre on the dome plate. **Design compression is 4 to 6 mm**, which is 8 to 12 N at the ear and 4 to 6 N at the tyre. research/loads.md section 5 asks for 3 to 5 N.
- **Lift stop:** one M5 x 20 socket cap in an M5 insert in the arm's stop boss. Screwed fully home its tip stands 5 mm above the beam top and meets the base underside after 1 mm of lift. With the dome off, this is what stops the gearbox touching the base.
- **Release:** back the thumb nut off about 8 mm, which is 10 turns. The arm falls under its own weight until the bolt shank meets the end of the ear slot at 7 degrees. The wheel top is then 4 mm below the dome plate and the dome lifts off without dragging the tyre. The nut stays on the bolt.

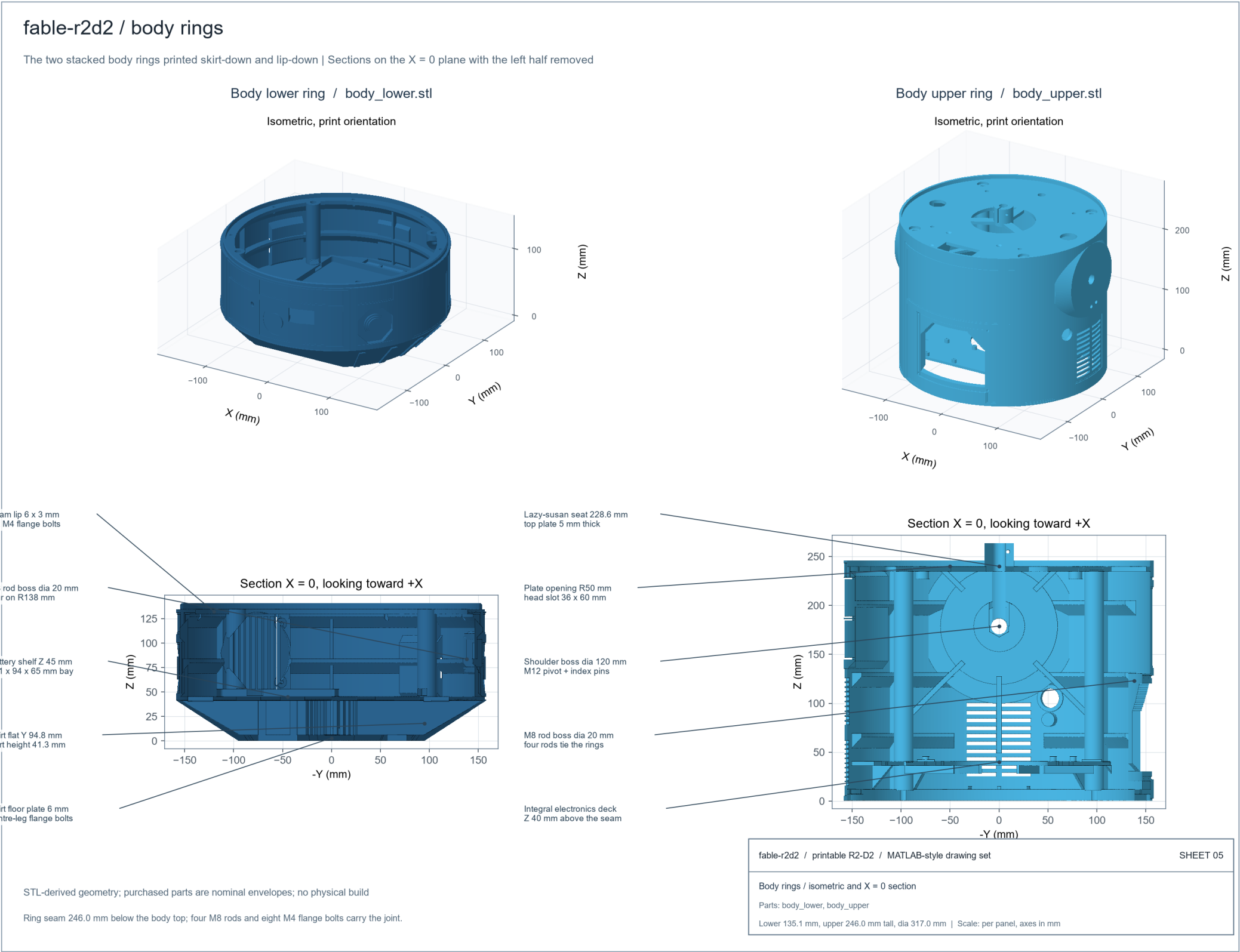
The wheel window is x +/-31.5, y -147.5 to -118.5. The slot in the body top plate must be at least 58 mm tangentially (hd_slot_min); params.scad sets head_slot = [36, 60], which meets it. The 36 mm radial size is what caps the release at 7 degrees.

Keep-out, honoured by cad/body.scad. Nothing of the body enters the swept volume on the rear side: |x| up to 66 mm, y -160 to -70 mm, body-frame z 160 to 378 mm, except the wall itself and the top plate. The plate underside is flat over x +/-56, y -148 to -74: no ribs, no bosses. The wheel sweeps to radius 150.9 mm at release and the body inner wall is at 154.9 mm.

The top plate also carries the slip-ring mount: a central opening of radius 50 mm (top_plate_opening_r) with a three-spoke spider to a 30 mm hub, bored 12.5 mm for the Adafruit 1195 body (slip_ring_d) and pinched by an M3 clamp screw. The bore is open top and bottom so the ring's two lead bundles pass straight through.

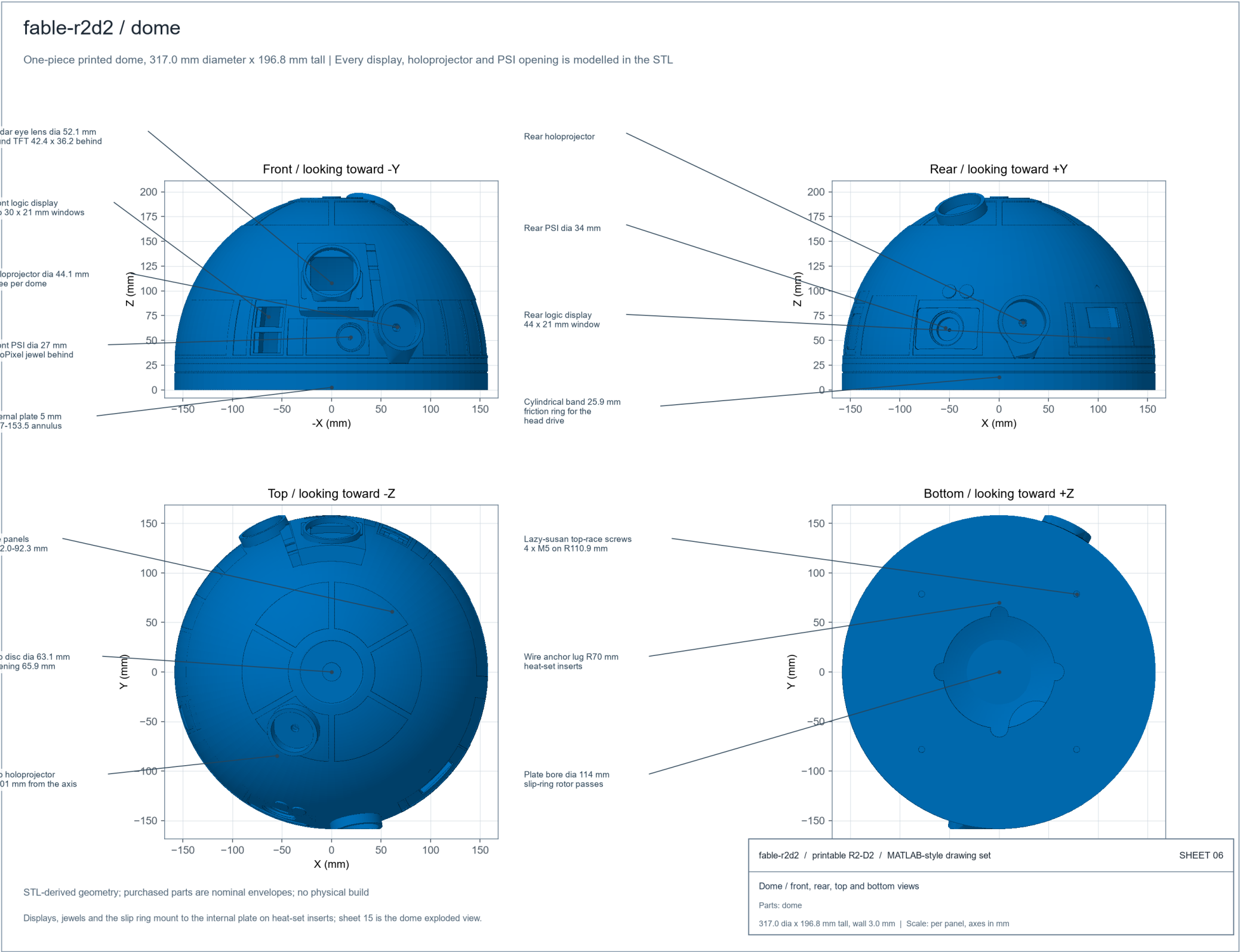
Drive numbers (research/loads.md section 5): ratio from the dome ring to the wheel is 4.6. Steady dome torque 0.10 N.m needs 0.69 N of tangential force and 0.86 N of normal force at mu 0.8, so a 3 to 5 N preload is 2 to 3x. Run the head motor at **50 percent PWM off the 6 V rail** for about 8.7 RPM loaded. Below 35 percent it stalls.

Figure 5 - the two body rings



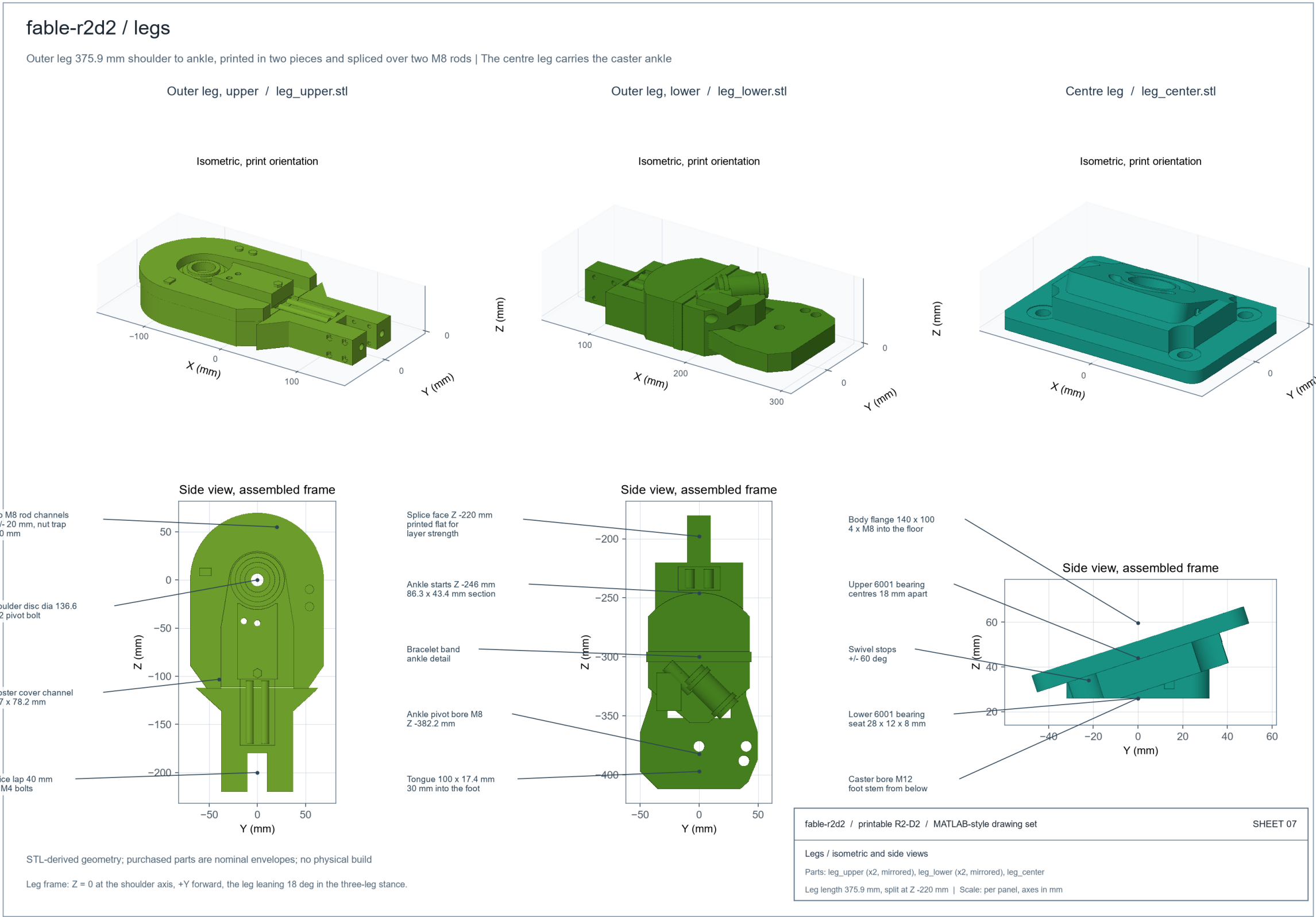
output/drawings/05_body_rings.png. Ring seam, shoulder bosses, integral electronics deck and the lazy-susan seat. Millimetre axes; purchased parts are nominal envelopes, not manufacturer CAD.

Figure 6 - the dome



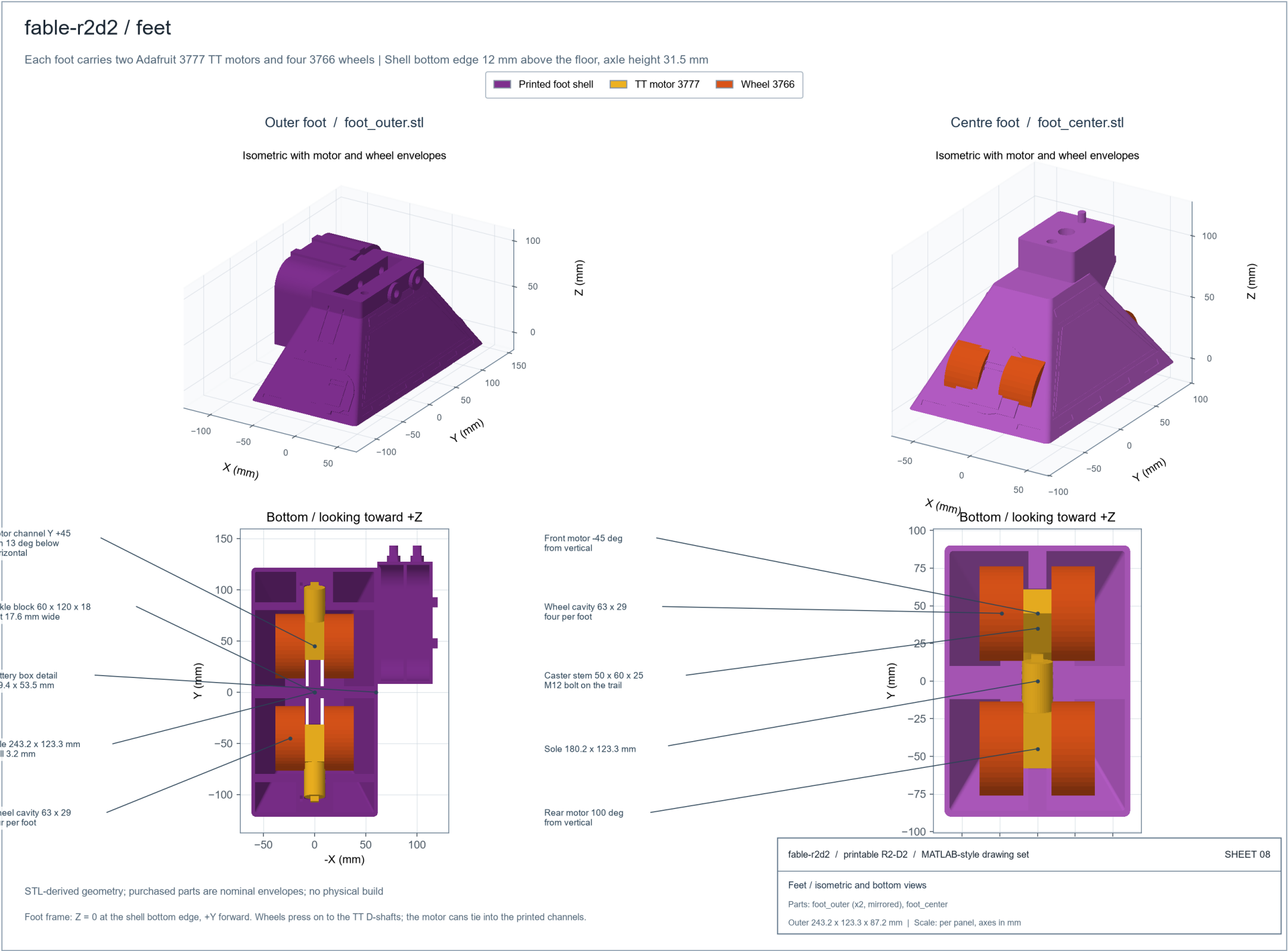
output/drawings/06_dome.png. One-piece dome with the display, holoprojector and PSI cut-outs. Millimetre axes; purchased parts are nominal envelopes, not manufacturer CAD.

Figure 7 - the legs



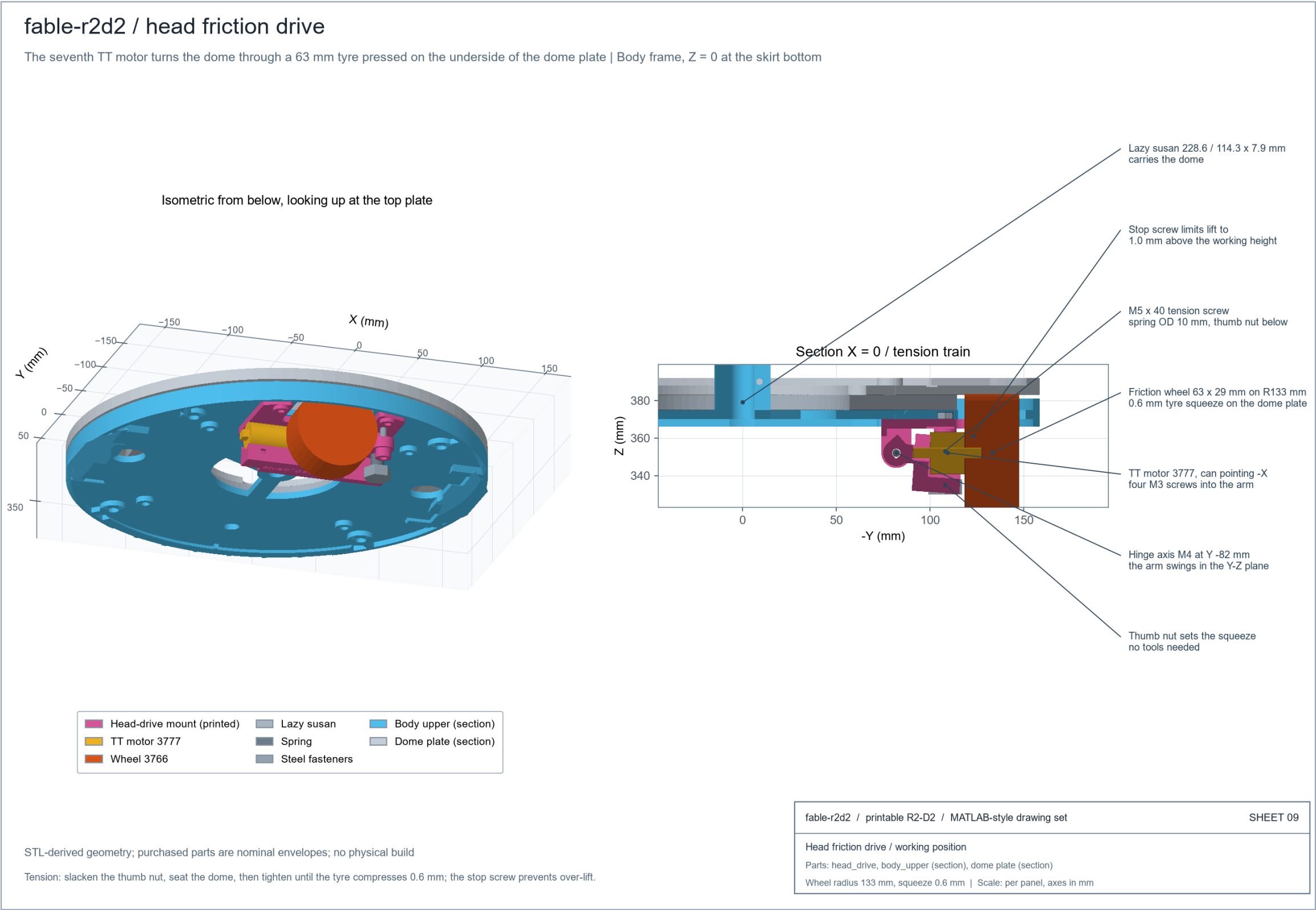
output/drawings/07_legs.png. Outer leg upper and lower, the comb splice and the centre leg. Millimetre axes; purchased parts are nominal envelopes, not manufacturer CAD.

Figure 8 - the feet



output/drawings/08_feet.png. Outer foot and centre foot with the motor pockets and wheel positions. Millimetre axes; purchased parts are nominal envelopes, not manufacturer CAD.

Figure 9 - the head friction drive



output/drawings/09_head_drive.png. Pivoting motor mount, tension screw and lift stop. Millimetre axes; purchased parts are nominal envelopes, not manufacturer CAD.

4. Fits and tolerances

Every clearance, insert pocket and nut pocket below comes from the functions in `cad/lib.scad`. If a hole size needs changing, change the function, not the part.

4.1 Screw clearance holes, `clearance_d(m)`

Thread	Hole mm
M3	3.4
M4	4.5
M5	5.5
M6	6.6
M8	9.0
M12	13.0

4.2 Heat-set insert pockets, `insert_d(m)` and `insert_depth(m)`

Insert	Pocket diameter mm	Pocket depth mm	Insert used
M2	3.2	4.0	ruthex, 4.0 mm tall
M2.5	3.7 (estimated from the same table)	5.0 (estimated)	any 2.5 mm brass
M3	4.0	6	Adafruit 4255, M3 x 4 mm
M4	5.6	8	ruthex RX-M4x8.1, 8.1 mm tall
M5	6.4	9.5	ruthex RX-M5x9.5, 9.5 mm tall

The ruthex table reads "Insert hole (mm) 3,2 / 4,0 / 5,6 / 6,4" for M2, M3, M4 and M5, which is what `insert_d()` returns. The design rule in `research/components-mechanical.md` is to model the pocket about 0.2 mm over the nominal insert hole and 1 mm deeper than the insert.

4.3 Nut pockets, `nut_af(m)` and `nut_h(m)`

Across flats includes 0.4 mm of clearance.

Thread	Across flats mm	Depth mm
M3	5.9	2.4
M4	7.4	3.2
M5	8.4	4.7
M8	13.4	6.8
M12	19.4	10.8

4.4 Bearing, wheel and slot fits

Fit	Value	Where
6001 bearing seat	$28.0 + 0.2 = \mathbf{28.2\text{ mm}}$ bore, 8.2 mm deep	lg_bearing_fit in cad/legs.scad
Bearing bore relief	24 mm through-hole below each seat	so the outer race seats on a ledge
Shoulder bushing	12 mm bore x 16 mm OD bronze, pressed into a 16 mm printed bore	research/components-mechanical.md
Index dowel	6.0 mm pin in a 6.2 mm hole	shoulder_index_d
Motor pocket	envelope +0.4 mm all round	motor_pocket_clear
Head-drive motor pocket	envelope +0.3 mm	hd_clear
Wheel cavity	3 mm clearance around each 63 x 29 mm wheel	ft_wheel_clear
Ankle tongue in the foot slot	17.4 mm tongue in a 17.6 mm slot = 0.2 mm	leg_tongue_t, foot_slot_w
Leg comb joint	0.2 mm each side	lg_finger_clear
Head-drive hinge boss to lug	0.2 mm each side	hd_boss_half = 20.8
Dome plate to body lip	1.5 mm radial	dome_plate_r_out 153.5 against 155
Dome band to body top edge	1.9 mm vertical	dome_gap
Acrylic lens seats	seat diameter +0.5 mm , glued	cad/dome.scad
Head-drive slot	36 x 60 mm against a 58 mm minimum	head_slot, hd_slot_min

The wheel press fit is the one fit you cannot control. The Adafruit 3766 presses onto a 5.40 mm D-shaft with a 3.70 mm flat and has no retaining screw. Wheel pages quote a 3.5 mm flat and the motor drawing quotes 3.7 mm; the press fit absorbs the difference. research/loads.md section 4 warns that lateral scrub of up to 6.8 N can walk a wheel off its shaft. Keep the wheel hub within 1 mm of the gearbox face, and check the wheel's axial position after the first hour of driving.

5. Load screening summary

Full working is in `research/loads.md`. This is the summary and the two things it changes.

5.1 Mass budget

Group	Nominal g	Low g	High g
Body	3,018	2,600	3,400
Dome	947	800	1,100
Head drive	440	350	600
Outer legs (2)	2,457	2,100	2,800
Centre leg	500	400	600
Feet (3) with motors and wheels	2,050	1,800	2,300
Electronics and wiring	750	600	900
Battery	2,260	2,260	2,260
Fasteners	851	700	1,000
Total	13,273	11,610	14,960

13.3 kg nominal, 130.2 N. That is above the 8-11 kg the note expected. The cause is wall thickness on a 317 mm cylinder. Every extra kilogram cuts the drive margin, cuts runtime and raises turning scrub, so **weigh every print as it comes off the bed and keep a running total**.

Two cross-checks now exist. The slice predictions in section 2.4 put the six sliced parts plus the dome at 5,028 g against a 5,894 g budget for the same groups, about 15 percent under. The whole-robot screening in `docs/stability.json` gives **11,705.5 g**, against the 13,273 g nominal above. Both say the same thing: the package is tracking at the **low end** of the budget range, which is the good direction.

5.2 Drive margin

Six drive motors, two per foot, four wheels per foot, 63 mm wheels.

Case	Rolling resistance N	Per-motor current A	Speed m/s	Margin at the 1 A limit	Margin at stall
13.3 kg, Crr 0.02	2.60	0.385	0.54	3.6x	5.7x
13.3 kg, Crr 0.04	5.21	0.620	0.43	1.8x	2.9x
11.0 kg, Crr 0.02	2.16	0.345	0.56	4.4x	6.9x
11.0 kg, Crr 0.04	4.31	0.540	0.47	2.2x	3.5x

Six motors move 13.3 kg on a level hard floor. Grade capability is 2-3 degrees at the 1 A driver limit and 4-5 degrees at stall. Carpet, thresholds above about 5 mm and ramps are outside the envelope.

Turning, not straight driving, is the limit. Spin-in-place needs 4.1 to 9.3 N per side against 4.7 N available at the 1 A limit. Command arc turns of 0.5 m radius or more by default, and allow spin only at full 6 V with a time limit.

5.3 Shoulder moment and hoop strength

- **Shoulder moment.** The design case is the robot rocked back onto the rear axle contacts with the centre foot lifted: 6.2 N.m, times a 3x dynamic factor, is **9.4 N.m per shoulder**, with 176 N of vertical shear. Every printed section has a 7x margin or better (section 3.2).
- **Body ring hoop strength.** Shoulder load enters the shell as in-plane shear at 0.35 MPa. The worst radial case over a 100 mm shell height gives 26.7 MPa, which is marginal against the 34 MPa PETG HF figure; over the full ring with the ring ribs it drops to 6.9 MPa. **The ring ribs are what make the shell work. Keep at least one ring rib within 30 mm of each shoulder boss.**

5.4 Stability, measured from the CAD

The tipping numbers in `research/loads.md` section 7 assumed the centre foot stood 290 mm **ahead** of the outer-foot line. The built geometry puts it 39.2 mm **behind** them (section 1.3), so those numbers do not apply and are not repeated here.

docs/stability.json replaces them. It is a screening calculation: STL volumes at a 0.45 fill fraction times 1.27 g/cm³, plus every purchased item as a point mass, in the three-leg stance. Nothing was weighed.

Measure	Value
Total mass	11,705.5 g (printed 6,046.9 g, purchased 5,658.6 g)
Centre of gravity (x, y, z)	(0.1, 52.3, 301.8) mm in the assembly frame
Wheel contact patch, fore-aft	y 32.0 to 161.2 mm
Wheel contact patch, across	x -205.5 to 205.5 mm
Outer foot axle mean	y 116.2 mm
Centre foot axle mean	y 77.0 mm
Centre-foot static share	1.63
Tip-back margin	20.3 mm, 3.9 degrees
Tip-forward margin	108.8 mm, 19.8 degrees
Side tip	34.2 degrees

Two things in that table need reading carefully.

- **The centre-foot share of 1.63 is not 163 percent of the weight in practice.** It is what the two-line axle model gives when the centre of gravity (y 52.3) lies *behind* the centre-foot axle line (y 77.0). The real robot rests on the centre-foot wheels and the rear wheels of the outer feet, and the front wheels of the outer feet carry almost nothing. The model is degenerate here; the bathroom-scale test in assembly.md stage 11 is the measurement that settles it.
- **The tip-back angle of 3.9 degrees is the headline risk.** The centre of gravity sits only 20.3 mm ahead of the rear contact edge, over a centre-of-gravity height of 301.8 mm. The acceptance criterion is 15 degrees. **The design as modelled does not meet it**, and the robot will rock back onto the centre foot under braking or over a bump.

The fix is to move the centre of gravity forward. The battery already sits battery_y = 15 mm forward of the body axis on its shelf, inside a printed curb, and docs/stability.json puts its centre at y 87.3 mm in the assembly frame. The remaining adjustments, in order of effect, are: rake the outer legs further forward, lower the shoulder axis in the body, and move the electronics deck load forward off the rear chord. Tip-forward (19.8 degrees) and side tip (34.2 degrees) both have room to give.

Total mass of 11.7 kg sits at the low end of the 11.6 to 15.0 kg range in research/loads.md section 1, so the drive-margin row to read in section 5.2 is the 11.0 kg case: **4.4x at Crr 0.02** and 2.2x on a poor floor.

5.5 Ranked risks

1. **Turning scrub exceeds drive force.** Retired by a 0.5 m arc at 50 percent PWM without a stall.
2. **Mass over target.** Retired by weighing every print against the section 5.1 budget.
3. **TT gearbox bores and press-fit wheels** under 11-14 N radial and 7 N lateral. Retired by an hour of figure-8 driving with under 0.3 mm of shaft play growth.
4. **Tip-back.** Now the **top** structural risk: the CAD screening in section 5.4 gives 3.9 degrees against a 15 degree criterion. Retired by moving the centre of gravity forward and then passing the tilt-board test and the centre-foot scale reading.
5. **Motor heating and runtime at the 1 A limit.** Retired by 20 minutes of driving with the motor cases below 70 C.

Lower ranked: head friction-wheel slip, retired by the rim pull test; PETG creep under the shoulder bolt, retired by the crush sleeve and a 48 hour loaded soak at 40 C.

5 Electrical

fable-r2d2 electrical design, harness and commissioning

Written 2026-09-12. This document, the four sheets in `electronics/` and `electronics/wiring.csv` are one drawing set; the numbers behind them are in `electronics/calculations.json`, which `electronics/generate.py` writes from `research/loads.md` section 6 and `research/components-electronics.md`.

Nothing here has been built or measured. Every current, drop and runtime is arithmetic on published ratings. Treat the commissioning section as the first real measurement, not as a confirmation.

Regenerate everything with:

```
python scripts/electronics.py
```

That script runs the generator and then fails loudly if any KB2040 pin in `wiring.csv` disagrees with the MOTORS table, `ENABLE_PIN`, `PIXEL_PIN` or `INDEX_PIN` in `firmware/kb2040/code.py`.

1. Power design

One 12 V sealed lead acid pack feeds one switched bus. Three switching regulators hang off that bus, each behind its own fuse, and nothing else touches the bus except the battery sense divider.

```
BT1 12 V 7 Ah SLA
+--F2 blade--> J10 Powerpole --> FH1 [F1 15 A] --> SW1 rocker --+--> 12 V BUS
|                                                                    |
|                                                                    |
|                FH2 [F2 3 A] |                +--> FH3 [F3 3 A] --> U5 12 V -> 5 V
|                |                +--> FH4 [F4 5 A] --> U6 12 V -> 6 V (rail A)
|                |                +--> FH5 [F5 5 A] --> U7 12 V -> 6 V (rail B)
|                J1 charge jack                +--> R2 100k / R3 15k --> U9 ADS1115 A0
|
+--F2 blade--> J10 Powerpole --> TB2 star ground
```

- **The charge branch joins the battery side of SW1.** The charger therefore reaches the battery with the robot switched off, and SW1 still isolates every load.
- **U5 (Pololu D36V50F5, 5 V)** feeds the Raspberry Pi through a USB-C plug breakout (J12), the PAM8302 amplifier (U8), and the whole dome through FH6 (F6 2 A) and the slip ring.
- **U6 (D36V50F6, 6 V, rail A)** feeds DRV8833 U1 (left foot) and U3 (centre foot).
- **U7 (D36V50F6, 6 V, rail B)** feeds DRV8833 U2 (right foot) and U4 (head drive). Splitting the two rails keeps a stalled foot from pulling the head drive down.
- **Never feed a DRV8833 from the 12 V bus.** Its maximum motor supply is 10.8 V.
- **The KB2040 is powered only by the Pi's USB port** (cable J11). Do not also connect its RAW pin to the 5 V rail.
- **The Pi is powered only through its USB-C input** (J12). Do not also feed 5 V into header pins 2 and 4.

Rail currents (from `electronics/calculations.json`)

Rail	Load	Current	Source
5 V	Pi 0.60 A, four matrices 0.15 A, NeoPixels 0.30 A average, TFT 0.10 A, KB2040 0.05 A, amplifier 0.20 A	1.40 A at 5 V = 7.0 W	loads.md section 6
12 V	the 5 V rail at 90 percent converter efficiency	0.648 A	calculated
12 V	driving: six drive motors at 0.385 A / 6 V plus the head motor	1.422 A on top of idle	loads.md section 2
12 V	worst case, seven motors at the DRV8833 1 A limit, plus idle	4.537 A	calculated

Runtime, to 50 percent depth of discharge (3.35 Ah usable)

Profile	12 V current	Runtime
Idle, displays running	0.65 A	5.2 h
Continuous driving	2.07 A	1.6 h
Mixed, half the time driving	1.36 A	2.5 h
Worst case, every motor at its limit	4.54 A	0.7 h

Running down to 11.2 V gives more minutes and fewer cycles. `firmware/pi/battery.py` calls 12.70 V full, 11.60 V empty and 11.20 V critical.

2. Charging procedure

The charger is a PowerStream PST-P2012-A12B: three stage, 1.5 A bulk, 14.75 V absorb, 13.75 V float, with a 5.5 mm OD x 2.1 mm ID **centre positive** barrel plug. The robot's jack is a Switchcraft L722A (J1) in the rear lower band of `body_lower`, next to the rocker switch.

1. **Switch the robot OFF at SW1.** The charge path does not need the switch, and a running robot confuses the charger's stage detection.
2. Stand the robot upright, or lay it on its back with the battery shelf horizontal. **Never charge it inverted** and never inside a closed box or bag.
3. Check the ambient temperature: **do not charge below 0 C**, and do not charge above 40 C. A cold SLA accepts charge badly and can be damaged.
4. Plug the charger into J1, then into the mains. Plugging into the mains first puts a live plug into your hand.
5. Read the charger LED: **red = bulk or absorption in progress, green = float**. Red that never turns green after about eight hours means the pack, the fuse F2, or the plug polarity is wrong; unplug and investigate rather than leaving it.
6. Expect **6 to 7 hours** from a half-flat pack: about 2.6 h of bulk at 1.5 A, then the absorption tail. Leaving it on float overnight is safe and is how an SLA likes to live.
7. Unplug the mains first, then J1.
8. Optional check: with the charger off and the robot off, the pack should settle to 12.7-12.9 V within an hour. Below 12.4 V after a full charge means the pack is tired.

A reversed plug blows F2 immediately. Check the plug with a meter before the first charge: centre pin positive with respect to the sleeve.

3. Fuse table

Every fuse is an ATO/ATC blade in a Littelfuse FHAC0001ZXJ in-line holder.

Fuse	Holder	Rating	Circuit	Worst-case load	Part
F1	FH1	15 A	Main, battery positive lead, within 100 mm of the terminal	4.54 A	Littelfuse 0287015.PXCN
F2	FH2	3 A	Charge jack J1 to the battery side of SW1	1.5 A (charger)	0287003.PXCN
F3	FH3	3 A	12 V bus to U5 (5 V regulator)	0.65 A idle, 1.4 A peak	0287003.PXCN
F4	FH4	5 A	12 V bus to U6 (6 V rail A, four motors)	2.2 A	0287005.PXCN
F5	FH5	5 A	12 V bus to U7 (6 V rail B, three motors)	1.7 A	0287005.PXCN
F6	FH6	2 A	5 V rail to the slip ring (the whole dome)	1.62 A	0287002.PXCN

Notes.

- F1 is sized for the 16 AWG lead and the 20 A switch, not for the load. `loads.md` section 6 would allow 7.5 A; the larger fuse still protects the cable (22 A chassis rating) and avoids nuisance blows on motor inrush.
- F6 replaces the 1.5 A polyfuse `loads.md` proposed. The dome worst case (1.02 A of NeoPixels at full white, 0.50 A of matrices, 0.10 A of TFT) is 1.62 A, which a 1.5 A device would eventually trip. The firmware caps NeoPixel brightness at 0.4, so the normal figure is nearer 0.4 A.
- Keep two spares of each value in the robot. They cost \$0.44 each.

4. Wire gauge table

Chassis ratings are the Powerstream table quoted in `loads.md` section 6.

Gauge	Chassis rating	Used for	Worst case	Margin
16 AWG silicone	22 A	Battery leads, F1, SW1, charge branch, the three regulator inputs, regulator grounds, the 5 V feed to the Pi	4.54 A	4.8x
22 AWG silicone	7 A	6 V branches to the drivers, motor extensions, dome feed, display and NeoPixel signals, audio	4.0 A	1.8x
28 AWG (factory)	1.4 A	The TT motors' own leads and the slip-ring leads	1.0 A (DRV8833 limit)	1.4x, keep them short

Voltage drop that matters:

- **Pi 5 V feed:** 16 AWG under 250 mm gives 31 mV of loop drop at 3 A. The Pi wants more than 4.75 V at its connector, so this run must stay short and must not be daisy-chained through anything.
- **Motor leads:** the stock 28 AWG pair drops 85 mV over 200 mm at 1 A. Splice to 22 AWG inside the foot shell and keep the thin part as short as the foot allows.
- **Slip ring:** 28 AWG leads, 2 A per ring. Two rings carry 5 V and two carry the return, so the dome has 4 A of capacity against a 1.62 A worst case.

Colour code used in `wiring.csv`, matching the six-colour 22 AWG kit:

Colour	Meaning
red	any positive supply (12 V, 6 V, 5 V, 3V3)
black	ground, every return
yellow	motor OUT1, SCL, SLP, TFT RST, audio right, speaker plus
blue	motor OUT2, SPI MOSI and SCLK, speaker minus
green	PWM inputs, NeoPixel data, TFT CS, index sensor
white	direction inputs, SDA, TFT DC, audio left and summed

Because colours repeat across functions, flag both ends of every signal wire with a printed or written heat-shrink label. In the motor pairs, **yellow is OUT1 and blue is OUT2**.

5. Connector list

Ref	Connector	Where	Notes
J10	Anderson Powerpole 15 A, two pairs (red and black housings)	Battery lead to the harness	16-20 AWG contacts; the only disconnect between the pack and the robot
TB1	Six insulated 0.250 inch female spades (TE 3-520408-2)	Battery terminals and SW1	Two battery, two switch, two spare
J1	Switchcraft L722A panel jack	Rear lower band of body_lower	5.5 x 2.1 mm, centre positive
SW1	Carling rocker, 20 A	Rear lower band, cutout 36.83 x 21.08 mm	0.250 inch quick connects
FH1-FH6	In-line ATO holders	On the tray, except FH1 which sits at the battery	Label each holder with its fuse value
J2-J9	JST-XH 2 pin pairs (Adafruit 4872)	One per motor at its DRV8833 output, plus one spare	Male half at the driver
J11	USB A to USB C, 1 m (Adafruit 4474)	Pi USB 2.0 port to the KB2040	Power and the CDC serial link
J12	USB Type C plug breakout (Adafruit 5978)	U5 VOUT to the Pi USB-C power input	Its 5.1 k CC1 resistor is what makes the Pi accept the supply
SR1	Adafruit 1195 slip ring, 12 wires	Clamped in the printed post on the body top plate	See the wire table below
TB2	Star ground stud	Electronics tray	Every return lands here and nowhere else

6. Slip-ring wire table (Adafruit 1195, 12 wires, 2 A each)

Wire	Net	Body (stator) side	Dome (rotor) side
1	5V-DOME	FH6 output (F6 2 A), 5 V rail	dome 5 V bus: DS5 VIN, LED1/LED3/LED5
2	5V-DOME	FH6 output, second leg	dome 5 V bus: DS1-DS4, LED2/LED4
3	GND	TB2 star ground	dome ground: DS5, LED1/LED3/LED5
4	GND	TB2 star ground, second leg	dome ground: DS1-DS4, LED2/LED4
5	NEOPIXEL	KB2040 A2	LED1 DIN (front PSI), then the chain
6	SDA	Pi GPIO2, header pin 3	DS1-DS4 SDA (0x70, 0x71, 0x72, 0x73)
7	SCL	Pi GPIO3, header pin 5	DS1-DS4 SCL
8	SPI-MOSI	Pi GPIO10, header pin 19	DS5 MOSI
9	SPI-SCLK	Pi GPIO11, header pin 23	DS5 SCK
10	TFT-CS	Pi GPIO8 (CE0), header pin 24	DS5 CS
11	TFT-DC	Pi GPIO25, header pin 22	DS5 DC
12	TFT-RST	Pi GPIO24, header pin 18	DS5 RST

The TFT backlight is **not** on the slip ring. LITE is tied to the dome 5 V bus through R1 (100 ohm) inside the dome, which gives a steady, slightly reduced brightness and one less wire to lose. `firmware/pi/displays.py` still drives GPIO18 as its backlight pin; with R1 fitted that pin controls nothing and can be left unconnected.

Mechanical: body 12.4 mm diameter x 19.5 mm, no flange, 150 mm of 28 AWG lead each side, 300 RPM maximum. Clamp the body in the printed post on the body top plate and cable-tie both bundles within 40 mm of the ring so no ring lead ever takes a pull.

7. Harness assembly order

Build the harness on the bench, in this order, before anything goes into the body. Do not connect the battery until step 11.

1. **Star ground first.** Fit TB2 on the electronics tray. Everything below returns here.
2. **Tray boards.** Mount U5, U6, U7, the four DRV8833 boards (U1-U4), U9 (ADS1115) and U8 (PAM8302) on M3 nylon standoffs. Leave room for the fuse holders.

3. **12 V bus.** Make up the 16 AWG bus: SW1 load side to TB2's 12 V stud, then FH3, FH4 and FH5 from that stud to the three regulator inputs. Fit no fuses yet.
4. **Battery branch.** Crimp the Powerpole pair (J10), the spade terminals (TB1) and FH1. Keep FH1 within 100 mm of where the battery's positive terminal will be.
5. **Charge branch.** J1 centre pin to FH2 to the *battery* side of SW1; J1 sleeve to TB2. Check the polarity with a meter now, while it is easy to reach.
6. **Regulator outputs.** U5 to J12 (Pi), U5 to FH6, U5 to U8; U6 and U7 to their two DRV8833 boards each. Ground each regulator back to TB2 with 16 AWG.
7. **Signal harness.** KB2040 to the four drivers: seven PWM wires, seven direction wires and the shared SLP wire, per `wiring.csv` and sheet 02. Label both ends.
8. **Motor leads.** Crimp a JST-XH pair per motor. Route through the legs with the ankle bolts loose so the wires are not trapped. Splice 28 AWG to 22 AWG inside each foot.
9. **Dome harness.** Solder the slip ring's twelve stator leads to the body side per the table in section 6, then the twelve rotor leads to the dome devices. Do the dome soldering with the dome off the robot and the backpack address jumpers already bridged.
10. **Audio.** Pi jack to R4A and R4B into U8 A+, jack sleeve to U8 A-, U8 to LS1. The two speaker leads go nowhere near ground.
11. **Rail check.** Run section 8 before the Pi, the drivers or the dome are connected.
12. **Close up.** Cable-tie the harness to the adhesive mounts, check nothing crosses a wheel, a rod or the lazy susan, and fit the dome last.

8. Commissioning: rail checks with a multimeter

Do this with the loads disconnected. Each step ends with the power off.

Step	Do this	Expect	If not
1	All fuses out, battery connected, SW1 off. Meter across the bus studs, then at FH1's input	0 V at the bus, 12.4-13.0 V at FH1	A reading at the bus means SW1 is wired through, or a fuse holder is bridged
2	Fit F1 only, SW1 on. Meter at the bus studs	12.0-13.0 V	0 V: check F1, SW1 terminals, the Powerpole crimps
3	SW1 off. Fit F3. SW1 on. Meter U5 VOUT to ground	5.00 V +/- 0.2 V	Over 5.5 V or unstable: stop, the regulator is wrong or damaged
4	SW1 off. Fit F4 and F5. SW1 on. Meter U6 VOUT and U7 VOUT to ground	6.0 V +/- 0.25 V each	More than 6.5 V at a DRV8833 input is out of spec
5	SW1 off. Connect J12 (Pi), the driver VMOTOR leads and FH6 (dome). SW1 on	Pi boots, no rail sags below its tolerance under load	A sagging 5 V rail usually means the Pi feed is too long or too thin
6	With the Pi running, compare the battery gauge to the meter at the battery terminals	Within 0.15 V	Check R2 (100 k) and R3 (15 k) and the ADS1115 address (0x48) before trusting the gauge
7	Wheels off the ground. E 1 then M 300 0 0 0 on the KB2040's serial port	Only the left foot turns, forward	If a motor turns the wrong way, swap its two leads at the driver output

Do not skip step 1. A bridged fuse holder on a pack that can deliver roughly 457 A into a short is how harnesses catch fire.

9. Safety

- **The battery is the hazard, not the electronics.** A 12 V 7 Ah SLA delivers about 457 A into a dead short (108 A for 5 s is the datasheet figure). Fuse first, then wire.
- **F1 goes within 100 mm of the battery's positive terminal.** A fuse further down the lead does not protect the lead.
- **Insulate both F2 blades** with the supplied boots or heat-shrink. A dropped spanner across the terminals welds itself in place.

- **An SLA vents hydrogen when it is charged.** Charge it upright, in free air, never in a sealed box, never inverted, never below 0 C.
- **Never charge through the main switch.** The charge branch is separate on purpose.
- **SLP low is coast, not a brake.** The only guaranteed disconnect is SW1 off and, for any work inside the body, the Powerpole pair unplugged.
- **The PAM8302 output is a bridge.** Grounding either speaker lead destroys it.
- **Do not connect USB to the KB2040 while the 5 V rail also feeds it.** The Pi's port is its only supply.
- **PETG softens at 62-68 C.** Do not leave the robot in a hot car; the battery dislikes it even more than the prints do.
- This is a hobby robot, not a certified machine. Nothing in this set has been validated on hardware.

10. Where each number comes from

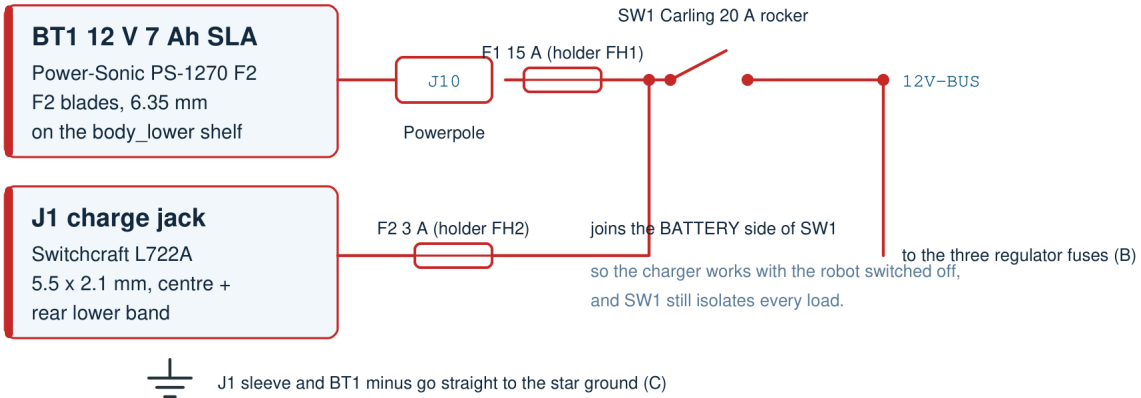
Number	Source
Motor current 0.385 A at 6 V, head 0.5 A at 3 V	research/loads.md section 2
5 V load table, fuse minimums, wire ampacity, runtime	research/loads.md section 6
Power tree, part numbers, prices, stock	research/components-electronics.md
KB2040 pin map, protocol, stop behaviour	firmware/kb2040/code.py, docs/firmware.md
Matrix addresses, SPI pins, backlight pin	firmware/pi/displays.py
Battery divider (100 k / 15 k, ADS1115 at 0x48)	firmware/pi/battery.py
Where each part sits in the robot	cad/params.scad
Wire-by-wire connections and length estimates	electronics/wiring.csv

Circuit sheet 01 - power and charging

01 Power and charging

12 V 7 Ah SLA, one main fuse, one switch, three regulators, one star ground. Charging reaches the battery with the switch off.

A. Battery, main fuse, switch and the 12 V bus



B. Regulators, each behind its own fuse

U5 Pololu D36V50F5 12 V -> 5 V 5.5 A

12V-BUS -> FH3 (F3 3 A) -> VIN
VOUT -> J12 USB-C plug breakout -> Pi USB-C power
VOUT -> FH6 (F6 2 A) -> slip ring wires 1 and 2 (dome)
VOUT -> U8 PAM8302 amplifier
Load: 0.65 A from 12 V idle, 1.40 A at 5 V worst case
Keep the Pi feed under 250 mm of 16 AWG: 31 mV drop at 3 A

U6 D36V50F6 12 V -> 6 V 5.5 A (rail A)

12V-BUS -> FH4 (F4 5 A) -> VIN
VOUT -> U1 VMOTOR (left foot) and U3 VMOTOR (centre foot)
Four motors, DRV8833 limited to 1 A each: 4 A worst case
24 W at 6 V is 2.2 A from 12 V at 90 percent efficiency
Never feed a DRV8833 from the 12 V bus: 10.8 V maximum
Pololu stock is rationed; order this part early

U7 D36V50F6 12 V -> 6 V 5.5 A (rail B)

12V-BUS -> FH5 (F5 5 A) -> VIN
VOUT -> U2 VMOTOR (right foot) and U4 VMOTOR (head drive)
Three motors at the 1 A limit: 3 A worst case
18 W at 6 V is 1.7 A from 12 V at 90 percent efficiency
Splitting the rails keeps a stalled foot off the head drive
Both 6 V rails share the star ground, nothing else

C. Star ground, charger polarity and the battery sense divider

TB2 star ground on the electronics tray

One stud. Every return lands here and nowhere else:
BT1 minus, J1 sleeve, U5/U6/U7 GND, U1-U4 GND, U8 GND,
slip-ring wires 3 and 4, the Pi (through J12) and the KB2040
(through the USB cable J11).
16 AWG for battery and regulator returns, 22 AWG for the rest.
Never daisy-chain a motor return through a logic board.

G1 charger, PowerStream PST-P2012-A12B

Bulk 1.5 A, absorb 14.75 V, float 13.75 V, 3 stage.
Plug: 5.5 mm OD x 2.1 mm ID barrel, CENTRE POSITIVE.
J1 centre pin = +12 V, sleeve = ground. Check with a meter
before the first plug-in: a reversed plug blows F2 at once.
Charge 6-7 h from 50 percent. Never below 0 C, never inverted,
never in a sealed box: an SLA vents hydrogen.

Battery sense divider -> U9 ADS1115

12.7 V pack -> 1.657 V at A0; ratio 7.6667; address 0x48 on Pi I2C-1.

12V-BUS

R2 100k

U9.A0

R3 15k

Legend

- 12 V, fused and switched (16 AWG red)
- 6 V motor rails A and B (22 AWG red)
- 5 V logic, dome and audio rail (16/22 AWG red)
- ground, star connected (16/22 AWG black)
- analogue sense

Fuses, in order from the battery

F1	15 A	main, at the battery lead, holder FH1
F2	3 A	charge port branch, holder FH2
F3	3 A	5 V regulator U5 input, holder FH3
F4	5 A	6 V regulator U6 input, holder FH4
F5	5 A	6 V regulator U7 input, holder FH5
F6	2 A	5 V dome feed to the slip ring, holder FH6

Safety, before the battery goes in

1. F1 within 100 mm of the battery positive terminal.
2. Short-circuit current of this pack is about 457 A. Fuse first, then wire: never test a rail with a bridged fuse.
3. Insulate every F2 blade; a dropped spanner welds itself on.
4. Charge upright, in free air, above 0 C.
5. Meter every rail before any load is connected (sheet 04).

Circuit sheet 02 - motor drive

02 Motor drive: KB2040, four DRV8833 boards, seven TT motors

Pin names are exactly those in firmware/kb2040/code.py. Each motor has ONE PWM input and ONE plain digital input.

A2 Adafruit KB2040 - every external connection

D3	U1 AIN2 PWM -> M1 lf-front
D2	U1 AIN1 direction (high = slow decay)
D5	U1 BIN2 PWM -> M2 lf-rear
D4	U1 BIN1 direction (high = slow decay)
D7	U2 AIN2 PWM -> M3 rf-front
D6	U2 AIN1 direction (high = slow decay)
D9	U2 BIN2 PWM -> M4 rf-rear
D8	U2 BIN1 direction (high = slow decay)
D10	U3 AIN2 PWM -> M5 cf-front
MOSI	U3 AIN1 direction (high = slow decay)
SCK	U3 BIN2 PWM -> M6 cf-rear
MISO	U3 BIN1 direction (high = slow decay)
A1	U4 AIN2 PWM -> M7 head
A0	U4 AIN1 direction (high = slow decay)
D1	SLP on all four DRV8833 boards, high = awake
A2	NeoPixel data out, 17 pixels, to slip ring wire 5
A3	dome index sensor SQ1, active low, internal pull-up
D0	spare, no connection
SDA/SCL	STEMMA QT, GPIO12/13, reserved and unused

Power into the KB2040

USB-C from the Pi (cable J11):
5 V and the USB CDC serial link.
Do NOT also feed RAW from U5.
3V3 pad feeds only SQ1.

Why one PWM pin per motor

RP2040 GPIO n uses PWM slice
(n >> 1) & 7, channel n & 1, so
GP2/GP18, GP3/GP19, GP4/GP20 and
GP10/GP26 collide. Fourteen PWM
outputs are impossible here.
The seven PWM pins above sit on
seven different slice/channels.

DRV8833 states reachable with one PWM and one digital pin

digital low + duty 0 -> both inputs low -> coast
digital high + duty FULL*(1-m) -> forward at m -> slow decay
digital low + duty FULL*m -> reverse at m -> fast decay
digital high + duty FULL -> both inputs high -> brake
Forward is the slow-decay direction, so wire each motor so that direction
drives the robot forward; if not, swap its two leads at the driver output.

Driver boards, motor rails and the seven motors

U1 DRV8833 #1 left foot

VMOTOR <- U6 (6 V rail A)
GND <- TB2 star ground
SLP <- KB2040 D1
AIN1 <- D2 / AIN2 <- D3
AOUT1/AOUT2 -> M1 lf-front
BIN1 <- D4 / BIN2 <- D5
BOUT1/BOUT2 -> M2 lf-rear
Keep both 0.2 ohm sense resistors

U2 DRV8833 #2 right foot

VMOTOR <- U7 (6 V rail B)
GND <- TB2 star ground
SLP <- KB2040 D1
AIN1 <- D6 / AIN2 <- D7
AOUT1/AOUT2 -> M3 rf-front
BIN1 <- D8 / BIN2 <- D9
BOUT1/BOUT2 -> M4 rf-rear
Keep both 0.2 ohm sense resistors

U3 DRV8833 #3 centre foot

VMOTOR <- U6 (6 V rail A)
GND <- TB2 star ground
SLP <- KB2040 D1
AIN1 <- MOSI / AIN2 <- D10
AOUT1/AOUT2 -> M5 cf-front
BIN1 <- MISO / BIN2 <- SCK
BOUT1/BOUT2 -> M6 cf-rear
Keep both 0.2 ohm sense resistors

U4 DRV8833 #4 head drive

VMOTOR <- U7 (6 V rail B)
GND <- TB2 star ground
SLP <- KB2040 D1
AIN1 <- A0 / AIN2 <- A1
AOUT1/AOUT2 -> M7 head
BIN1/BIN2 tied to ground
BOUT1/BOUT2 not connected
Keep both 0.2 ohm sense resistors



M1 lf-front



M2 lf-rear



M3 rf-front



M4 rf-rear



M5 cf-front



M6 cf-rear



M7 head

Motor wiring rules

Each output pair goes to one JST-XH 2 pin connector (J2-J9) and then to the motor's own leads.
Every motor lead pair is twisted 22 AWG, yellow = OUT1, blue = OUT2.
Leave the stock 28 AWG motor leads as short as the foot allows: they drop 85 mV over 200 mm at 1 A. Splice to 22 AWG inside the foot shell.
One 100 nF ceramic across each motor's own tabs if brush noise upsets the NeoPixel chain; it is not needed for the drivers themselves.
Bridge outputs never join each other and never see the 12 V bus.

Legend

- 6 V motor rail
- PWM input
- direction / SLP input
- ground

Stop behaviour

KB2040 drops SLP on E 0, on S,
and after 500 ms with no line
from the Pi: every motor coasts.
SLP low is coast, not a brake.
Power off at SW1 is the only
guaranteed disconnect.

Circuit sheet 03 - head electronics

03 Head electronics: slip ring, PSIs, holoprojectors, logic matrices, radar eye

Twelve slip-ring wires carry everything into the dome. No multiplexer: the four backpacks use four addresses on Pi I2C-1.

SR1 Adafruit 1195 slip ring, 12 wires, 2 A each

1	5V-DOME	5 V dome feed, leg 1 of 2 <- F6 fuse output (5 V rail)
2	5V-DOME	5 V dome feed, leg 2 of 2 <- F6 fuse output (5 V rail)
3	GND	dome return, leg 1 of 2 <- star ground TB2
4	GND	dome return, leg 2 of 2 <- star ground TB2
5	NEOPIXEL	NeoPixel data to LED1 <- KB2040 A2
6	SDA	I2C-1 data, four backpacks <- Pi GPIO2 (header pin 3)
7	SCL	I2C-1 clock, four backpacks <- Pi GPIO3 (header pin 5)
8	SPI-MOSI	SPI0 MOSI to the TFT <- Pi GPIO10 (header pin 19)
9	SPI-SCLK	SPI0 clock to the TFT <- Pi GPIO11 (header pin 23)
10	TFT-CS	SPI0 CE0 chip select <- Pi GPIO8 (header pin 24)
11	TFT-DC	TFT data/command <- Pi GPIO25 (header pin 22)
12	TFT-RST	TFT reset <- Pi GPIO24 (header pin 18)

SR1 mechanical

Body 12.4 mm dia x 19.5 mm, no flange.
Clamped in the printed post on the body top plate; rotor stub 4 mm.
Stator (body) leads: 6 in / 150 mm.
Rotor (dome) leads: 6 in / 150 mm.
300 RPM maximum; the head drive turns the dome far slower than that.
Strain-relieve both bundles with a cable tie within 40 mm of the body.

Current budget in the dome

17 NeoPixels, worst case white at 60 mA each: 1.02 A (firmware caps brightness at 0.4, so about 0.41 A in practice).
Four HT16K33 backpacks: 0.50 A.
Radar-eye TFT: 0.10 A.
Total worst case 1.62 A on F6 2 A.
Two wires carry 5 V and two carry the return: 4 A of slip-ring capacity.

A. NeoPixel chain, one data wire, five devices



DIN to DOUT in that order. 5 V and ground for every device come from the dome 5 V bus (slip-ring wires 1-4), never from the data wire's neighbour.

B. Logic matrices on Pi I2C-1 and the radar-eye TFT on Pi SPI0

DS1-DS4 Adafruit 0.8 inch 8x8 HT16K33 backpacks

0x70	DS1	front logic, upper window - address jumpers: none bridged
0x71	DS2	front logic, lower window - address jumpers: A0 bridged
0x72	DS3	rear logic, left half - address jumpers: A1 bridged
0x73	DS4	rear logic, right half - address jumpers: A0 and A1 bridged

Matrix wiring

All four share slip-ring wires 6 (SDA) and 7 (SCL), daisy-chained backpack to backpack inside the dome, plus 5 V (wire 2) and ground (wire 4).
The HT16K33 has its own 3.3 V-tolerant I2C and runs from 5 V: no level shifter.
Bridge the address pads with solder BEFORE the backpacks go behind the windows.
firmware/pi/displays.py drives 0x70/0x71 as two independent front windows and 0x72/0x73 as one 16 x 8 rear strip that scrolls across the seam.

DS5 Adafruit 6178 round TFT (GC9A01A) behind the radar lens

VIN	slip-ring wire 1, dome 5 V bus
GND	slip-ring wire 3, dome ground
MOSI	slip-ring wire 8 <- Pi GPIO10, header pin 19
SCK	slip-ring wire 9 <- Pi GPIO11, header pin 23
CS	slip-ring wire 10 <- Pi GPIO8 (CE0), header pin 24
DC	slip-ring wire 11 <- Pi GPIO25, header pin 22
RST	slip-ring wire 12 <- Pi GPIO24, header pin 18
LITE	R1 100 ohm to the dome 5 V bus: always on

Backlight: tied on, not switched

There is no thirteenth slip-ring wire, so the backlight is not switched from the Pi. LITE goes to 5 V through R1 (100 ohm), which sets a steady, slightly reduced brightness and survives the dome turning. firmware/pi/displays.py still drives GPIO18 as its backlight pin; with the resistor fitted that pin controls nothing and can be left unconnected. If a dimmable backlight is wanted later, give up the dome index sensor wire or move to a 2-wire dome controller.

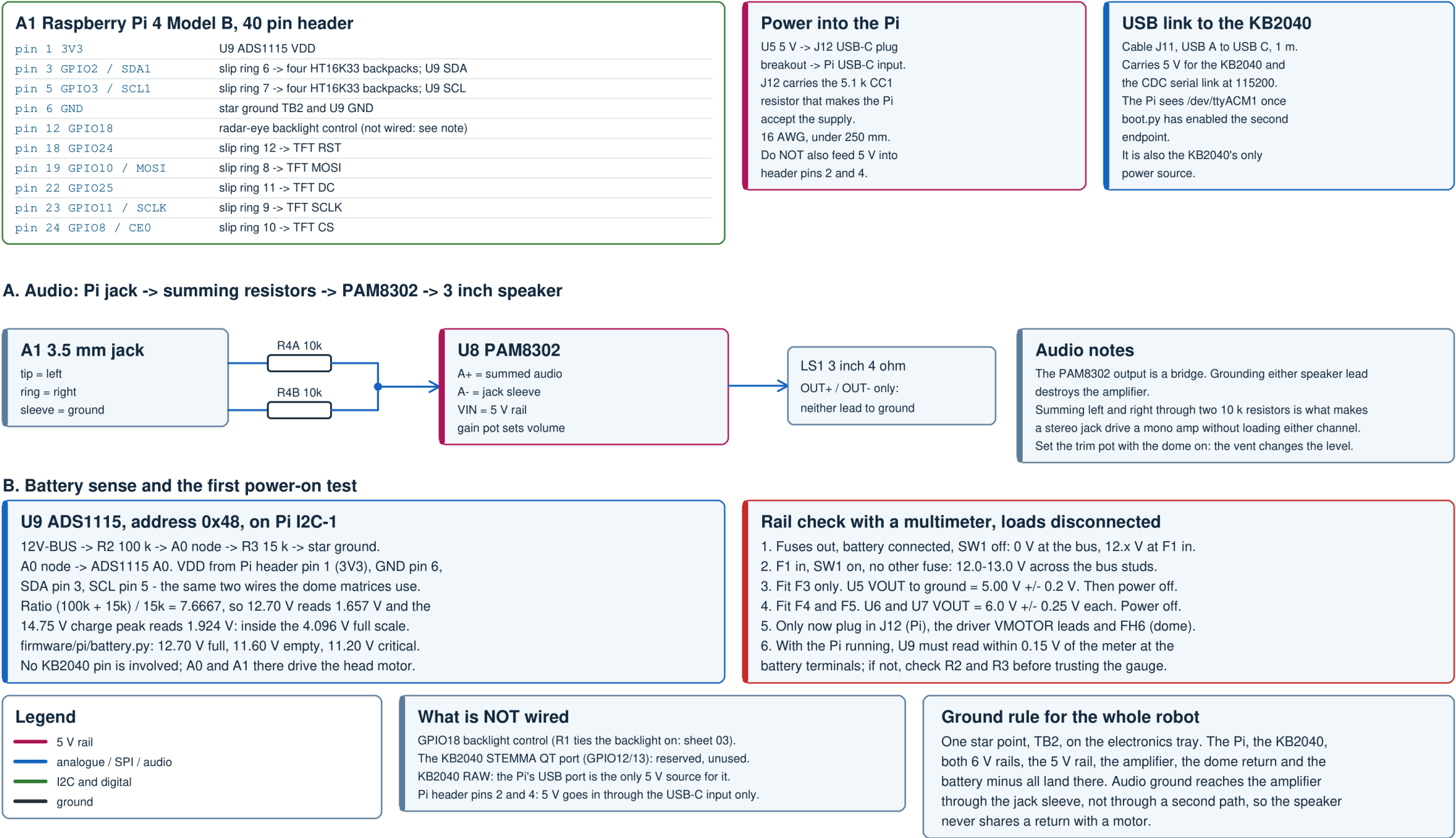
fable-r2d2 electrical set, 2026-09-12 | sheets 01-04 and electronics/wiring.csv are one drawing set | pin names are the final firmware names in firmware/kb2040/code.py | not yet validated on hardware

electronics/03-head-electronics.svg. Named nets join across all four sheets; the wire-by-wire list is electronics/wiring.csv. Nothing here has been built or measured.

Circuit sheet 04 - pi audio and links

04 Raspberry Pi 4: header pins in use, USB link, audio and battery sense

The Pi runs the web server, the head displays, the sounds and the battery gauge. It never drives a motor directly.



5.1 Point-to-point wiring schedule

One row is one connection; 136 in total. Identical net names join across the four sheets.

wire id	net	source	source pin	target	target pin	gauge awg	colour	length mm estimate	route
W001	BAT-POS	BT1	+ (F2 blade)	J10	red housing	16	red	200	body_lower (battery shelf)
W002	BAT-POS	J10	red housing	FH1	in	16	red	150	body_lower
W003	BAT-FUSED	FH1	out (F1 15 A)	SW1	term 1 (battery side)	16	red	260	body_lower -> rear lower band
W004	12V-BUS	SW1	term 2 (load side)	TB2	12V+ stud	16	red	420	body_lower -> body_upper tray
W005	BAT-NEG	BT1	- (F2 blade)	J10	black housing	16	black	200	body_lower (battery shelf)
W006	GND	J10	black housing	TB2	star ground stud	16	black	420	body_lower -> body_upper tray
W007	CHG-POS	J1	centre pin (+)	FH2	in	16	red	180	rear lower band of body_lower
W008	CHG-FUSED	FH2	out (F2 3 A)	SW1	term 1 (battery side)	16	red	160	rear lower band of body_lower
W009	GND	J1	sleeve (-)	TB2	star ground stud	16	black	430	body_lower -> body_upper tray
W010	12V-BUS	TB2	12V+ stud	FH3	in	16	red	120	body_upper tray
W011	5V-REG-IN	FH3	out (F3 3 A)	U5	VIN	16	red	120	body_upper tray
W012	GND	TB2	star ground stud	U5	GND	16	black	120	body_upper tray
W013	12V-BUS	TB2	12V+ stud	FH4	in	16	red	120	body_upper tray
W014	6VA-REG-IN	FH4	out (F4 5 A)	U6	VIN	16	red	120	body_upper tray
W015	GND	TB2	star ground stud	U6	GND	16	black	120	body_upper tray
W016	12V-BUS	TB2	12V+ stud	FH5	in	16	red	120	body_upper tray
W017	6VB-REG-IN	FH5	out (F5 5 A)	U7	VIN	16	red	120	body_upper tray
W018	GND	TB2	star ground stud	U7	GND	16	black	120	body_upper tray
W019	5V	U5	VOOUT	J12	VBUS	16	red	220	body_upper tray
W020	GND	U5	GND	J12	GND	16	black	220	body_upper tray
W021	5V-USB	J12	USB-C plug	A1	USB-C power input	16	red	0	body_upper tray (the plug is the connector, not a wire)
W022	5V	U5	VOOUT	FH6	in	22	red	140	body_upper tray
W023	5V	U5	VOOUT	U8	VIN	22	red	260	body_upper tray
W024	GND	TB2	star ground stud	U8	GND	22	black	260	body_upper tray
W025	6V-U6	U6	VOOUT	U1	VMOTOR	22	red	150	body_upper tray
W026	GND	TB2	star ground stud	U1	GND	22	black	150	body_upper tray

wire id	net	source	source pin	target	target pin	gauge awg	colour	length mm estimate	route
W027	6V-U6	U6	VOUT	U3	VMOTOR	22	red	150	body_upper tray
W028	GND	TB2	star ground stud	U3	GND	22	black	150	body_upper tray
W029	6V-U7	U7	VOUT	U2	VMOTOR	22	red	150	body_upper tray
W030	GND	TB2	star ground stud	U2	GND	22	black	150	body_upper tray
W031	6V-U7	U7	VOUT	U4	VMOTOR	22	red	150	body_upper tray
W032	GND	TB2	star ground stud	U4	GND	22	black	150	body_upper tray
W033	12V-BUS	TB2	12V+ stud	R2	100k leg 1	22	red	150	body_upper tray
W034	VBAT-SENSE	R2	100k leg 2	R3	15k leg 1	22	white	20	body_upper tray
W035	VBAT-SENSE	R3	15k leg 1	U9	A0	22	white	120	body_upper tray
W036	GND	R3	15k leg 2	TB2	star ground stud	22	black	150	body_upper tray
W037	3V3	A1	header pin 1 (3V3)	U9	VDD	22	red	180	body_upper tray
W038	GND	A1	header pin 6 (GND)	U9	GND	22	black	180	body_upper tray
W039	SDA	A1	header pin 3 (GPIO2)	U9	SDA	22	white	180	body_upper tray
W040	SCL	A1	header pin 5 (GPIO3)	U9	SCL	22	yellow	180	body_upper tray
W041	USB-LINK	A1	USB 2.0 type A port	A2	USB-C	0	cable J11	1000	body_upper tray (5 V power and the USB CDC serial link)
W042	SIG-lf-front-PWM	A2	D3	U1	AIN2	22	green	140	body_upper tray
W043	SIG-lf-front-DIR	A2	D2	U1	AIN1	22	white	140	body_upper tray
W044	SIG-lf-rear-PWM	A2	D5	U1	BIN2	22	green	140	body_upper tray
W045	SIG-lf-rear-DIR	A2	D4	U1	BIN1	22	white	140	body_upper tray
W046	SIG-rf-front-PWM	A2	D7	U2	AIN2	22	green	140	body_upper tray
W047	SIG-rf-front-DIR	A2	D6	U2	AIN1	22	white	140	body_upper tray
W048	SIG-rf-rear-PWM	A2	D9	U2	BIN2	22	green	140	body_upper tray
W049	SIG-rf-rear-DIR	A2	D8	U2	BIN1	22	white	140	body_upper tray
W050	SIG-cf-front-PWM	A2	D10	U3	AIN2	22	green	140	body_upper tray
W051	SIG-cf-front-DIR	A2	MOSI	U3	AIN1	22	white	140	body_upper tray
W052	SIG-cf-rear-PWM	A2	SCK	U3	BIN2	22	green	140	body_upper tray
W053	SIG-cf-rear-DIR	A2	MISO	U3	BIN1	22	white	140	body_upper tray

wire id	net	source	source pin	target	target pin	gauge awg	colour	length mm estimate	route
W054	SIG-head-PWM	A2	A1	U4	AIN2	22	green	140	body_upper tray
W055	SIG-head-DIR	A2	A0	U4	AIN1	22	white	140	body_upper tray
W056	SIG-SLP	A2	D1	U1	SLP	22	yellow	140	body_upper tray
W057	SIG-SLP	A2	D1	U2	SLP	22	yellow	140	body_upper tray
W058	SIG-SLP	A2	D1	U3	SLP	22	yellow	140	body_upper tray
W059	SIG-SLP	A2	D1	U4	SLP	22	yellow	140	body_upper tray
W060	MOT-lf-front-A	U1	AOUT1	M1	terminal + (JST-XH pin 1)	22	yellow	820	body_upper tray -> left leg -> left foot
W061	MOT-lf-front-B	U1	AOUT2	M1	terminal - (JST-XH pin 2)	22	blue	820	body_upper tray -> left leg -> left foot
W062	MOT-lf-rear-A	U1	BOUT1	M2	terminal + (JST-XH pin 1)	22	yellow	820	body_upper tray -> left leg -> left foot
W063	MOT-lf-rear-B	U1	BOUT2	M2	terminal - (JST-XH pin 2)	22	blue	820	body_upper tray -> left leg -> left foot
W064	MOT-rf-front-A	U2	AOUT1	M3	terminal + (JST-XH pin 1)	22	yellow	820	body_upper tray -> right leg -> right foot
W065	MOT-rf-front-B	U2	AOUT2	M3	terminal - (JST-XH pin 2)	22	blue	820	body_upper tray -> right leg -> right foot
W066	MOT-rf-rear-A	U2	BOUT1	M4	terminal + (JST-XH pin 1)	22	yellow	820	body_upper tray -> right leg -> right foot
W067	MOT-rf-rear-B	U2	BOUT2	M4	terminal - (JST-XH pin 2)	22	blue	820	body_upper tray -> right leg -> right foot
W068	MOT-cf-front-A	U3	AOUT1	M5	terminal + (JST-XH pin 1)	22	yellow	760	body_upper tray -> body_lower -> centre leg -> centre foot
W069	MOT-cf-front-B	U3	AOUT2	M5	terminal - (JST-XH pin 2)	22	blue	760	body_upper tray -> body_lower -> centre leg -> centre foot
W070	MOT-cf-rear-A	U3	BOUT1	M6	terminal + (JST-XH pin 1)	22	yellow	760	body_upper tray -> body_lower -> centre leg -> centre foot
W071	MOT-cf-rear-B	U3	BOUT2	M6	terminal - (JST-XH pin 2)	22	blue	760	body_upper tray -> body_lower -> centre leg -> centre foot
W072	MOT-head-A	U4	AOUT1	M7	terminal + (JST-XH pin 1)	22	yellow	300	body_upper tray -> under the body top plate
W073	MOT-head-B	U4	AOUT2	M7	terminal - (JST-XH pin 2)	22	blue	300	body_upper tray -> under the body top plate
W074	GND	TB2	star ground stud	U4	BIN1 and BIN2 (tied low)	22	black	150	body_upper tray (driver 4 channel B is unused)
W075	3V3	A2	3V3 pad	SQ1	pin 1 (VCC)	22	red	420	body_upper tray -> body top plate
W076	GND	A2	GND pad	SQ1	pin 2 (GND)	22	black	420	body_upper tray -> body top plate
W077	SIG-DOME-IN-DEX	A2	A3	SQ1	pin 3 (open drain out)	22	green	420	body_upper tray -> body top plate (magnet on the dome plate)
W078	5V-DOME	FH6	out (F6 2 A)	SR1	stator wire 1	22	red	520	body_upper tray -> body top plate -> slip-ring post

wire id	net	source	source pin	target	target pin	gauge awg	colour	length mm estimate	route
W079	5V-DOME	FH6	out (F6 2 A)	SR1	stator wire 2	22	red	520	body_upper tray -> body top plate -> slip-ring post
W080	GND	TB2	star ground stud	SR1	stator wire 3	22	black	520	body_upper tray -> body top plate -> slip-ring post
W081	GND	TB2	star ground stud	SR1	stator wire 4	22	black	520	body_upper tray -> body top plate -> slip-ring post
W082	NEOPIXEL	A2	A2	SR1	stator wire 5	22	green	520	body_upper tray -> body top plate -> slip-ring post
W083	SDA	A1	header pin 3 (GPIO2)	SR1	stator wire 6	22	white	520	body_upper tray -> body top plate -> slip-ring post
W084	SCL	A1	header pin 5 (GPIO3)	SR1	stator wire 7	22	yellow	520	body_upper tray -> body top plate -> slip-ring post
W085	SPI-MOSI	A1	header pin 19 (GPIO10)	SR1	stator wire 8	22	blue	520	body_upper tray -> body top plate -> slip-ring post
W086	SPI-SCLK	A1	header pin 23 (GPIO11)	SR1	stator wire 9	22	blue	520	body_upper tray -> body top plate -> slip-ring post
W087	TFT-CS	A1	header pin 24 (GPIO8)	SR1	stator wire 10	22	green	520	body_upper tray -> body top plate -> slip-ring post
W088	TFT-DC	A1	header pin 22 (GPIO25)	SR1	stator wire 11	22	white	520	body_upper tray -> body top plate -> slip-ring post
W089	TFT-RST	A1	header pin 18 (GPIO24)	SR1	stator wire 12	22	yellow	520	body_upper tray -> body top plate -> slip-ring post
W090	NEOPIXEL	SR1	rotor wire 5	LE D1	DIN	22	green	260	dome (after the slip ring)
W091	NEOPIXEL	LE D1	DOUT	LE D2	DIN	22	green	300	dome (after the slip ring)
W092	NEOPIXEL	LE D2	DOUT	LE D3	DIN	22	green	260	dome (after the slip ring)
W093	NEOPIXEL	LE D3	DOUT	LE D4	DIN	22	green	150	dome (after the slip ring)
W094	NEOPIXEL	LE D4	DOUT	LE D5	DIN	22	green	150	dome (after the slip ring)
W095	5V-DOME	SR1	rotor wire 1	LE D1	5V	22	red	240	dome (after the slip ring)
W096	GND	SR1	rotor wire 3	LE D1	GND	22	black	240	dome (after the slip ring)
W097	5V-DOME	SR1	rotor wire 2	LE D2	5V	22	red	270	dome (after the slip ring)
W098	GND	SR1	rotor wire 4	LE D2	GND	22	black	270	dome (after the slip ring)
W099	5V-DOME	SR1	rotor wire 1	LE D3	5V	22	red	300	dome (after the slip ring)
W100	GND	SR1	rotor wire 3	LE D3	GND	22	black	300	dome (after the slip ring)
W101	5V-DOME	SR1	rotor wire 2	LE D4	5V	22	red	330	dome (after the slip ring)
W102	GND	SR1	rotor wire 4	LE D4	GND	22	black	330	dome (after the slip ring)
W103	5V-DOME	SR1	rotor wire 1	LE D5	5V	22	red	360	dome (after the slip ring)
W104	GND	SR1	rotor wire 3	LE D5	GND	22	black	360	dome (after the slip ring)
W105	SDA	SR1	rotor wire 6	DS1	SDA (0x70)	22	white	220	dome (after the slip ring)

wire id	net	source	source pin	target	target pin	gauge awg	colour	length mm estimate	route
W106	SCL	SR1	rotor wire 7	DS1	SCL (0x70)	22	yellow	220	dome (after the slip ring)
W107	5V-DOME	SR1	rotor wire 2	DS1	+	22	red	220	dome (after the slip ring)
W108	GND	SR1	rotor wire 4	DS1	-	22	black	220	dome (after the slip ring)
W109	SDA	SR1	rotor wire 6	DS2	SDA (0x71)	22	white	260	dome (after the slip ring)
W110	SCL	SR1	rotor wire 7	DS2	SCL (0x71)	22	yellow	260	dome (after the slip ring)
W111	5V-DOME	SR1	rotor wire 2	DS2	+	22	red	260	dome (after the slip ring)
W112	GND	SR1	rotor wire 4	DS2	-	22	black	260	dome (after the slip ring)
W113	SDA	SR1	rotor wire 6	DS3	SDA (0x72)	22	white	300	dome (after the slip ring)
W114	SCL	SR1	rotor wire 7	DS3	SCL (0x72)	22	yellow	300	dome (after the slip ring)
W115	5V-DOME	SR1	rotor wire 2	DS3	+	22	red	300	dome (after the slip ring)
W116	GND	SR1	rotor wire 4	DS3	-	22	black	300	dome (after the slip ring)
W117	SDA	SR1	rotor wire 6	DS4	SDA (0x73)	22	white	340	dome (after the slip ring)
W118	SCL	SR1	rotor wire 7	DS4	SCL (0x73)	22	yellow	340	dome (after the slip ring)
W119	5V-DOME	SR1	rotor wire 2	DS4	+	22	red	340	dome (after the slip ring)
W120	GND	SR1	rotor wire 4	DS4	-	22	black	340	dome (after the slip ring)
W121	SPI-MOSI	SR1	rotor wire 8	DS5	MOSI	22	blue	300	dome (after the slip ring)
W122	SPI-SCLK	SR1	rotor wire 9	DS5	SCK	22	blue	300	dome (after the slip ring)
W123	TFT-CS	SR1	rotor wire 10	DS5	CS	22	green	300	dome (after the slip ring)
W124	TFT-DC	SR1	rotor wire 11	DS5	DC	22	white	300	dome (after the slip ring)
W125	TFT-RST	SR1	rotor wire 12	DS5	RST	22	yellow	300	dome (after the slip ring)
W126	5V-DOME	SR1	rotor wire 1	DS5	VIN	22	red	300	dome (after the slip ring)
W127	GND	SR1	rotor wire 3	DS5	GND	22	black	300	dome (after the slip ring)
W128	5V-DOME	SR1	rotor wire 1	R1	100 ohm leg 1	22	red	300	dome (after the slip ring)
W129	TFT-BL	R1	100 ohm leg 2	DS5	LITE (backlight)	22	white	30	dome (after the slip ring)
W130	AUDIO-L	A1	3.5 mm jack, tip (left)	R4A	10k leg 1	22	white	260	body_upper tray (short shielded lead)
W131	AUDIO-R	A1	3.5 mm jack, ring (right)	R4B	10k leg 1	22	yellow	260	body_upper tray
W132	AUDIO-SUM	R4A	10k leg 2	U8	A+	22	white	40	body_upper tray

wire id	net	source	source pin	target	target pin	gauge awg	colour	length mm estimate	route
W133	AUDIO-SUM	R4B	10k leg 2	U8	A+	22	yellow	40	body_upper tray
W134	GND	A1	3.5 mm jack, sleeve	U8	A-	22	black	260	body_upper tray
W135	SPK+	U8	OUT+	LS1	+ tab	22	yellow	420	body_upper tray -> front vent of body_lower
W136	SPK-	U8	OUT-	LS1	- tab	22	blue	420	body_upper tray -> front vent of body_lower

Source: electronics/wiring.csv.

6 Assembly

fable-r2d2 assembly sequence

Written 2026-09-12. Eleven stages, in order. Each stage ends with a check you can fail.

Read mechanical.md first. It has the print settings, the support notes, the print order and every fit. This document does not repeat them.

The harness is built and checked separately, on the bench, before it goes in the body. That procedure is electrical.md section 7, and it is not repeated here. The software setup is firmware.md sections 5 and 6, and that is not repeated either.

Nothing in this package has been built. No torque below has been measured on hardware. Torques marked "estimated" are design judgement, not test results.

Tools

Tool	Used for
Soldering iron with heat-set insert tips, M2 to M5	Pressing 21 brass inserts
Hex keys 1.5, 2, 2.5, 3, 4 mm	M2 to M5 socket screws
Torque screwdriver, 0.3 to 3 N.m	M3, M4 and M5 fasteners
Torque wrench, 2 to 20 N.m	M8 and M12 fasteners
Sockets and spanners 5.5, 7, 8, 13, 19 mm	M3, M4, M5, M8 and M12 nuts
Deep 13 mm socket	M8 rod nuts in the 22 mm ankle counterbores
Hacksaw or threaded-rod cutter, and a flat file	Cutting and chamfering eight M8 rods
Drill with 6.0, 8.0 and 12.0 mm bits, and a round file	Clearing printed bores that came out tight
Sharp craft knife	Cutting the two head-drive sacrificial webs
Digital calipers	Checking every bore before pressing anything into it
Paint pen or fine marker	Marking wheel axial positions
Crimp tool, wire strippers, soldering iron	The harness, per electrical.md
Multimeter	The rail checks, per electrical.md section 8
Kitchen scale	Weighing every print as it comes off the bed
Bathroom scale	The centre-foot load test
Spring scale, 0 to 20 N and 0 to 50 N	Pull, yaw and dome rim tests
Dial indicator	Motor shaft play, before and after
IR thermometer	Motor case temperature
Tilt board and protractor, or a phone inclinometer	The tip-back test

Consumables

Item	Quantity	Note
Bambu PETG Basic 1 kg	6 spools	About 8 kg of printed mass is predicted; buy six
Bambu PLA Basic 1 kg	1 spool, optional	Dome only
Threaded rod M8 x 1.25, 1 m	4	Cut plan in stage 2 and stage 7
Cable ties, 4.8 mm	8	Two per foot, in the printed saddle grooves
Cable ties, 3.6 mm	2	Head-drive motor can, plus one spare
Cable ties, 8 inch	100 pack	Harness bundling and slip-ring strain relief
Adhesive cable-tie mounts, 3/4 inch	100 pack	Harness routing inside the rings and legs
Closed-cell foam tape, 3/4 x 3/8 inch	1 roll	Under the battery, behind the speaker, round the dome plate
Medium-strength threadlocker	1 bottle	Ankle bolts and the caster bolt (estimated need)
Isopropyl alcohol	500 ml	Cleaning before paint and before gluing lenses
Acrylic cement or clear epoxy	1 tube	Three lens discs
Filler primer, white, blue, silver paint	as needed	See mechanical.md section 7

The full purchased list with part numbers, prices and sources is `bom/hardware.csv` and `bom/electronics.csv`.

Insert pressing schedule

Press every insert before any painting and before any wiring. Twenty-one inserts in four sizes. Pocket sizes come from `cad/lib.scad` and are listed in `mechanical.md` section 4.2.

Insert	Count	Part	Where exactly	Pocket
M5 x 9.5 brass	4	body_upper top plate	Lazy-susan bottom race, radius 110.9 mm at 45, 135, 225, 315 degrees	6.4 mm x 9.5 mm
M5 x 9.5 brass	4	dome plate	Lazy-susan top race, same radius and angles, pressed from below into 12 mm tall bosses	6.4 mm x 9.5 mm
M5 x 9.5 brass	1	head_drive arm	Lift-stop boss, faces up in the print	6.4 mm x 9.5 mm
M4 x 8.1 brass	4	head_drive base	Mating face, in the 12 mm x 3 mm pads at (+/-45, -84) and (+/-42, -139)	5.6 mm x 8 mm
M2.5 brass	2	dome radar eye housing	Inner face beside the TFT pocket, 54.4 mm apart, for the retaining strap	3.7 mm x 5 mm (estimated)
M2 brass	6	dome	Two per logic-display block: lower front, upper front, rear. Inner faces, for the backpack straps	3.2 mm x 4 mm
M3 brass	4	body_upper	Rear access cover, in the 10 mm lip round the 120 x 90 mm opening	4.0 mm x 6 mm
M2.5 brass	4	body_upper deck	Raspberry Pi 4 bosses, on the <code>pi4_holes</code> pattern	3.7 mm x 5.2 mm
M2.5 brass	2	body_upper deck	ADS1115 bosses	3.7 mm x 4.0 mm
M2 brass	10	body_upper deck	KB2040 (2) and the four DRV8833 boards (2 each)	3.2 mm x 4.0 mm

That is 41 inserts: 9 x M5, 4 x M4, 4 x M3, 8 x M2.5 and 16 x M2. Press each one square, with the iron set to about 250 C for PETG, and stop when the insert face is flush. An insert pressed at an angle cannot be straightened.

No inserts go in the feet, the legs, the centre leg or body_lower. Those parts use through-bolts with captive nuts in printed pockets. This is deliberate: a nut in a pocket does not pull out of a printed part under preload the way an insert can. It is also why the four M8 body rods and the four centre-leg bolts land in hex pockets, not inserts.

Stage 1 - feet, motors and wheels

Six Adafruit 3777 motors and twelve Adafruit 3766 wheels go in here. Two motors and four wheels per foot. **Do this stage on one outer foot first, before printing anything else** (mechanical.md section 6).

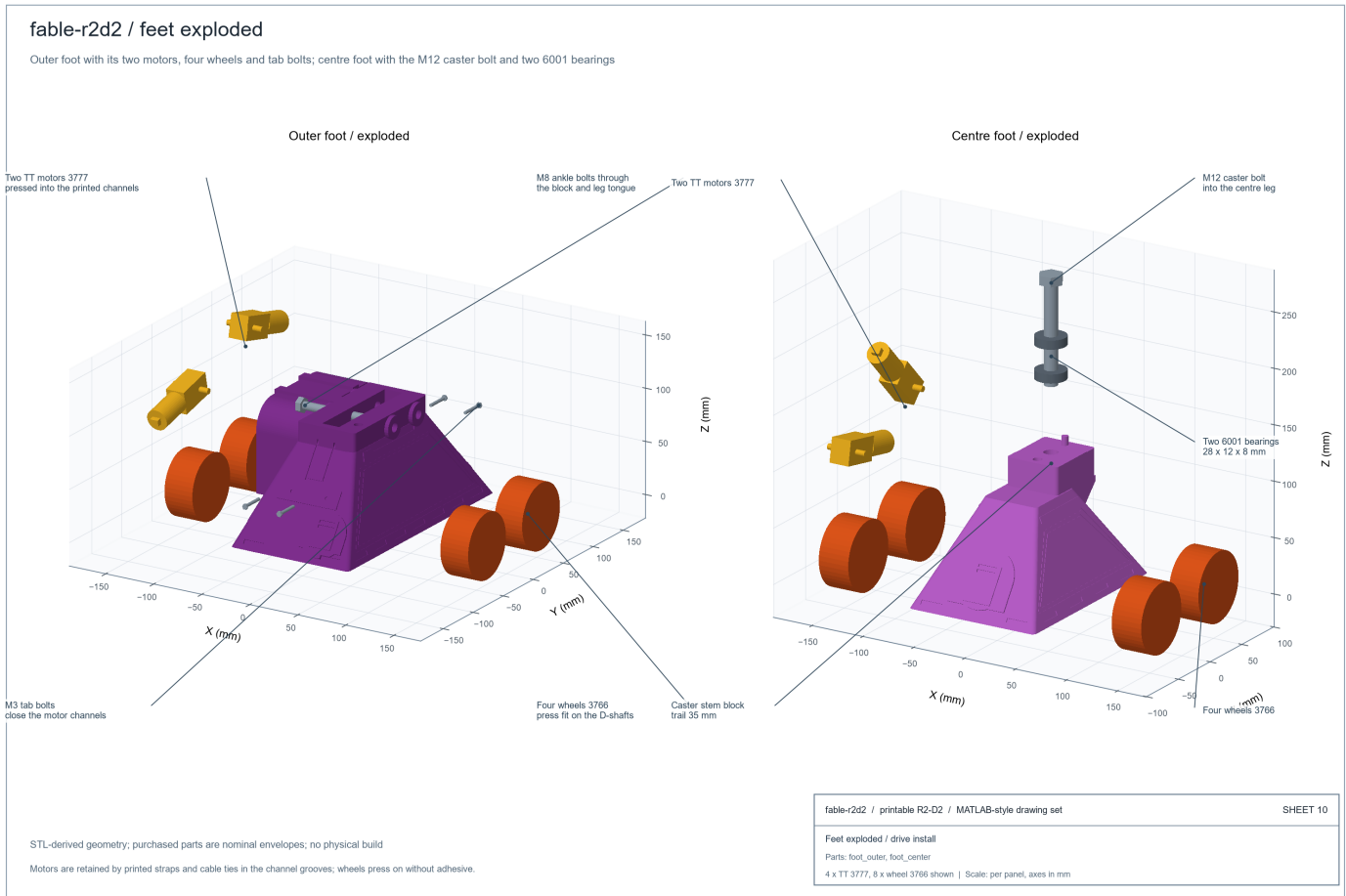


Figure 10 - outer foot exploded: shell, two TT motors, four wheels, tab bolts, cable ties.

- Centre foot only, and first:** slide the M12 hex nut into its slot under the top plate, entering from the +X side. The slot is 19.4 mm across flats and 8.4 mm deep. Once the motors are in you cannot reach it. Turn the foot upright and check the nut stays put.
- Check every motor pocket with calipers** before pushing a motor in. Each pocket is the motor envelope plus 0.4 mm (motor_pocket_clear). Clear any support residue.
- Push each motor up into its pocket from the open sole.** Every pocket is a vertical drop-in channel, so the motor and its wheels go in as one unit from below. The gearbox top face must land flat on the tilted pocket ceiling. That face contact is what carries the wheel load.
 - Outer feet:** both cans point outward and sit 13 degrees below horizontal.
 - Centre foot:** the front can points inward and **up 45 degrees**; the rear can points inward and 10 degrees below horizontal.
- Outer feet only:** fit the two M3 x 35 tab bolts, one per motor, across the end-block channel. The head goes on the -X cheek and the nut drops into the slot in the +X cheek. Tighten to **0.6 N.m (estimated)**. These pin the motor's rear tab.
- Cable-tie each can** in its saddle groove with a 4.8 mm tie. Two ties per foot. The tie holds the motor when the foot is lifted; the pocket ceiling holds it when the foot is down.
- Press the four wheels on.** Two per motor, at 23.8 mm each side of the motor centre plane (wheel1_x). Press the hub down until it is within 1 mm of the gearbox face. Support the gearbox, not the shaft, while pressing. **Mark the hub position on the shaft with a paint pen** - this mark is what the one-hour wear test in stage 11 reads.

7. **Splice the motor leads.** The stock 28 AWG leads are short. Splice each to 22 AWG inside the foot, per electrical.md section 7 step 8, and crimp a JST-XH pair on the far end. Route through the 8 mm wire hole in the top plate: on the outer foot at (X 20, Y -38), on the centre foot per the note in stage 3. Both motors reach it through the 8 mm pass in the centre rib.
8. **Check:** every wheel turns freely in its 3 mm cavity and nothing rubs the shell. Spin each motor by hand from the wheel. Both motors in a foot should feel the same.

Repeat for the second outer foot and the centre foot.

Stage 2 - outer legs

Two legs. Each is leg_upper plus leg_lower, spliced and rodded, then bolted to its foot.

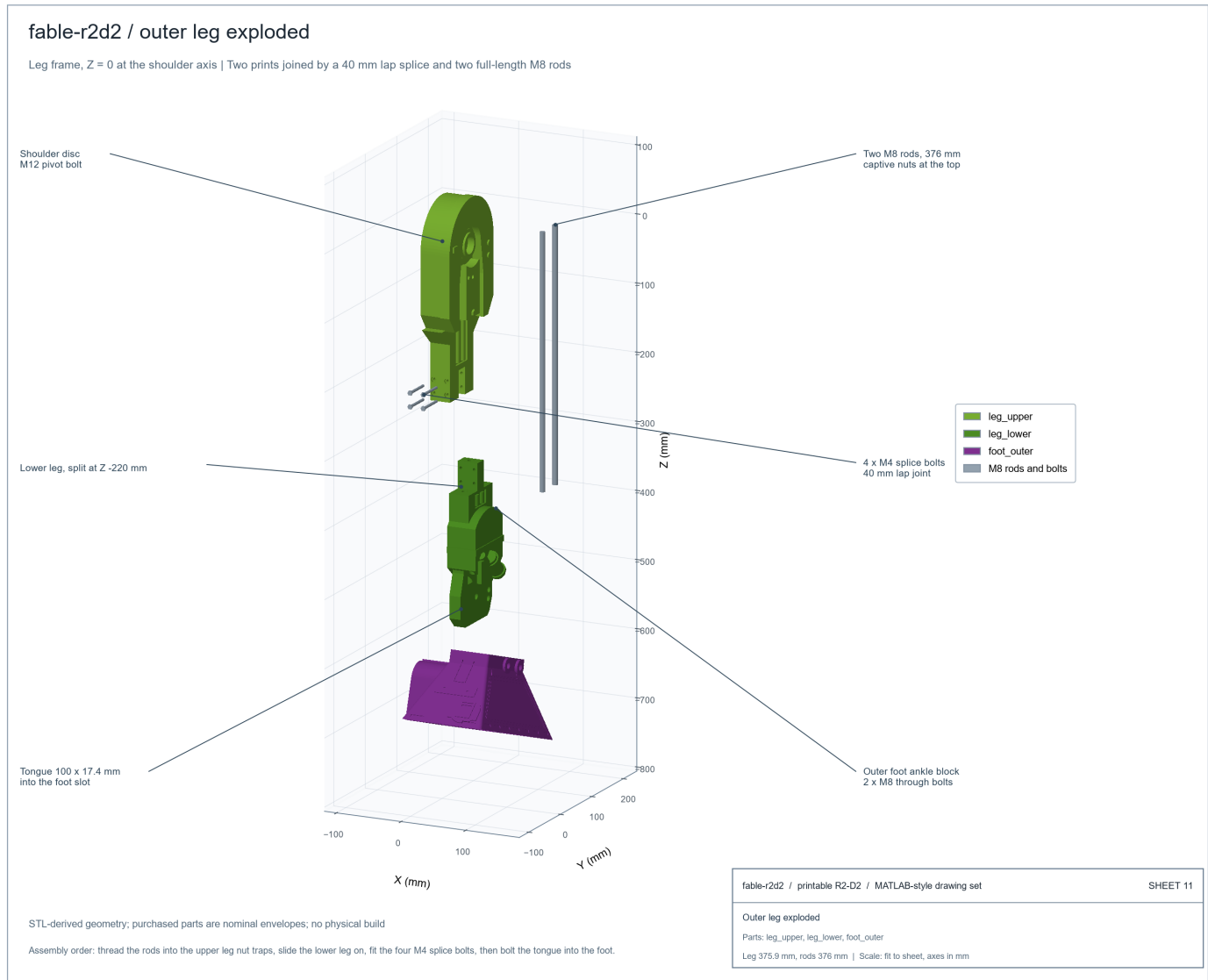


Figure 11 - outer leg exploded: upper, lower, comb joint, four M4 x 80, two M8 rods, ankle bolts.

1. **Cut four M8 rods to 390 mm.** The leg bore runs from z -336 to z +55, which is 391 mm. cad/legs.scad names M8 x 400 stock, so cut 10 mm off each. **File a chamfer on both ends of every rod** or it will not start in the captive nut. Two 1 m rods give all four leg rods with a 220 mm offcut. (bom/hardware.csv H09 plans 425 mm cuts. That does not fit the bore. Use 390 mm.)
2. **Load the captive top nuts.** Each leg_upper has two hex nut traps that open on the inboard face at z 40 to 48.4 mm. Slide an M8 nut into each from the inboard face. Check it sits square and cannot rotate.

3. **Dry-fit the comb joint.** The tongue on `leg_lower` is 20 mm wide and 40 mm tall and goes into the slot between the two fingers on `leg_upper`. Clearance is 0.2 mm each side. If it is tight, file the tongue, never the fingers - the fingers carry the rod bores.
4. **Fit the four M4 x 80 splice bolts.** They pass through the whole strut width, front to rear, at $x = 5$ and 25.4 mm, $z = -210$ and -190 mm. Heads go in the 8 mm counterbores on the **front** face; M4 nuts go in the pockets on the **rear** face. Add an M4 washer under each head and nut. Torque to **1.5 N.m (estimated)**.
5. **Feed the two M8 rods in from the ankle end.** Push each rod up through `leg_lower`, across the splice, into `leg_upper`, and thread it into the captive nut at the top. Turn the rod, not the leg.
6. **Tighten the rods.** At the ankle underside, fit an M8 fender washer (24 mm) and an M8 nut in each 22 mm counterbore. Use a deep 13 mm socket. Torque each to **1.6 N.m**, which is about 1 kN of preload (`research/loads.md` section 7). Work them up in two passes so the splice closes evenly.
7. **Check the splice:** no visible gap at the joint line, and the leg does not flex at the splice when you twist the two halves against each other by hand.
8. **Bolt the leg to its foot.** Drop the tongue into the foot slot. Line up the pivot bore at 6.3 mm above the shell top and push the **M8 x 80 pivot bolt** through both block cheeks and the tongue, head outboard. The nut sits in the pocket in the inboard boss. Run it down until the play is gone **but the tongue still swings by hand**. Do not torque it.
9. **Set the stance with the lock bolt.** The tongue's two lock holes lie on the same 40 mm arc about the pivot as the foot's single lock bore, so swinging the leg brings each in turn onto it. Swing to the 18 degree lean and fit the **M8 x 90** lock bolt through the foot's lock bore into tongue hole **B** at $(y +38.04, z -12.36)$. Hole **A** at $(y +40, z 0)$ is the two-leg display stance. Add threadlocker. Tighten until firm, **about 6 N.m (estimated)**; this bolt takes the ankle moment.
10. **Pull the motor leads up the leg.** The wire route enters the ankle underside at $(35, -38)$, runs diagonally into the strut centre, up the 9 mm bore, and exits the inboard face at $z -75$, which is below the shoulder pad. Leave 300 mm of slack at the top.
11. **Check:** the finished leg and foot weigh about 2.8 kg. The ankle has no play. The leg is straight when sighted down the strut. Repeat, mirrored, for the second leg.

Stage 3 - centre leg and caster

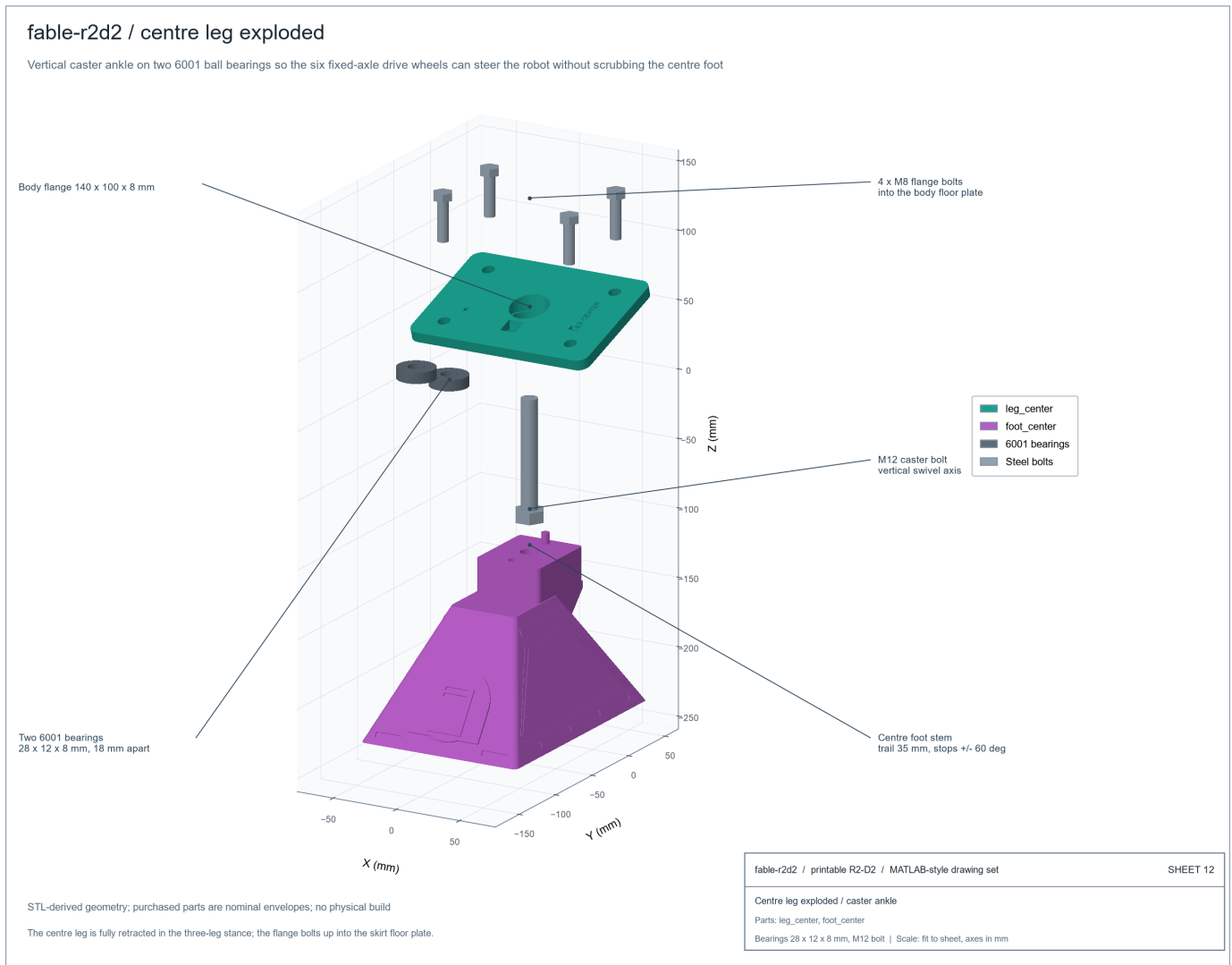


Figure 12 - centre leg exploded: leg, two 6001-2RS bearings, spacer tube, M12 x 70, stop pin, centre foot.

1. **Check both bearing seats with calipers.** They are 28.2 mm (caster_bearing_od plus 0.2 mm). The lower seat opens to the bottom face at z 26 to 34.2 mm; the upper opens into the 28 mm head counterbore. Centres are 18 mm apart (caster_bearing_gap).
2. **Press in the two 6001-2RS bearings,** 12 x 28 x 8 mm. Press on the **outer** race only. Each must sit flat on its ledge.
3. **Fit the 9.8 mm spacer tube** (12 mm ID) between the two inner races. This is what stops the M12 preload crushing the bearings.
4. **Fit the 6 x 8 mm swivel-stop pin** in the centre foot's stem-block top, at (X 0, Y caster_trail + 22), which is 22 mm ahead of the caster axis. That is the radius of the leg's arc groove (ft_stop_r = lg_stop_r = 22), and it lands on solid stem-block material.
5. **Set the leg on the foot** and drop the **M12 x 70 hex bolt** down from above: head into the 28 mm counterbore, washer, upper bearing, spacer tube, lower bearing, washer, 1 mm thrust washer, into the M12 nut already captive in the foot.
6. **Tighten the M12 from above, now.** Once body_lower is on the flange you cannot reach it. Tighten until the spacer tube is solidly clamped, **about 8 N.m (estimated)**. Add threadlocker.
7. **Check the swivel:** the leg must turn freely and stop at 60 degrees each way against the pin. If the caster is stiff, the spacer tube is too short or a bearing is not square.
8. **Route the centre-foot motor leads** up through the foot's 8 mm wire hole at (X 0, Y caster_trail - 21) = (0, -1), into the arc slot in the leg's bottom face (radius 16.5 to 22.5 mm), into the 12 x 12.5 mm vertical channel, and out of the flange top at

body-frame (0, -22). The hole sits at leg-frame radius 21, on the slot at **every** swivel angle. Leave 400 mm of slack: this bundle must survive 120 degrees of swivel.

Stage 4 - lower body ring

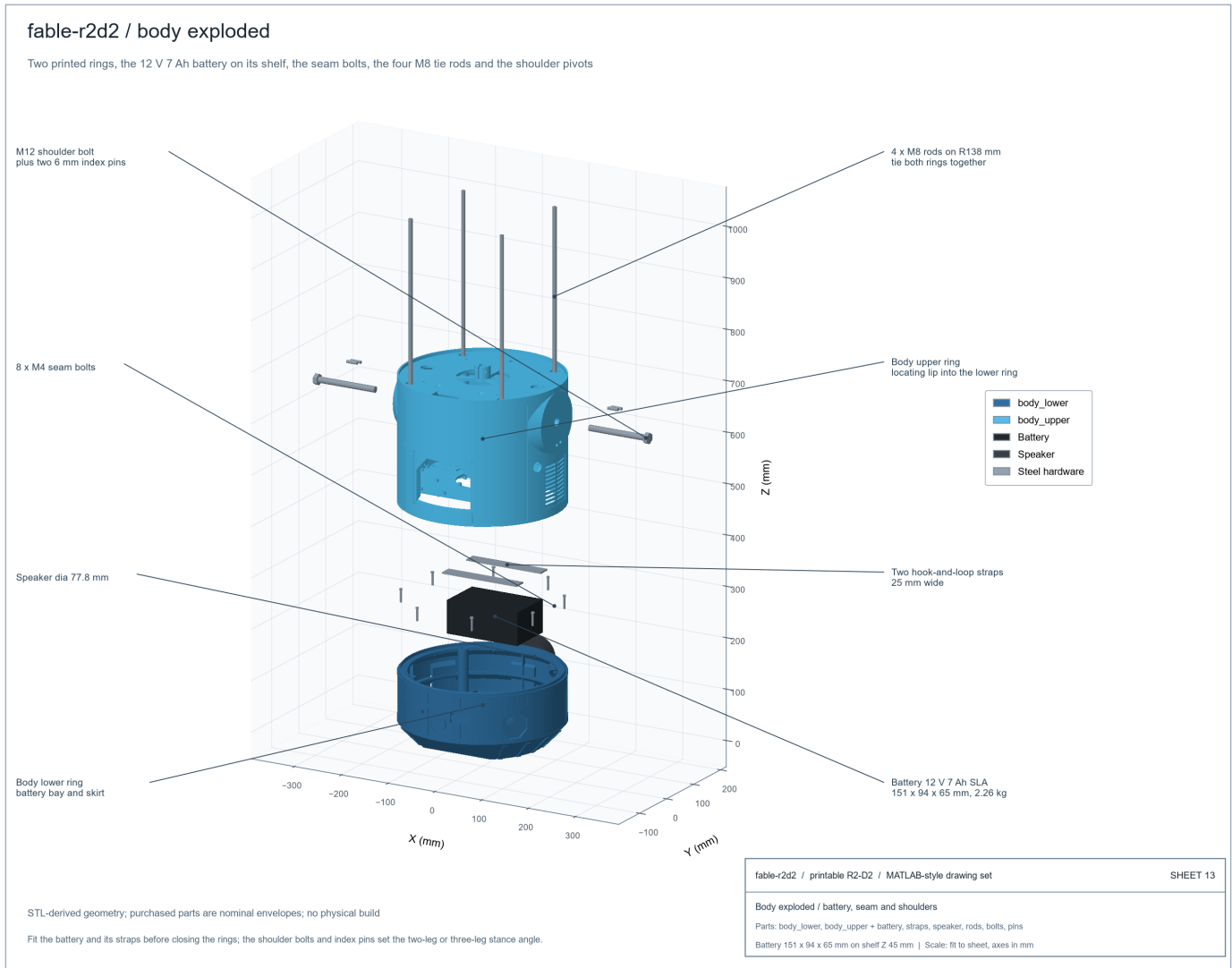


Figure 13 - body exploded: lower ring, upper ring, seam lip, eight M4, four M8 rods, battery, centre leg.

Work with the ring on the bench, skirt down.

Order matters here. The four M8 rod nuts and the four centre-leg nuts are both reached from inside, and the battery shelf covers part of the floor. Do steps 1 to 3 before the battery goes in.

- Load the four captive M8 rod nuts.** Each rod column starts on the **battery shelf**, not on the floor: radius 138 mm lies outside the skirt-bottom plate, whose maximum radius is 116.6 mm. The nut is a hex trap at **z 46**, opened downward with a slot facing the body axis. Slide an M8 nut into each of the four traps and check it cannot rotate.
- Bolt the centre leg on.** The floor has four M8 clearance holes at (+/-50, +/-32) mm with **hex nut pockets on its top face**. Drop an M8 nut into each pocket. Fit four **M8 x 30** bolts from below, heads in the 19 mm socket notches through the collar corners. Torque to **2 N.m (estimated)** - about 3 MPa on the printed floor, under the 6.6 MPa at which research/loads.md warns of creep. Go round twice. The battery shelf carries **four 26 mm socket windows** over these nuts, so you can reach them again later with a socket.

3. **Feed the centre-leg wire bundle** up through the 60 x 40 mm cable opening in the floor, centred at Y = -60, and out through the rear harness slot in the battery shelf.
4. **Line the battery shelf with foam tape**, inside the printed curb. The shelf is a 4 mm plate at z 41 to 45 and its curb stands 8 mm proud, 3 mm clear of the battery all round.
5. **Lay the battery on its side** inside the curb: 151 mm along X, 94 mm along Y, 65 mm tall. The pocket centre is at battery_y = **15 mm forward** of the body axis, which is deliberate - see mechanical.md section 5.4. Terminals must face where the harness reaches them and must not touch a rod column.
6. **Strap the battery down** with two 25 mm hook-and-loop cinch straps, threaded through the four 3 mm strap slots in the shelf and cinched across the top. One strap does not reach round the battery on its own. The straps and the curb are the only things holding 2.26 kg when the robot is tipped.
7. **Fit F1 within 100 mm of the battery's positive terminal**. This is not optional. A 12 V 7 Ah SLA delivers roughly 457 A into a short.
8. **Fit the speaker** behind the front pocket vent door, which is a grille of **nine 5 mm vertical louvres**. The speaker is 77.8 mm in diameter and 25.5 mm deep and lands on the printed bosses behind the door. Use foam tape round the rim so it does not buzz.
9. **Fit the charge port and the switch** on the rear lower band pad: the Switchcraft L722A panel jack (5.5 x 2.1 mm, centre positive) and the Carling rocker in its 36.83 x 21.08 mm cutout. The charge branch joins the **battery** side of the switch, so the charger works with the robot off. Wiring is electrical.md sections 1 and 5.
10. **Stand the four M8 body rods** in the captive nuts from step 1, at radius 138 mm, at 45, 135, 225 and 315 degrees. Cut them to **385 mm** each. Two 1 m rods give all four with a 230 mm offcut. Chamfer both ends. Thread each a few turns into its nut and leave it loose: all the tightening happens at the top plate in stage 7.
11. **Check:** the battery does not move when the ring is tipped 45 degrees in any direction. All four rods stand parallel and clear the shelf, the curb and the skin.

Stage 5 - upper body ring, shoulders and legs

The legs go on the upper ring while it is still separate and open at both ends. Every shoulder fastener is reachable from inside. **Do not do this after the rings are joined.**

1. **Press the inserts** listed in the schedule: four M5 into the top plate from **above**, for the lazy-susan bottom race; four M3 into the rear access lip; and the deck inserts (four M2.5 for the Pi, two M2.5 for the ADS1115, ten M2 for the KB2040 and the four drivers).
2. **Fit a steel crush sleeve** through each 30 mm shoulder boss. research/loads.md section 7 is explicit: do not clamp printed faces with a torqued M12. The sleeve makes the clamp steel to steel.
3. **Fit the two flanged bronze bushings** (12 mm bore, 16 mm OD, 20 mm long, 20 mm flange) into the leg hub bore, one from each side.
4. **Support the leg**. With its foot it weighs about 2.8 kg and it will fall.
5. **Offer the leg up to the pad** with the 8 mm spacer stack (bushing flange plus washer) between the flat pad and the flat inboard face of the leg plate.
6. **Fit the M12 x 130 class 8.8 bolt** from the leg side. The head goes into the 26 mm counterbore in the leg hub, 10 mm deep. Add M12 fender washers (37 mm) under the head and the nut. The M12 nyloc drops into its hex pocket inside the ring, 13 mm deep.
7. **Torque to 5 N.m**. That is about 2.1 kN of preload and about 1.9 MPa of face pressure. Do not go higher: a smaller washer or more torque makes the PETG creep.
8. **Set the stance with the index dowels**. The body pad carries **two** 6.2 mm holes, 20 mm deep, at radius 45 mm, placed toward the **front**; the leg carries two at the same radius placed toward the **rear**. Rotate the leg forward 18 degrees. Two pairs line up at once: leg hole 18 on body hole 0, and leg hole 0 on body hole 18. **Fit both 6 x 30 mm steel dowels**, pushed in from **outboard** through the booster cover and the plate. Light taps, not a hammer. In the two-leg display stance only one pair lines up, and one dowel is used.

9. **Check:** with both dowels in, the leg cannot rotate. With them out and the M12 at 5 N.m, the leg turns with firm hand pressure. If it turns freely with a dowel in, the hole has been drilled oversize and the pin will not carry the 9.4 N.m shoulder moment.
10. Repeat for the second leg.
11. **Pull both leg wire bundles** in through the shoulder area and up into the ring.

Stage 6 - electronics deck

The deck is integral to body_upper: a 4 mm annulus at local z 36 to 40, from radius 50 to 154.9 mm, cut off by a chord at y = -114.9 that leaves a **40 mm harness gap** at the rear. Build the whole harness on the bench first, in the order in electrical.md section 7. Do not connect the battery yet.

Each board has its own printed seat, and each seat is labelled in the print.

Seat	Label	Fixing
Raspberry Pi 4	PI4	Four M2.5 boss inserts on the pi4_holes pattern
KB2040	KB2040	Two M2 boss inserts
Four DRV8833	DRV8833	A 26 x 18 mm pad each, two M2 boss inserts each
Three regulators	5V / 12V	A 25.4 x 25.4 mm pad each with four 2.2 mm holes
ADS1115	ADC	Two M2.5 boss inserts
Fuse holders	FUSE	One 46 x 20 mm seat

1. **Fit TB2, the star ground stud, first.** Every return in the robot lands there and nowhere else.
2. **Mount each board on its own seat**, using the inserts above. Use M3 nylon standoffs only for the PAM8302 amplifier, which has no printed seat.
3. **Route the 12 V bus, the regulator outputs and the signal harness** per electrical.md section 7 and electronics/wiring.csv. Label both ends of every signal wire. The deck has **eight 6 x 2.5 mm cable slots** through it; use them rather than running wire round the deck edge.
4. **Bring the seven motor pairs up** through the 40 mm rear harness gap: two from each outer leg, two from the centre leg, and the head-drive pair. Land each at its driver output on a JST-XH pair.
5. **Cable-tie everything to adhesive mounts** on the inside of the ring. Keep the harness out of the head-drive keep-out: |x| up to 66 mm, y -160 to -70 mm, body-frame z 160 to 378 mm. A cable tie in that space will be cut by the friction wheel.
6. **Fit no fuses yet.** They go in during the rail checks in stage 10.
7. **The rear access opening** (120 x 90 mm, body-frame z 165.1 to 255.1) is how you reach this deck once the robot is together. Leave its cover off until stage 10 is finished, then fit it with four M3 screws into the lip inserts.

Stage 7 - join the two rings

1. **Check the seam lip is clean.** It is at radius 151.6 to 154.6 mm on the lower ring, 6 mm tall, with a 15 degree lead-in chamfer on its top outer edge. The socket in the upper ring is radius 151.3 to 154.9, so there is 0.3 mm of clearance each side. Do not sand it.
2. **Lower the upper ring onto the lower ring**, feeding the four M8 rods through the upper ring's rod bosses as it goes. Two people, or a strap. Watch the leg wire bundles.
3. **Seat the lip fully.** No gap anywhere round the joint line.
4. **Fit the eight M4 x 20 socket cap screws from above**, through both internal flanges on the radius 148.9 mm bolt circle at 22.5, 67.5, 112.5 and so on. The hex nut pockets open downward under the lower flange, so the nuts drop in and stay put. Add a washer under each head. Torque to **1.5 N.m (estimated)**, going round in a star pattern, twice.
5. **Fit the four M8 rod nuts at the top plate**, each in its 18 mm x 4 mm counterbore with a washer. Torque each to **1.6 N.m** (about 1 kN of preload). Work them in a cross pattern in two passes. These rods put the whole body in compression from the battery shelf to the top plate and carry the dome, the head drive and the shoulder reaction. Each nut then stands 2.8 mm proud and clears the dome plate by 5.1 mm.

6. **Do this before the lazy susan goes on.** The bottom race covers the rod counterbores.
7. **Check:** the body is one column. Lift it by the top plate - nothing at the seam moves. Re-check the eight M4 after the rods are torqued; the rods will have pulled the seam tighter.

Stage 8 - head drive

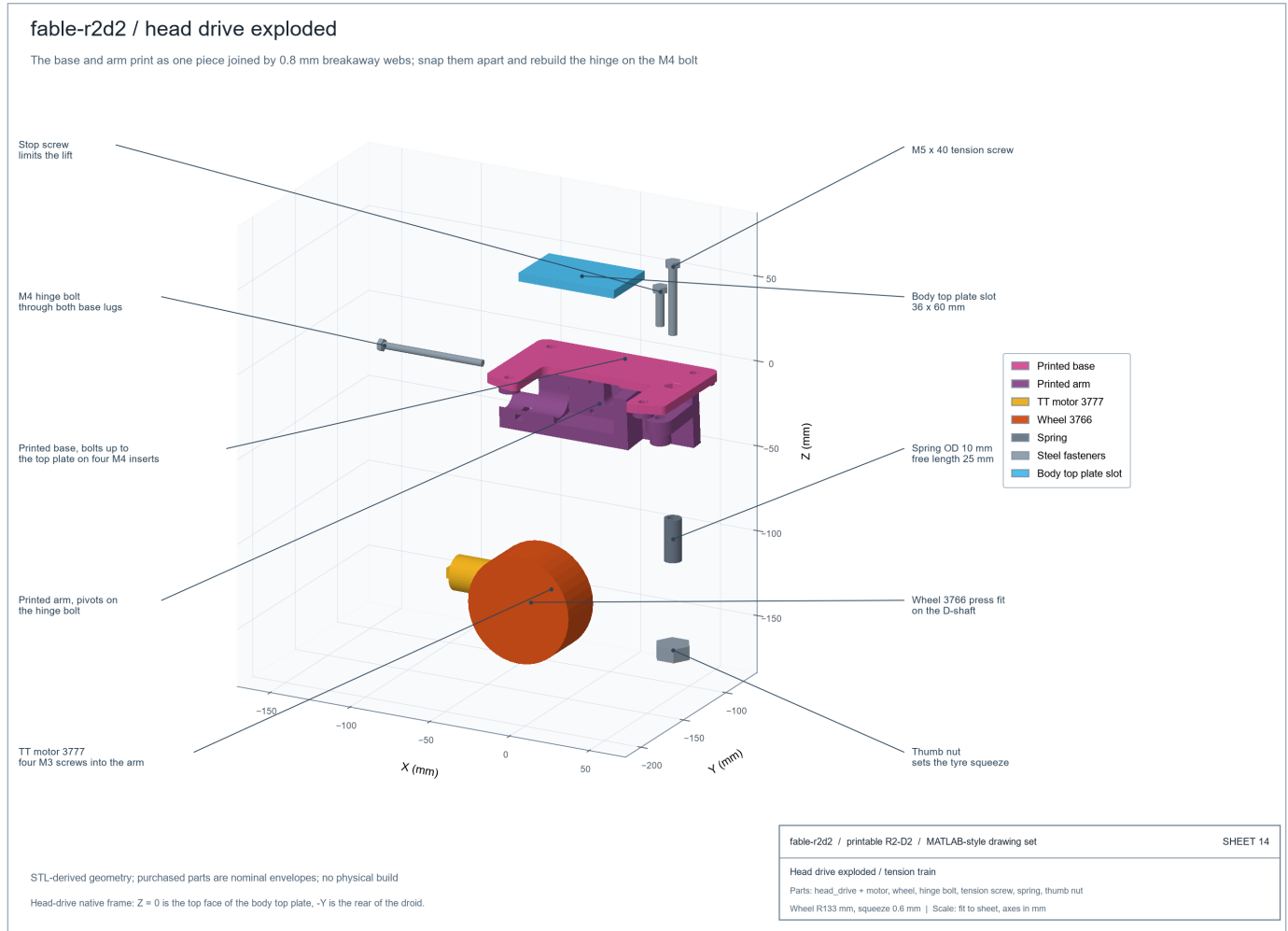


Figure 14 - head-drive exploded: base, arm, motor, wheel, M5 tension bolt, spring, thumb nut, lift stop.

The order below is the order in the `cad/head_drive.scad` header. It exists because each screw is only reachable at one point in the sequence.

1. **Cut the two sacrificial webs.** They are 0.8 mm thick, 6 mm long, at $x = \pm 15$ mm, joining the arm's hinge-boss ridge to the base plate underside. Cut both with a knife and **file the stubs flush**. The arm needs 3 mm of clearance at the ridge.
2. **Remove the tree supports** from inside the motor pocket, under the side beam and under the boss-to-wall bridge.
3. **Press the four M4 inserts** into the base mating face and the **one M5 insert** into the arm's stop boss.
4. **Drop the M5 x 60 hex head into its pocket** on the base's mating face. It hangs head-up and the top plate captures the head. It must be a hex head (DIN 933): a socket cap head spins in the pocket.
5. **Hold the base under the body top plate** and fit the **four M4 x 12 countersunk screws** from the **top face**, into the inserts. Torque to **1.2 N.m (estimated)**. The two inner heads sit under the lazy-susan bottom race, so they must be flush.
6. **Fit the arm.** Slide its ear slot over the hanging M5. Push the **M4 x 70 socket head** through the base lugs and the arm boss from inside the open body ring. Washer each side, then the M4 nyloc. **Snug it only. Do not torque it.** The arm must swing freely.

7. **Drop the motor into the saddle.** Fit the **two M3 x 30 button heads from the wheel side**, with their nuts in the blind pocket and the top slot. **They must go in before the wheel is pressed on.** Torque to **0.6 N.m (estimated)**.
8. **Press the wheel on** and **cable-tie the can** through the tunnel with a 3.6 mm tie.
9. **Fit the tension stack** on the M5: 15 mm OD washer, compression spring (10 mm OD, 25 mm free, 1.0 mm wire), plain washer, M5 knurled thumb nut. Leave it loose.
10. **Screw the M5 x 20 lift stop fully home** in the arm boss. Its tip then stands 5 mm proud and limits the arm's lift to 1 mm above working. With the dome off, this is what stops the gearbox hitting the base.
11. **Leave the drive released.** Tensioning happens in stage 9, after the dome is on.
12. **Check:** the arm swings from the lift stop down to the 7 degree release without touching anything. The wheel comes up through the 36 x 60 mm slot and nothing else does.

Release procedure, for every time the dome comes off later: back the thumb nut off about 8 mm, which is 10 turns. The arm falls until the bolt shank meets the end of the ear slot at 7 degrees. The wheel top is then 4 mm below the dome plate. The nut stays on the bolt.

Stage 9 - dome electronics and the lazy susan

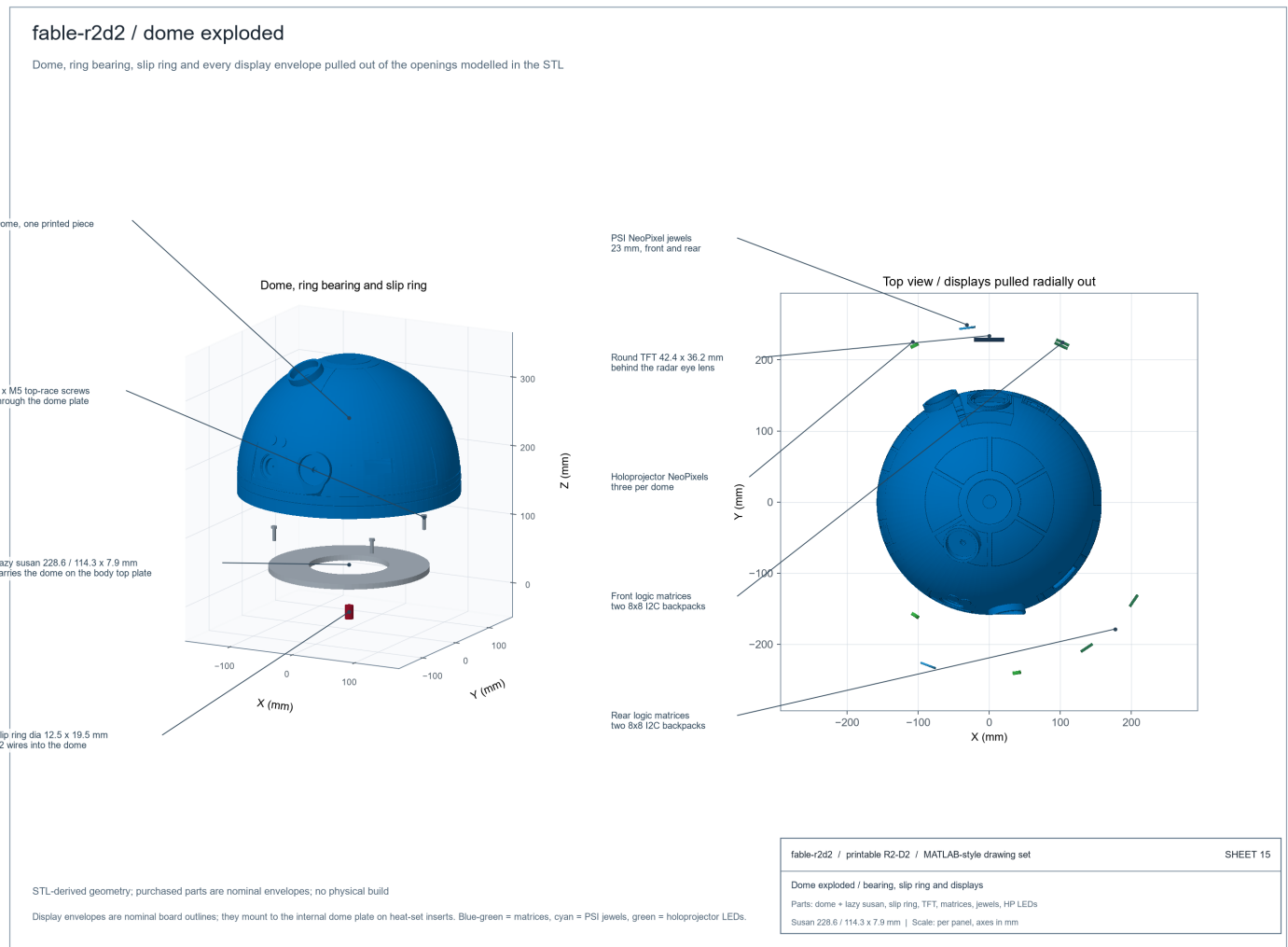


Figure 15 - dome exploded: dome, TFT, four matrix backpacks, two jewels, three holo projector pixels, lens discs, lazy susan, slip ring.

Do all the dome soldering with the dome off the robot and the backpack address jumpers already bridged (electrical.md section 7 step 9).

1. **Press the dome inserts:** four M5 from below into the plate bosses, two M2.5 in the radar eye housing, six M2 in the three logic-display blocks.
2. **Fit the round TFT** (Adafruit 6178, 42.4 x 36.2 x 5.4 mm, two mounting holes 22.8 mm apart). Its 55.7 mm diagonal will not pass the 52 mm lens seat, so **it loads from inside**, against the seat lip. Hold it with a strap across the two M2.5 bosses, 54.4 mm apart.
3. **Fit the four 8x8 matrix backpacks:** two in the front logic pockets (20.6 x 28.6 mm each, one shifted down and one shifted up so each 20 x 20 matrix centres in its window), and two side by side in the rear pocket (40.6 x 28.6 mm). Strap each block with the M2 screws. Addresses are 0x70, 0x71, 0x72 and 0x73 (electrical.md section 6).
4. **Fit the two NeoPixel Jewels** in the front and rear PSI pockets, 24 mm x 4.5 mm, on their 3.3 mm floors, with the leads through the 3.2 mm wire holes.
5. **Fit the three holoprojector NeoPixels**, one per barrel, in the 8.4 x 6 mm pockets with the leads through the 8 x 3.5 mm slots.
6. **Glue the three acrylic lens discs** into their seats: radar eye 52.1 x 2 mm, front PSI 29 x 1.5 mm, rear PSI 36 x 1.5 mm. Every seat has 0.5 mm of clearance. Use a thin bead and keep cement off the face.
7. **Wire the dome side of the slip ring.** Twelve rotor leads to the dome devices, per the table in electrical.md section 6. Pull the bundle through the cable notches in the plate inner edge and **anchor it with a zip tie through the wire-anchor lug** at radius 70 mm. The anchor is what stops a pull ever reaching a slip-ring lead.
8. **Fit the slip-ring body** in the hub at the centre of the top plate's three-spoke spider. The bore is 12.5 mm and is open top and bottom; an M3 pinch clamp closes it. Cable-tie **both** bundles within 40 mm of the ring.
9. **Fit the lazy-susan bottom race first.** Set the Triangle 9C on the body top plate, concentric, and drive four M5 x 12 screws down into the top-plate inserts. Torque to **2 N.m (estimated)**. Check the dome turns smoothly by hand on the race before the dome goes on.
10. **Lower the dome on.** Line its plate bosses up over the race's top plate.
11. **Drive the top-race screws up through the access holes.** There are **three** 12 mm access holes, at 0, 90 and 270 degrees; the top-race screw holes are at 45, 135, 225 and 315 degrees. Rotate the dome to bring each screw hole over an access hole, then drive an M5 x 12 up into the dome-plate insert. *cad/body.scad omits the 180 degree access hole altogether*, because the head-drive base covers it. Three holes reach all four screws.
12. **Tension the head drive.** Turn the thumb nut **up** until the spring is compressed 4 to 6 mm, which is a spring length of about 20 mm. That is 8 to 12 N at the ear and 4 to 6 N at the tyre. *research/loads.md* section 5 asks for 3 to 5 N.
13. **Check:** the dome turns by hand with light, even drag and no notch. The gap at the body top edge is even all the way round, about 1.9 mm. The dome does not lift off the race anywhere.

Stage 10 - firmware and first power-on

Do the rail checks **before** the Pi, the drivers or the dome are connected. Follow electrical.md section 8 step by step and do not skip step 1: a bridged fuse holder on a pack that can deliver about 457 A into a short is how harnesses catch fire.

1. **Rail checks**, electrical.md section 8 steps 1 to 4. Fuses go in one at a time. Each step ends with the power off.
2. **Flash the KB2040**, firmware.md section 5.
3. **Set up the Raspberry Pi**, firmware.md section 6: buses and groups, code and dependencies, controller port, sound, the R2-FABLE access point, and the service.
4. **Connect the loads and boot**, electrical.md section 8 step 5. The Pi must boot and no rail may sag below its tolerance.
5. **Check the battery gauge** against a meter at the battery terminals, within 0.15 V (electrical.md section 8 step 6).
6. **Drive test with the wheels off the ground.** Put the robot on blocks. Send `E 1` then `M 300 0 0 0` on the KB2040's serial port. Only the left foot should turn, forward. If a motor turns the wrong way, swap its two leads at the driver output.
7. **Repeat for every motor**, one at a time, before any of them touch the floor. Seven motors: six drive and one head.
8. **Head drive:** run it at 50 percent PWM. The dome should turn at about 8 to 9 RPM. Below 35 percent it stalls.

9. Check the control page, firmware.md section 7.

Figures

The manual builder inserts these exploded diagrams. The names are fixed; the PDF build renders them from output/drawings/.

Figure	File	Caption
10	output/drawings/10_exploded_feet.png	Outer foot exploded: shell, two TT motors, four wheels, tab bolts, cable ties
11	output/drawings/11_exploded_leg.png	Outer leg exploded: upper, lower, comb joint, four M4 x 80, two M8 rods, ankle bolts
12	output/drawings/12_exploded_center_leg.png	Centre leg exploded: leg, two 6001-2RS bearings, spacer tube, M12 x 70, stop pin, centre foot
13	output/drawings/13_exploded_body.png	Body exploded: lower ring, upper ring, seam lip, eight M4, four M8 rods, battery, centre leg
14	output/drawings/14_exploded_head_drive.png	Head-drive exploded: base, arm, motor, wheel, M5 tension bolt, spring, thumb nut, lift stop
15	output/drawings/15_exploded_dome.png	Dome exploded: dome, TFT, four matrix backpacks, two jewels, three holoprojector pixels, lens discs, lazy susan, slip ring
16	output/drawings/16_exploded_robot.png	Whole robot exploded: dome, two body rings, head drive, two legs, centre leg, three feet

Torque summary

Every value below is collected from the stages above. Values marked "estimated" are design judgement; the two unmarked values come from research/loads.md section 7.

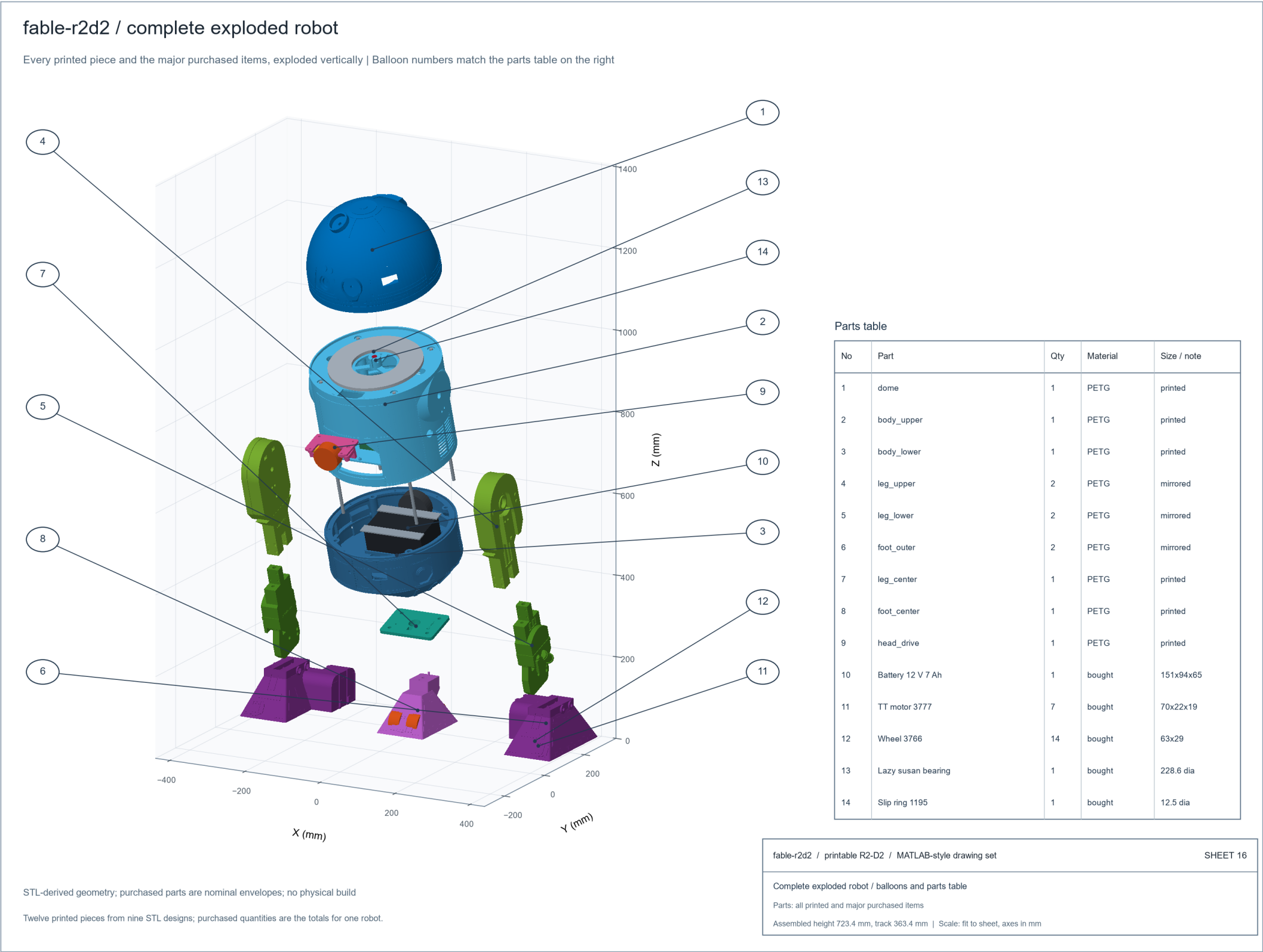
Fastener	Where	Torque
M12 x 130 class 8.8	Shoulder pivot, with a crush sleeve and 37 mm fender washers	5 N.m
M8 nut on a threaded rod	Four body rods, four leg rods	1.6 N.m (about 1 kN preload)
M12 x 70	Centre caster, against the spacer tube	8 N.m (estimated)
M8 x 80	Ankle lock bolt	6 N.m (estimated)
M8 x 80	Ankle pivot bolt	Snug only, the tongue must swing
M8 x 30	Centre-leg flange, four off	2 N.m (estimated)
M4 x 80	Leg splice, four per leg	1.5 N.m (estimated)
M4	Ring seam, eight off	1.5 N.m (estimated)
M5 x 12	Lazy susan, eight off, into inserts	2 N.m (estimated)
M4 x 12 countersunk	Head-drive mount, four off	1.2 N.m (estimated)
M4 x 70	Head-drive hinge, nyloc	Snug only, never torqued
M3 x 35	Foot motor tab bolts	0.6 N.m (estimated)
M3 x 30	Head-drive motor	0.6 N.m (estimated)
M2, M2.5	Dome display straps	Finger tight

Fastener count for the whole robot

Fastener	Count	Where
M12 x 130 class 8.8 hex bolt, nyloc, 2 fender washers	2	Shoulders
M12 x 70 hex bolt, washers, 1 mm thrust washer	1	Centre caster
M12 hex nut, captive	1	Centre foot stem
6001-2RS bearing, 12 x 28 x 8	2	Centre caster
12 mm ID x 9.8 mm spacer tube	1	Centre caster
Flanged bronze bushing, 12 x 16 x 20	4	Shoulders, two each
Steel crush sleeve, 30 mm	2	Shoulder bosses
6 x 30 mm steel dowel	4	Shoulder index pins, two per shoulder in the three-leg stance
6 x 8 mm pin	1	Caster swivel stop
M8 threaded rod, 385 mm	4	Body
M8 threaded rod, 390 mm	4	Legs, two each
M8 x 80 hex bolt and nut	2	Ankle pivot, one per outer foot
M8 x 90 hex bolt and nut	2	Ankle lock, one per outer foot
M8 x 30 bolt and nut	4	Centre-leg flange
M8 nut, captive in a trap	4	Leg rod tops
M8 nut and fender washer	12	Rod ends
M5 x 12 socket cap	8	Lazy susan, four per race
M5 x 60 hex head (DIN 933), spring, 2 washers, thumb nut	1	Head-drive tension
M5 x 20 socket cap	1	Head-drive lift stop
M5 heat-set insert	9	Top plate 4, dome plate 4, stop boss 1
M4 x 80 bolt and nut	8	Leg splices, four per leg
M4 x 20 socket cap and nut	8	Ring seam, screws from above, nuts under the lower flange
M4 x 12 countersunk	4	Head-drive mount
M4 x 70 socket head and nyloc	1	Head-drive hinge
M4 heat-set insert	4	Head-drive base
M3 x 35 bolt and nut	4	Outer-foot motor tabs
M3 x 30 button head and nut	2	Head-drive motor
M3 heat-set insert and screw	4	Rear access cover
M2.5 heat-set insert and screw	2	Dome TFT strap
M2.5 heat-set insert and screw	6	Deck: Raspberry Pi 4 (4), ADS1115 (2)
M2 heat-set insert and screw	6	Dome display straps
M2 heat-set insert and screw	10	Deck: KB2040 (2), four DRV8833 (8)
Cable tie, 4.8 mm	6	Motor cans, two per foot
Cable tie, 3.6 mm	1	Head-drive motor can

Quantities, part numbers, prices and sources are in `bom/hardware.csv`.

Figure 16 - the whole robot exploded



7 Firmware and first power-on

Reproduced from docs/firmware.md sections 1, 2 and 3 (what runs where, the serial protocol and the drive mixing) and sections 5 and 6 (flashing the KB2040 and setting up the Raspberry Pi).

1. Files

KB2040 - firmware/kb2040/

File	Purpose
boot.py	Enables the second USB CDC endpoint (the Pi's data port).
code.py	Controller: motors, pixels, index sensor, serial loop, heartbeat.
protocol.py	Hardware-free parser, ramp and duty maths. Imported by code.py.
test_protocol.py	CPython unit tests for protocol.py. Not copied to the board.
README.md	Exact copy list, bundle libraries, pin table, bench check.

Raspberry Pi - firmware/pi/

File	Purpose
server.py	aiohttp server: static page, WebSocket, control state, workers.
mixing.py	Differential drive maths, clamps, speed cap, minimum duty, ramp.
serial_link.py	Background thread owning the USB CDC port; 20 Hz command stream.
displays.py	Four 8x8 HT16K33 logic matrices and the radar eye.
audio.py	WAV playback through aplay, or pygame.mixer as a fallback.
battery.py	Optional pack voltage through an ADS1115.
test_mixing.py	CPython unit tests for mixing.py.
requirements.txt	Pi Python dependencies.
r2d2.service	systemd unit.
static/index.html, static/app.js, static/style.css	The control page.
sounds/*.wav	Twelve generated clips, 22.05 kHz 16 bit mono.

Elsewhere

File	Purpose
scripts/generate_audio.py	Synthesises the twelve clips and writes the catalogue.
audio/catalog.csv	name, file, seconds, sha256 for every clip.

Dome display and I2C map

Everything in this table is on the **Raspberry Pi's own I2C bus 1** (header pins 3 and 5, /dev/i2c-1) except the radar eye, which is on the Pi's SPI0. The KB2040 drives no display at all; its STEMMA QT bus stays unused.

Device	Bus	Address	Driven as
Front logic, upper window	I2C-1	0x70	Adafruit 0.8 inch 8x8 mini matrix backpack, Matrix8x8, its own animation
Front logic, lower window	I2C-1	0x71	Adafruit 0.8 inch 8x8 mini matrix backpack, Matrix8x8, its own animation
Rear logic, left (seen from behind)	I2C-1	0x72	8x8 backpack, columns 0-7 of one 16x8 strip
Rear logic, right (seen from behind)	I2C-1	0x73	8x8 backpack, columns 8-15 of the same strip
Battery ADC, optional	I2C-1	0x48	ADS1115, channel A0/P0
Radar eye	SPI0	CE0	GC9A01A 240x240, DC GPIO25, RST GPIO24, backlight GPIO18

Four 0.8 inch **8x8** backpacks, not three 8x16 units. Set each backpack's address with its A0/A1 solder jumpers: none bridged gives 0x70, A0 bridged 0x71, A1 bridged 0x72, both bridged 0x73.

The two **front** windows are independent displays and get independent animations with separate random seeds, so the upper and lower windows never fall into the same rhythm. The two **rear** backpacks are treated as one 16 column by 8 row strip sharing a single pattern that scrolls sideways across the seam, so the movement carries from one unit to the other instead of stopping between them.

Degradation is per device. A backpack that does not answer at start up, or that stops answering later, is logged with its address and dropped while the others keep running. If only one rear backpack answers, the strip still runs on that half.

Battery sensing does not touch the KB2040

Stated here so the wiring author does not route a divider to the microcontroller:

- `firmware/pi/battery.py` reads **only** an ADS1115 on the **Pi's** I2C-1 bus at 0x48, single-ended channel A0 (`ads1115.P0`), through a 100 kilohm over 15 kilohm divider across the 12 V pack (`DIVIDER_TOP_OHMS` and `DIVIDER_BOTTOM_OHMS` in that file).
- It never opens a serial port and never speaks to the KB2040.
- The KB2040 serial protocol carries **no** voltage field: `st` reports `<enabled>` `<l>` `<r>` `<c>` `<h>` `<index>` and nothing else.
- No KB2040 pin is free for this anyway. Its only ADC pins are A0-A3 (GP26-GP29): A0 and A1 drive the head motor, A2 is the NeoPixel data line and A3 is the dome index sensor. The one spare pin, D0/GP0, has no ADC on the RP2040.

So the divider's tap wire goes to the **ADS1115 A0 input on the Pi's I2C bus**, and nowhere near the KB2040.

2. Serial protocol

ASCII, newline terminated, 115200 8N1 on the KB2040's second USB CDC endpoint (`/dev/ttyACM1` on a Pi with nothing else plugged in).

Pi to KB2040

Line	Meaning	
M <code><l></code> <code><r></code> <code><c></code> <code><h></code>	Drive permille, each -1000..1000: left foot, right foot, centre foot, head. Both motors in a foot get that foot's value. Out of range values are clamped, not rejected.	
L <code><r></code> <code><g></code> <code></code>	Set all 17 pixels, each channel 0..255.	
P <code><i></code> <code><r></code> <code><g></code> <code></code>	Set one pixel, <code>i</code> in 0..16.	
E <code><0\</code>	1>	DRV8833 SLP line: 1 wakes the drivers, 0 sleeps them and zeroes the motors.
S	Stop: targets and outputs to zero at once.	
?	Status request.	

KB2040 to Pi

Line	Meaning
ok	The previous command was executed.
st <code><enabled></code> <code><l></code> <code><r></code> <code><c></code> <code><h></code> <code><index></code>	Status. <code>enabled</code> and <code>index</code> are 0/1; the four channels are the post-ramp permille actually applied. Sent on ? and unsolicited every 200 ms.
err <code><text></code>	The line could not be executed, or the heartbeat expired.

Rates and limits

- **Tick:** 10 ms on the KB2040.
- **Ramp:** at most 40 permille per tick per channel, so a full -1000 to +1000 reversal takes 50 ticks (500 ms). The Pi applies the same limit at 200 permille per 50 ms command, so the value it reports is the value the motors are really being given.
- **Heartbeat:** no line for 500 ms and every motor coasts, with `err heartbeat timeout`, `motors coasted` sent once. The Pi's 20 Hz M stream is the heartbeat; it never stops while the server is up.

- **Pixel index range:** 0..16. Front PSI jewel 0-6, rear PSI jewel 7-13, holoprojectors 14-16.

3. Drive mixing

The web page sends a joystick position x , y each in $-1..1$, with y forward and x to the right. `firmware/pi/mixing.py` computes:

```
left  = y + x * turn_gain      turn_gain = 0.70
right = y - x * turn_gain
centre = y                     the centre foot is a driven caster
```

Each channel is then clamped to $-1..1$, run through the deadband (0.05), multiplied by the global speed cap (default 0.60, slider range 0.20 to 1.00), and raised to the minimum duty (0.15 of full) if it is still moving, so a gearbox never sits buzzing below its stiction. The result is converted to permille and sent as M at 20 Hz.

Worked examples at the default cap:

Joystick	M line
full forward $x=0$, $y=1$	<code>M 600 600 600 <head></code>
full reverse $x=0$, $y=-1$	<code>M -600 -600 -600 <head></code>
spin right $x=1$, $y=0$	<code>M 420 -420 0 <head></code>
centred	<code>M 0 0 0 <head></code>

Head: the spin buttons hold the dome motor at the head-duty slider value (default 0.45) while pressed; a nudge runs it for 200 ms. The same 0.15 minimum duty applies so the friction wheel does not stall against the dome ring.

Safety. The page sends a drive message or a ping at least five times a second. If the server hears nothing for 500 ms it zeroes the drive; if the page closes, the server sends `S`; if the server dies, the KB2040's own heartbeat coasts the motors 500 ms later. The STOP button zeroes the drive and drops the enable line in one press.

5. Flashing the KB2040

1. Download CircuitPython 9.x for the KB2040 from <https://circuitpython.org/board/adafruit_kb2040/> - a .uf2 file.
 2. Plug the KB2040 into a computer with USB-C. Press the BOOT button, tap RESET while holding it, then release BOOT. A drive named RPI-RP2 appears.
 3. Copy the .uf2 onto RPI-RP2. The board reboots and a drive named CIRCUITPY appears.
 4. Copy firmware/kb2040/boot.py, code.py and protocol.py to the root of CIRCUITPY. Do **not** copy test_protocol.py.
 5. Download the CircuitPython **9.x** library bundle from <<https://circuitpython.org/libraries>>, and copy neopixel.mpy from its lib/ folder into CIRCUITPY/lib/. That is the only bundle file needed; everything else code.py uses is built into the firmware. In particular the four HT16K33 logic matrices and the radar eye are driven from the **Pi**, so adafruit_ht16k33 and adafruit_rgb_display are pip packages in firmware/pi/requirements.txt and must **not** be copied into CIRCUITPY/lib/.
 6. Unplug and replug the board, or press RESET. A power cycle is required for boot.py to take effect; a soft reload is not enough.
- firmware/kb2040/README.md has the full pin table and a terminal bench check that needs no Pi.

6. Setting up the Raspberry Pi

Raspberry Pi OS Bookworm, 64 bit, with the user pi.

6.1 Buses and groups

```
sudo raspi-config nonint do_i2c 0      # enable I2C  (logic displays)
sudo raspi-config nonint do_spi 0      # enable SPI  (radar eye)
sudo usermod -aG dialout,i2c,spi,gpio,audio pi
sudo reboot
```

After the reboot, confirm the displays answer:

```
i2cdetect -y 1      # expect 70, 71, 72, 73, and 48 if the ADS1115 is fitted
ls /dev/spidev0.*   # expect /dev/spidev0.0 and /dev/spidev0.1
```

6.2 Code and dependencies

```
git clone <this repository> /home/pi/fable-r2d2
cd /home/pi/fable-r2d2
python3 -m venv --system-site-packages /home/pi/r2d2-venv
/home/pi/r2d2-venv/bin/pip install -r firmware/pi/requirements.txt
```

Check the parts that matter:

```
/home/pi/r2d2-venv/bin/python -c "import aiohttp, serial; print('core ok')"
/home/pi/r2d2-venv/bin/python -c "from adafruit_rgb_display import gc9a01a; print('radar ok')"
/home/pi/r2d2-venv/bin/python -c "from adafruit_ht16k33.matrix import Matrix8x8; print('logic ok')"
which aplay
```

Anything missing here only disables that one feature: the server logs the exact reason and the pip command that fixes it, then carries on.

6.3 Controller port

With the KB2040 plugged in:

```
ls -l /dev/ttyACM*   # ACM0 is the REPL console, ACM1 is the data endpoint
```

serial_link.find_port() picks the data endpoint on its own. To pin it, set R2D2_SERIAL_PORT=/dev/ttyACM1 in the systemd unit.

6.4 Sound

The clips are committed, so nothing needs generating on the Pi. To rebuild them:

```
/home/pi/r2d2-venv/bin/python scripts/generate_audio.py
```

Pick the output device with `sudo raspi-config` (System Options, Audio) and test with `aplay firmware/pi/sounds/greet.wav`.

6.5 Wi-Fi access point named R2-FABLE

Bookworm uses NetworkManager, so one command creates the hotspot. Replace the password with your own; it must be at least eight characters.

```
sudo nmcli device wifi hotspot ifname wlan0 ssid R2-FABLE password <your-password>
sudo nmcli connection modify Hotspot connection.autoconnect yes \
    connection.autoconnect-priority 100
sudo nmcli connection modify Hotspot ipv4.method shared ipv4.addresses 192.168.4.1/24
sudo nmcli connection up Hotspot
```

The phone then joins R2-FABLE and opens **http://192.168.4.1:8080/**.

hostapd is not needed on Bookworm. If you are on an older image that does not have NetworkManager, install hostapd and dnsmasq instead and give wlan0 the static address 192.168.4.1/24.

6.6 Service

```
sudo cp firmware/pi/r2d2.service /etc/systemd/system/r2d2.service
sudo systemctl daemon-reload
sudo systemctl enable --now r2d2.service
systemctl status r2d2.service
journalctl -u r2d2.service -f
```

The unit runs as pi, restarts on failure at most five times a minute, and is stopped with SIGINT so the aiohttp cleanup handlers stop the motors, drop the enable line and blank the displays before the process exits.

To run it by hand instead:

```
/home/pi/r2d2-venv/bin/python firmware/pi/server.py --host 0.0.0.0 --port 8080
```

Useful switches: `--serial-port`, `--no-displays`, `--no-battery`, `--log-level DEBUG`.

7. Using the control page

Open **http://192.168.4.1:8080/** on a phone.

1. Tick **Motors armed**. That sends E 1 and wakes the four DRV8833 boards. Nothing moves until it is ticked.
2. Drag the round joystick. Forward is up; the knob returns to centre and the robot stops when you lift your finger.
3. **Speed cap** limits every channel; 60 % is the default and is sensible indoors.
4. **Spin left / Spin right** turn the dome while held. **Nudge** turns it for 200 ms.
5. The dome light buttons set all 17 pixels; the sound buttons are built from whatever the Pi actually found in `firmware/pi/sounds`.
6. The red **STOP** bar is always on screen. It zeroes the drive, sends s and un-arms the motors.

The header shows link state and, when an ADS1115 is fitted, pack voltage and a rough state of charge.

8 Tests and acceptance

These are the first physical measurements made on the robot. Every number elsewhere in this manual is a calculation waiting to be checked here. Do not skip a test because the calculation looked comfortable, and do not carry the robot by the dome before the tilt test has been done.

Stage 11 - acceptance tests

These are the five ranked tests from `research/loads.md` section 8, plus the head-drive rim test from section 5 and a stopping check. Run them in this order. Each has a pass criterion, and a failure is a real failure, not a note.

Figure 16 - whole robot exploded: dome, two body rings, head drive, two legs, centre leg, three feet. This drawing has a full-size landscape page of its own in this manual.

11.1 Mass

Weigh the whole robot on the bathroom scale.

- **Expected: 11.7 kg**, from the CAD screening in `docs/stability.json` (6,046.9 g printed plus 5,658.6 g purchased). The wider range from `research/loads.md` section 1 is 11.6 to 15.0 kg.
- **Pass:** the reading is inside that range and the running total of printed masses matches the group budgets in `mechanical.md` section 5.1.
- **If over:** every extra kilogram cuts drive margin and runtime and raises turning scrub. The lightening options are in `loads.md` section 1.

11.2 Centre-foot load, on the bathroom scale

Stand the robot in the three-leg stance with the centre foot on the bathroom scale and the two outer feet on packers of the same height.

- **Pass:** the centre foot carries **at least 28 percent of total mass**.
- **Expect to exceed it, by a lot.** `docs/stability.json` puts the centre of gravity at y 52.3 mm, which is *behind* the centre-foot axle line at y 77.0 mm, and gives a static share of **1.63** on the two-line model. In plain terms: the robot sits back on the centre foot and the outer feet are lightly loaded. That is the same problem as 11.3, seen from the other side.
- **If the outer feet read near zero:** the centre of gravity must move forward before the robot drives. Differential drive needs load on the outer wheels.
- **Note:** the built geometry puts the centre foot 39.2 mm *behind* the outer-foot line, where `research/loads.md` assumed it 290 mm ahead. This test is the first real measurement of that share.

11.3 Tilt test

Put the robot on a tilt board and raise the front until the centre foot lifts.

- **Pass:** the robot reaches **15 degrees or more** of backward tilt before the centre foot lifts.
- **Expect to fail as modelled.** `docs/stability.json` gives a tip-back margin of **20.3 mm** and an angle of **3.9 degrees**, against the 15 degree criterion. Tip-forward is 19.8 degrees and side tip 34.2 degrees, so only backward tipping is at risk.
- **If under:** the centre of gravity must move forward. The battery is already at `battery_y = 15 mm` forward inside its curb. The remaining moves, in order of effect, are: rake the outer legs further forward, lower the shoulder axis in the body, and move the deck load forward off the rear chord. Re-run 11.2 after any change.
- **Do not skip to 11.4 on a failure.** A robot that rocks back 4 degrees will tip on its first threshold.

11.4 Traction and turning

1. **Pull test.** Tow the robot on the target floor with a spring scale, at a constant slow speed, motors unpowered. **Expected 2.6 to 5.2 N.**
2. **Push test.** Measure the force the powered robot delivers against the scale. **Expected 4.7 N per side at the 1 A driver limit.**

3. **Arc turn.** Command a 0.5 m radius arc at 50 percent PWM. **Pass: it completes without a stall.** This is the test that retires the number one risk.
4. **Back-drive check.** Push the robot with the motors unpowered. The gearboxes must back-drive, or a stalled motor drags.

11.5 Twenty-minute drive, current and temperature

Drive for 20 minutes continuously on the worst floor in the house.

Measure	Pass
Motor case temperature, IR thermometer	Below 70 C
Battery voltage after one hour of mixed use	Above 11.5 V
DRV8833 drivers	No thermal shutdown
Steady current per motor	About 0.385 A at 6 V (expected)
Total 12 V current while driving	About 2.07 A (expected)

Stop immediately at 70 C.

11.6 Stopping

Drive at full commanded speed, about 0.5 m/s, on the target floor, then command stop.

- **Pass:** the robot stops in **150 mm or less (estimated)** and does not rock forward far enough to lift the centre foot.
- **Remember:** SLP low is **coast, not a brake** (electrical.md section 9). The only guaranteed disconnect is the switch off and the Powerpole pair unplugged.

11.7 Head-drive rim pull

With the dome on its bearing and the drive released, pull the dome round by hand with a spring scale at the rim.

- **Expected:** about **0.7 N steady at a 0.158 m radius**, which is 0.1 N.m.
- **If over 1.5 N:** raise the head PWM to 60 percent or lighten the dome. Do not change the wheel first.

11.8 One-hour wear test

Run one loaded foot, or the whole robot, through an hour of figure-8 driving.

- **Pass:** shaft radial play, measured with a dial indicator before and after, grows by **0.3 mm or less**, and every wheel is still on its paint-pen mark.
- **Then:** re-torque every M8 nut. New prints bed in and the preload drops.

8.1 Ranked risks and the test that retires each

Reproduced from research/loads.md section 8.

8. Ranked risks and the test that retires each

1. **Turning scrub exceeds drive force.** Spin-in-place needs 4.1-9.3 N per side against 4.7 N available at the 1 A limit. Test: one loaded foot (46 N) on the target floor, measure yaw torque with a spring scale at a 0.2 m arm (expect 0.6-1.5 N-m), then measure the powered push force (expect 4.7 N at 1 A). Retire if a 0.5 m arc at 50 % PWM completes without a stall.
2. **Mass over the 11 kg target.** Every extra kilogram cuts the drive margin and runtime and raises scrub. Test: weigh every print on a kitchen scale as it comes off the bed, keep a running total in bom/, stop and lighten when the running total passes the group budget in section 1.
3. **TT gearbox shaft bores and press-fit wheels under 11-14 N radial plus 7 N lateral.** Test: one hour figure-8 on a loaded foot, dial-indicator shaft play before and after (reject > 0.3 mm), wheel axial position unchanged.
4. **Tip-back with the CG 29 mm ahead of the outer-foot line.** Test: tilt board to $\geq 15^\circ$ backward without lifting the centre foot; bathroom scale under the centre foot reads $\geq 28\%$ of total mass. If not, move the battery forward and lower the shoulder axis.
5. **Motor heating and runtime at the 1 A limit.** Test: 20 minutes of continuous driving on the worst floor in the house; motor case temperature $< 70^\circ\text{C}$ by IR thermometer, battery above 11.5 V after one hour of mixed use, no DRV8833 thermal shutdown.

Lower-ranked: head friction-wheel slip (retired by the rim pull test in section 5); PETG creep under the shoulder bolt (retired by the crush-sleeve design and a 48 h loaded soak at 40°C).

9 Verification record

What was actually checked by a program, and what was not. Every row below was produced by a script in scripts/ or electronics/ and written to a file in this package. No row is a physical measurement.

9.1 Package checks

Check	Result	Detail
images	pass	59 listed files present, 54 distinct by hash (≥ 30 required)
firmware	pass	OK; OK; app.js syntax OK
audio	pass	12 clips, mismatches=[]
bom	pass	purchased subtotal \$1,052.34; xlsx=present; problems=[]
wiring	pass	136 wires; KB2040 pins in wiring not in code.py: []

docs/verification.json: all_pass = True, physical_validation = False.

9.2 CAD and mesh validation

cad/validation.json: 9 unique STL files, 12 printed pieces, print envelope [322, 317, 315] mm, solid-material upper bound 13.69 kg. all_pass = True. physical_fit_verified = False; physical_strength_verified = False.

Part	Qty	Watertight	Winding	Solids	Fits plate	Result
dome	1	yes	yes	1	yes	pass
body_upper	1	yes	yes	1	yes	pass
body_lower	1	yes	yes	1	yes	pass
leg_upper	2	yes	yes	1	yes	pass
leg_lower	2	yes	yes	1	yes	pass
foot_outer	2	yes	yes	1	yes	pass
leg_center	1	yes	yes	1	yes	pass
foot_center	1	yes	yes	1	yes	pass
head_drive	1	yes	yes	1	yes	pass

9.3 Interference checks

Each row is a boolean intersection between two mating parts in the assembled model. An empty result means the two bodies do not overlap. A check that produced no solid at all was not evaluated and is not evidence of clearance.

Check	Verdict	Raw output
check_center	no solid produced; check not evaluated	check_center volume 2261.1347 mm3 INTERFERENCE
check_seam	no solid produced; check not evaluated	check_seam volume 0.0000 mm3 OK contact plane
check_dome	no solid produced; check not evaluated	check_dome EMPTY
check_head_drive	no solid produced; check not evaluated	check_head_drive volume 0.0000 mm3 OK contact plane
check_shouldler	no solid produced; check not evaluated	check_shouldler EMPTY
check_ankle	no solid produced; check not evaluated	check_ankle EMPTY
check_leg_split	no solid produced; check not evaluated	check_leg_split volume 0.0000 mm3 OK contact plane

Source: cad/interference-final.txt.

9.4 Mass and stability screening

docs/stability.json. Printed mass 6160 g at fill fraction 0.45, purchased mass 5659 g, total 11.82 kg. Centres are millimetres in the model frame (X across, Y forward of the shoulder axis, Z above the floor).

Item	Kind	Mass g	Centre X, Y, Z mm
body_lower	printed	964.6	1.1, 80.7, 213.3
body_upper	printed	1550.4	0.1, 17.8, 429.4
dome	printed	586.9	-1.2, -38.2, 590.9
head_drive	printed	48.1	0.5, -106.1, 470.4
leg_upper_right	printed	650.4	187.2, 15.7, 414.8
leg_lower_right	printed	299.7	186.9, 93.6, 174.5
foot_outer_right	printed	384.7	161.9, 132.1, 59.0
leg_upper_left	printed	650.4	-187.2, 15.7, 414.8
leg_lower_left	printed	299.7	-186.9, 93.6, 174.5
foot_outer_left	printed	384.7	-161.9, 132.1, 59.0
leg_center	printed	111.1	-0.0, 59.9, 141.9
foot_center	printed	229.5	0.0, 26.7, 63.8
battery	purchased	2260.0	0.0, 115.8, 252.1
electronics_and_harness	purchased	750.0	0.0, 38.3, 345.3
lazy_susan	purchased	250.0	0.0, -20.1, 525.0
dome_electronics	purchased	200.0	0.0, -39.3, 584.0
body_rods_and_fasteners	purchased	850.0	0.0, 38.1, 345.7
foot_outer_right_motor	purchased	30.6	181.7, 71.2, 31.5
foot_outer_right_motor	purchased	30.6	181.7, 161.2, 31.5
foot_outer_right_wheel	purchased	38.0	157.9, 71.2, 31.5
foot_outer_right_wheel	purchased	38.0	205.5, 71.2, 31.5
foot_outer_right_wheel	purchased	38.0	157.9, 161.2, 31.5
foot_outer_right_wheel	purchased	38.0	205.5, 161.2, 31.5
foot_outer_left_motor	purchased	30.6	-181.7, 71.2, 31.5
foot_outer_left_motor	purchased	30.6	-181.7, 161.2, 31.5
foot_outer_left_wheel	purchased	38.0	-157.9, 71.2, 31.5
foot_outer_left_wheel	purchased	38.0	-205.5, 71.2, 31.5
foot_outer_left_wheel	purchased	38.0	-157.9, 161.2, 31.5
foot_outer_left_wheel	purchased	38.0	-205.5, 161.2, 31.5
foot_center_motor	purchased	30.6	0.0, -25.8, 31.5
foot_center_motor	purchased	30.6	0.0, 64.2, 31.5
foot_center_wheel	purchased	38.0	-23.8, -25.8, 31.5
foot_center_wheel	purchased	38.0	23.8, -25.8, 31.5
foot_center_wheel	purchased	38.0	-23.8, 64.2, 31.5
foot_center_wheel	purchased	38.0	23.8, 64.2, 31.5
head_drive_motor_wheel	purchased	69.0	0.0, -137.7, 456.3
leg_rods_-1	purchased	320.0	-181.5, 46.4, 320.3
leg_rods_1	purchased	320.0	181.5, 46.4, 320.3
Total		11818.7	0.0, 55.2, 303.6

9.5 Slicer check

cad/h2d-slice-check.json, all_pass = True. Local Bambu Studio CLI slicing of package meshes; no printer connection or physical print. A FAIL row means the slicer would not produce G-code for that mesh in this run; slice it yourself before printing.

Part	Result	Filament g	Hours	Slicer message
dome	pass	1243.4	35.36	Success.
body_upper	pass	3115.1	107.62	Success.
body_lower	pass	1939.4	59.24	Success.
leg_upper	pass	612.4	20.04	Success.
leg_lower	pass	314.5	11.20	Success.
foot_outer	pass	680.9	19.59	Success.
leg_center	pass	128.3	4.25	Success.
foot_center	pass	330.7	9.48	Success.
head_drive	pass	81.3	3.00	Success.

9.6 What is not verified

No part has been printed. No assembly has been made. No motor has been run, no current measured, no temperature taken, no distance driven and no mass weighed. Fit between mating printed parts is checked only as CAD geometry at nominal dimensions, with no allowance for printer tolerance, warp or shrinkage. Strength, traction, runtime, stopping distance and dome-drive slip are calculations against published ratings. Chapter 8 is where those become measurements.

Appendix A Package file list

502 files, 211.7 MB in total, as built on 2026-09-12. Generated caches, the git database and the page-render directory used by the manual build are excluded.

./ - 4 files

File	Size	File	Size
.gitattributes	105 B	.gitignore	164 B
README.md	3.0 kB	plan.md	11.8 kB

audio/ - 1 files

File	Size	File	Size
catalog.csv	1.3 kB		

bom/ - 4 files

File	Size	File	Size
bill-of-materials.xlsx	29.2 kB	electronics.csv	13.4 kB
hardware.csv	9.6 kB	printed-parts.csv	1.7 kB

cad/ - 50 files

File	Size	File	Size
assembly.png	177.6 kB	body.scad	56.3 kB
dome.scad	34.4 kB	exploded.png	179.3 kB
feet.scad	34.0 kB	feet_front.png	47.3 kB
h2d-slice-check.json	5.5 kB	head_drive.scad	23.8 kB
integration-round1.md	16.6 kB	integration-round2.md	18.0 kB
interference-final.txt	319 B	legs.scad	30.5 kB
lib.scad	4.7 kB	params.scad	10.9 kB
preview_body_assembly_section.png	126.3 kB	preview_body_lower_front.png	37.5 kB
preview_body_lower_iso.png	90.9 kB	preview_body_lower_rear.png	39.1 kB
preview_body_lower_section.png	53.5 kB	preview_body_upper_front.png	24.6 kB
preview_body_upper_iso.png	73.2 kB	preview_body_upper_rear.png	26.6 kB
preview_body_upper_section.png	49.2 kB	preview_dome_bottom.png	38.4 kB
preview_dome_front.png	92.0 kB	preview_dome_left.png	87.5 kB
preview_dome_rear.png	90.5 kB	preview_dome_top.png	144.2 kB
preview_foot_center_bottom.png	38.6 kB	preview_foot_center_iso.png	62.0 kB
preview_foot_center_side.png	58.8 kB	preview_foot_outer_bottom.png	27.1 kB
preview_foot_outer_front.png	40.5 kB	preview_foot_outer_iso.png	73.3 kB
preview_foot_outer_side.png	45.3 kB	preview_head_drive_iso.png	40.2 kB
preview_head_drive_print.png	57.6 kB	preview_head_drive_side.png	10.7 kB
preview_head_drive_working.png	58.9 kB	preview_leg_center.png	50.8 kB
preview_leg_lower.png	31.1 kB	preview_leg_upper.png	45.0 kB
preview_legs_assembly.png	72.0 kB	r2d2.scad	6.8 kB
rear.png	172.7 kB	review-body-findings.md	7.2 kB
section.png	160.2 kB	side.png	81.5 kB
slice_all.log	1.2 kB	validation.json	5.4 kB

docs/ - 7 files

File	Size	File	Size
assembly.md	40.1 kB	electrical.md	16.4 kB
firmware.md	25.9 kB	mechanical.md	55.1 kB
pdf-check.json	2.0 kB	stability.json	7.3 kB
verification.json	701 B		

electronics/ - 7 files

File	Size	File	Size
01-power-and-charging.svg	23.0 kB	02-motor-drive.svg	30.9 kB
03-head-electronics.svg	23.1 kB	04-pi-audio-and-links.svg	21.5 kB
calculations.json	5.3 kB	generate.py	55.6 kB
wiring.csv	11.1 kB		

firmware/kb2040/ - 5 files

File	Size	File	Size
README.md	5.6 kB	boot.py	477 B
code.py	13.2 kB	protocol.py	9.7 kB
test_protocol.py	9.1 kB		

firmware/pi/ - 9 files

File	Size	File	Size
audio.py	6.2 kB	battery.py	4.4 kB
displays.py	20.5 kB	mixing.py	6.5 kB
r2d2.service	1.4 kB	requirements.txt	917 B
serial_link.py	11.6 kB	server.py	13.5 kB
test_mixing.py	9.4 kB		

firmware/pi/sounds/ - 12 files

File	Size	File	Size
acknowledge.wav	8.4 kB	alarm.wav	31.0 kB
chirp-double.wav	6.9 kB	greet.wav	23.7 kB
happy.wav	21.1 kB	power-off.wav	36.6 kB
power-on.wav	43.1 kB	question.wav	17.7 kB
sad.wav	36.6 kB	scan.wav	43.1 kB
whistle-down.wav	17.3 kB	whistle-up.wav	17.3 kB

firmware/pi/static/ - 3 files

File	Size	File	Size
app.js	10.4 kB	index.html	2.2 kB
style.css	3.7 kB		

output/ - 1 files

File	Size	File	Size
delivery-preview.html	4.0 kB		

output/drawings/ - 19 files

File	Size	File	Size
01_robot_assembled.png	560.0 kB	02_robot_exploded.png	496.4 kB
03_robot_orthographic.png	423.2 kB	04_printed_components.png	460.3 kB
05_body_rings.png	462.2 kB	06_dome.png	553.8 kB
07_legs.png	446.5 kB	08_feet.png	419.2 kB
09_head_drive.png	322.7 kB	10_exploded_feet.png	499.4 kB
11_exploded_leg.png	312.7 kB	12_exploded_center_leg.png	306.3 kB
13_exploded_body.png	437.0 kB	14_exploded_head_drive.png	337.3 kB
15_exploded_dome.png	417.5 kB	16_exploded_robot.png	594.0 kB
build.log	1.5 kB	drawing-manifest.json	6.5 kB
r2d2-matlab-style-drawings.zip	8.8 MB		

output/drawings/components/ - 9 files

File	Size	File	Size
body_lower.png	308.8 kB	body_upper.png	293.1 kB
dome.png	376.2 kB	foot_center.png	275.6 kB
foot_outer.png	274.7 kB	head_drive.png	240.6 kB
leg_center.png	233.4 kB	leg_lower.png	232.8 kB
leg_upper.png	233.8 kB		

output/pdf/ - 2 files

File	Size	File	Size
build.log	0 B	r2d2-assembly-manual.pdf	13.6 MB

output/video/ - 5 files

File	Size	File	Size
assembly-captions.srt	2.9 kB	build.log	3.9 kB
r2d2-assembly-and-operation.mp4	50.6 MB	video-manifest.json	28.9 kB
video-poster.png	914.3 kB		

output/video/review/ - 32 files

File	Size	File	Size
frame-0000.0.png	939.4 kB	frame-0004.0.png	874.8 kB
frame-0014.0.png	653.2 kB	frame-0020.0.png	792.4 kB
frame-0028.0.png	935.9 kB	frame-0036.0.png	910.6 kB
frame-0044.0.png	794.1 kB	frame-0050.0.png	807.7 kB
frame-0053.5.png	783.8 kB	frame-0054.0.png	742.1 kB
frame-0058.0.png	978.8 kB	frame-0066.0.png	917.7 kB
frame-0072.0.png	1.0 MB	frame-0079.0.png	1.0 MB
frame-0086.0.png	900.6 kB	frame-0092.0.png	952.8 kB
frame-0099.0.png	1.0 MB	frame-0104.0.png	1007.0 kB
frame-0110.0.png	919.7 kB	frame-0116.0.png	917.8 kB
frame-0120.0.png	637.2 kB	frame-0126.0.png	831.2 kB
frame-0133.0.png	1000.9 kB	frame-0139.0.png	743.8 kB
frame-0145.0.png	919.6 kB	frame-0152.0.png	989.5 kB
frame-0158.0.png	978.5 kB	frame-0160.0.png	975.9 kB
frame-0162.0.png	977.6 kB	frame-0166.0.png	958.7 kB
frame-0169.0.png	842.4 kB	motion-checks.json	22.8 kB

research/ - 8 files

File	Size	File	Size
components-electronics.md	28.0 kB	components-mechanical.md	27.8 kB
critique.md	6.8 kB	image-review-body.md	25.0 kB
image-review-dome.md	28.9 kB	image-review-legs.md	33.0 kB
loads.md	29.6 kB	proportions.md	43.9 kB

research/components/ - 1 files

File	Size	File	Size
3777_diagram.jpg	16.4 kB		

research/drawings/ - 39 files

File	Size	File	Size
MC.Hex.Center.Ankle.doc.set.v2.zip	50.3 kB	csr.anh.skins.rev2.zip	207.2 kB
curiousmarc-2-3-2-overview.pdf	1.6 MB	dome-p1.png	524.1 kB
dome-p2.png	505.0 kB	dome-p3.png	228.7 kB
dome-p4.png	196.5 kB	dome-p5.png	197.4 kB
dome-p6.png	179.1 kB	dome.pdf	244.0 kB
mc.2leg.fixed.shoulder.zip	80.8 kB	mc.3leg.fixed.shoulder.zip	106.5 kB
mc.NextGen.Frame.v2.zip	719.8 kB	mc.SAK.CSR.ctr.foot.v1.zip	3.7 MB
mc.battery.box.3D.printed.zip	94.0 kB	mc.battery.box.zip	94.0 kB
mc.box.beam.legs.v4.zip	182.6 kB	mc.center.ankle.zip	53.5 kB
mc.center.foot.zip	1.8 MB	mc.csr.anh.skins.v2.zip	207.2 kB
mc.csr.horseshoes.zip	375.8 kB	mc.dome.motivator.zip	43.1 kB
mc.eggcrate.frame.zip	291.3 kB	mc.octagon.port.zip	35.2 kB
mc.outer.foot.drive.zip	70.0 kB	mc.outer.foot.shell.zip	131.1 kB
mc.shoulder.hub.zip	26.9 kB	mc.skirt.v2.zip	99.3 kB
mc.util.arm.box.zip	37.2 kB	mc.util.arm.zip	52.7 kB
outer-ellipse.jpg	342.9 kB	packages-index.html	3.7 kB
radar-eye-p1.png	125.0 kB	radar-eye-p2.png	146.8 kB
radar-eye-p3.png	119.3 kB	radar-eye-p4.png	169.2 kB
radar-eye-p5.png	124.6 kB	radar-eye.pdf	77.3 kB
skins-page.html	8.9 kB		

research/drawings/MC.Hex.Center.Ankle.doc.set.v2/ - 2 files

File	Size	File	Size
MC.Hex.Center.Ankle.188.V2.pdf	26.8 kB	MC.Hex.Center.Ankle.v2.dwg	55.0 kB

research/drawings/csr.anh.skins.rev2/ - 8 files

File	Size	File	Size
anh.front.inner.p.44s.v2.dwg	32.3 kB	anh.front.inner.p.44s.v2.pdf	46.5 kB
anh.front.outer.plus.v2.dwg	36.0 kB	anh.front.outer.plus.v2.pdf	49.9 kB
anh.rear.inner.p.s44.v2.dwg	26.8 kB	anh.rear.inner.p.s44.v2.pdf	23.6 kB
anh.rear.outer.plus.v2.dwg	30.0 kB	anh.rear.outer.plus.v2.pdf	25.9 kB

research/drawings/mc.2leg.fixed.shoulder/ - 3 files

File	Size	File	Size
copyright.notice.txt	456 B	mc.2leg.fixed.shoulder.dwg	73.8 kB
mc.2leg.fixed.shoulder.pdf	52.2 kB		

research/drawings/mc.3leg.fixed.shoulder/ - 3 files

File	Size	File	Size
copyright.notice.txt	456 B	mc.3leg.fixed.shoulder.dwg	79.3 kB
mc.3leg.fixed.shoulder.pdf	74.1 kB		

research/drawings/mc.NextGen.Frame.v2/ - 13 files

File	Size	File	Size
NextGen Frame document set release2.pdf	541.3 kB	copyright.notice.txt	456 B
core2.dwg	33.7 kB	dome.motivator.dwg	47.2 kB
read.door.dwg	34.4 kB	ribs.plates.dwg	56.7 kB
ring0.dwg	28.5 kB	ring1.dwg	31.8 kB
ring3.core3.dwg	38.2 kB	ring4.core4.dwg	33.4 kB
shoulder.hub.dwg	69.3 kB	slider.319.stp	15.9 kB
util.arm.box.dwg	28.9 kB		

research/drawings/mc.SAK.CSR.ctr.foot.v1/ - 8 files

File	Size	File	Size
SAK.ctr.foot.shell.v1.dwg	26.5 kB	SAK.ctr.foot.struct.v1.dwg	46.5 kB
SAK.ctr.foot.v1.drawings.pdf	86.9 kB	SAK.ctr.foot.wheel.v1.dwg	32.3 kB
end.trim.r.stl	3.7 MB	full.trim.set.stl	2.3 MB
nut.trap.hex.x16.s129.30.20.2P.2L.r1.stl	5.1 MB	side.trim.r.stl	6.7 MB

research/drawings/mc.battery.box/ - 6 files

File	Size	File	Size
copyright.notice.txt	460 B	mc.3D.Printed.Battery.Box.dwg	42.7 kB
mc.3D.Printed.Battery.Box.pdf	32.4 kB	mc.corner.filler.stp	14.7 kB
mc.curved.end.stp	73.7 kB	mc.curved.end.w.cutout.stp	133.3 kB

research/drawings/mc.battery.box.3D.printed/ - 6 files

File	Size	File	Size
copyright.notice.txt	460 B	mc.3D.Printed.Battery.Box.dwg	42.7 kB
mc.3D.Printed.Battery.Box.pdf	32.4 kB	mc.corner.filler.stp	14.7 kB
mc.curved.end.stp	73.7 kB	mc.curved.end.w.cutout.stp	133.3 kB

research/drawings/mc.box.beam.legs.v4/ - 11 files

File	Size	File	Size
copyright.notice.txt	456 B	mc.legs.1.5.v4.dwg	42.3 kB
mc.legs.1.5.v4.pdf	13.3 kB	mc.legs.2.5.v4.dwg	45.1 kB
mc.legs.2.5.v4.pdf	34.3 kB	mc.legs.3.5.v4.dwg	24.6 kB
mc.legs.3.5.v4.pdf	22.9 kB	mc.legs.4.5.v4.dwg	53.0 kB
mc.legs.4.5.v4.pdf	23.0 kB	mc.legs.5.5.v4.dwg	40.7 kB
mc.legs.5.5.v4.pdf	10.5 kB		

research/drawings/mc.center.ankle/ - 5 files

File	Size	File	Size
copyright.notice.txt	530 B	mc.center.ankle.1.2.dwg	43.5 kB
mc.center.ankle.1.2.pdf	22.6 kB	mc.center.ankle.2.2.dwg	17.4 kB
mc.center.ankle.2.2.pdf	6.7 kB		

research/drawings/mc.center.foot/ - 7 files

File	Size	File	Size
3D.Center.Foot.Trim.stp	163.8 kB	3DCenter.Foot.pdf	19.7 kB
center.foot.full.pdf	867.7 kB	center.foot.shell.pdf	53.4 kB
center.foot.structural.pdf	841.8 kB	mc.center.foot.dwg	38.0 kB
mc.center.foot.pdf	23.9 kB		

research/drawings/mc.csr.anh.skins.v2/ - 8 files

File	Size	File	Size
anh.front.inner.p.44s.v2.dwg	32.3 kB	anh.front.inner.p.44s.v2.pdf	46.5 kB
anh.front.outer.plus.v2.dwg	36.0 kB	anh.front.outer.plus.v2.pdf	49.9 kB
anh.rear.inner.p.s44.v2.dwg	26.8 kB	anh.rear.inner.p.s44.v2.pdf	23.6 kB
anh.rear.outer.plus.v2.dwg	30.0 kB	anh.rear.outer.plus.v2.pdf	25.9 kB

research/drawings/mc.csr.horseshoes/ - 4 files

File	Size	File	Size
CSR Horseshoes Dwg Set.pdf	104.4 kB	CSR SHOULDER - HORSESHOE 20140601.PDF	58.2 kB
horseshoe.188.expld.3d.pdf	150.1 kB	wedge.r.stl	420.5 kB

research/drawings/mc.dome.motivator/ - 2 files

File	Size	File	Size
legacy.dome.motivator.dwg	39.0 kB	legacy.dome.motivator.pdf	22.3 kB

research/drawings/mc.eggcrate.frame/ - 12 files

File	Size	File	Size
body.flat.fdim.dwg	154.8 kB	copyright.notice.txt	456 B
mc.frame parts.dwg	55.4 kB	mc.frame.parts.pdf	17.0 kB
mc.ring1.dwg	26.7 kB	mc.ring1.pdf	28.4 kB
mc.ring2.dwg	48.2 kB	mc.ring2.pdf	39.8 kB
mc.ring3.dwg	47.8 kB	mc.ring3.pdf	39.2 kB
mc.ring4.dwg	48.9 kB	mc.ring4.pdf	37.6 kB

research/drawings/mc.octagon.port/ - 3 files

File	Size	File	Size
copyright.notice.txt	456 B	mc.octagon.port.dwg	54.5 kB
mc.octagon.port.pdf	17.4 kB		

research/drawings/mc.outer.foot.drive/ - 4 files

File	Size	File	Size
mc.3D.Printed.Battery.Box.dwg	42.7 kB	mc.3d.printed.battery.box.pdf	32.7 kB
mc.battery.box.corner.filler.stp	14.7 kB	mc.battery.box.curved.end.stp	73.7 kB

research/drawings/mc.outer.foot.shell/ - 10 files

File	Size	File	Size
copyright.notice.txt	460 B	mc.foot.shell.curve.left.stp	25.2 kB
mc.foot.shell.curve.right.stp	24.7 kB	mc.foot.shell.inside.brace.w.nuttrap.stp	89.6 kB
mc.foot.shell.nut.trap.stp	24.1 kB	mc.foot.shell.pg1.2.dwg	31.5 kB
mc.foot.shell.pg1.2.pdf	21.4 kB	mc.foot.shell.pg2.2.dwg	43.3 kB
mc.foot.shell.pg2.2.pdf	20.6 kB	mc.outer.foot.apron.stp	170.9 kB

research/drawings/mc.shoulder.hub/ - 2 files

File	Size	File	Size
copyright.notice.txt	456 B	mc.shoulder.hub.dwg	68.0 kB

research/drawings/mc.skirt.v2/ - 5 files

File	Size	File	Size
copyright.notice.txt	456 B	mc.skirt.base.v2.dwg	54.6 kB
mc.skirt.base.v2.pdf	36.2 kB	mc.skirt.bottom.v2.dwg	57.4 kB
mc.skirt.bottom.v2.pdf	29.7 kB		

research/drawings/mc.util.arm/ - 2 files

File	Size	File	Size
utility.arm.igs	269.9 kB	utility.arm.stl	168.1 kB

research/drawings/mc.util.arm.box/ - 2 files

File	Size	File	Size
mc.arm.box.dwg	30.6 kB	mc.arm.box.pdf	25.3 kB

research/drawings/photos/ - 12 files

File	Size	File	Size
body-front.jpg	119.4 kB	body-rear.jpg	114.1 kB
dome-terms.jpg	94.6 kB	leg.png	335.7 kB
lucasfilm-r2.jpeg	92.0 kB	terminology-p1.png	187.1 kB
terminology-p2.png	550.3 kB	terminology-p3.png	550.8 kB
terminology-p4.png	1.1 MB	terminology-p5.png	1.2 MB
terminology-p6.png	1.2 MB	terminology.pdf	4.5 MB

research/drawings/png/ - 62 files

File	Size	File	Size
3DCenter.Foot-p1.png	49.0 kB	CSR_Horseshoes_Dwg_Set-p1.png	370.2 kB
CSR_Horseshoes_Dwg_Set-p2.png	344.7 kB	CSR_Horseshoes_Dwg_Set-p3.png	484.4 kB
CSR_SHOULDER_-_HORSESHOE_20140601-p1.png	359.0 kB	MC.Hex.Center.Ankle.188.V2-p1.png	408.1 kB
NextGen_Frame_document_set_release2-p1.png	231.4 kB	NextGen_Frame_document_set_release2-p10.png	583.9 kB
NextGen_Frame_document_set_release2-p2.png	191.9 kB	NextGen_Frame_document_set_release2-p3.png	276.9 kB
NextGen_Frame_document_set_release2-p4.png	280.5 kB	NextGen_Frame_document_set_release2-p5.png	355.2 kB
NextGen_Frame_document_set_release2-p6.png	358.3 kB	NextGen_Frame_document_set_release2-p7.png	271.0 kB
NextGen_Frame_document_set_release2-p8.png	269.8 kB	NextGen_Frame_document_set_release2-p9.png	254.0 kB
SAK.ctr.foot.v1.drawings-p1.png	293.6 kB	SAK.ctr.foot.v1.drawings-p2.png	255.2 kB
SAK.ctr.foot.v1.drawings-p3.png	177.2 kB	SAK.ctr.foot.v1.drawings-p4.png	436.4 kB
anh.front.inner.p.44s.v2-p1.png	283.6 kB	anh.front.outer.plus.v2-p1.png	409.6 kB
anh.rear.inner.p.s44.v2-p1.png	271.5 kB	anh.rear.outer.plus.v2-p1.png	388.7 kB
center.foot.full-p1.png	94.0 kB	center.foot.shell-p1.png	51.3 kB
center.foot.structural-p1.png	78.1 kB	dome-cc-stack.png	41.0 kB
dome-cc.png	12.8 kB	dome-front-centre.png	179.9 kB
dome-front-left.png	268.1 kB	dome-left-lower.png	150.9 kB
dome-rear-lower.png	107.9 kB	dome-top-left.png	358.2 kB
dome-top-right.png	357.0 kB	front-skin-crop.png	175.7 kB
front-skin-top.png	74.5 kB	horseshoe.188.expld.3d-p1.png	119.8 kB
legacy.dome.motivator-p1.png	91.9 kB	mc.2leg.fixed.shoulder-p1.png	402.2 kB
mc.3d.printed.battery.box-p1.png	391.6 kB	mc.3leg.fixed.shoulder-p1.png	569.7 kB
mc.arm.box-p1.png	280.5 kB	mc.center.ankle.1.2-p1.png	334.4 kB
mc.center.ankle.2.2-p1.png	104.5 kB	mc.center.foot-p1.png	408.6 kB
mc.foot.shell.pg1.2-p1.png	389.1 kB	mc.foot.shell.pg2.2-p1.png	379.1 kB
mc.frame.parts-p1.png	216.0 kB	mc.legs.1.5.v4-p1.png	248.8 kB
mc.legs.2.5.v4-p1.png	274.6 kB	mc.legs.3.5.v4-p1.png	236.2 kB
mc.legs.4.5.v4-p1.png	376.2 kB	mc.legs.5.5.v4-p1.png	238.9 kB
mc.octagon.port-p1.png	245.7 kB	mc.ring1-p1.png	212.7 kB
mc.ring2-p1.png	290.0 kB	mc.ring3-p1.png	293.0 kB
mc.ring4-p1.png	275.3 kB	mc.skirt.base.v2-p1.png	301.6 kB
mc.skirt.bottom.v2-p1.png	290.1 kB	rear-skin-crop.png	150.2 kB

research/images/ - 59 files

File	Size	File	Size
body-01.jpg	509.3 kB	body-02.jpg	628.3 kB
body-03.jpg	241.0 kB	body-04.jpg	291.8 kB
body-05.jpg	180.7 kB	body-06.jpg	236.1 kB
body-07.jpg	396.1 kB	body-08.jpg	194.0 kB
body-09.jpg	587.9 kB	body-10.jpg	509.2 kB
body-11.jpg	365.2 kB	body-12.jpg	483.1 kB
body-13.jpg	308.6 kB	body-14.jpg	37.9 kB
body-15.jpg	31.9 kB	body-16.jpg	17.6 kB
body-17.jpg	119.4 kB	body-18.jpg	113.8 kB
body-19.jpg	114.1 kB	body-20.jpg	161.5 kB
dome-01.jpg	97.9 kB	dome-02.jpg	94.0 kB
dome-03.jpg	92.5 kB	dome-04.jpg	89.8 kB
dome-05.jpg	94.6 kB	dome-06.jpg	509.3 kB
dome-07.jpg	597.4 kB	dome-08.jpg	343.2 kB
dome-09.jpg	463.3 kB	dome-10.jpg	786.0 kB
dome-11.jpg	377.9 kB	dome-12.jpg	1.1 MB
dome-13.jpg	1.1 MB	dome-14.jpg	1016.7 kB
dome-15.jpg	1.2 MB	dome-16.jpg	1.3 MB
dome-17.jpg	1.1 MB	dome-18.jpg	619.7 kB
dome-19.jpg	717.8 kB	legs-01.jpg	597.4 kB
legs-02.jpg	509.3 kB	legs-03.jpg	1.3 MB
legs-04.png	335.7 kB	legs-05.jpg	88.3 kB
legs-06.jpg	208.7 kB	legs-07.jpg	183.3 kB
legs-08.jpg	52.5 kB	legs-09.jpg	388.8 kB
legs-10.jpg	40.1 kB	legs-11.jpg	94.7 kB
legs-12.jpg	118.8 kB	legs-13.jpg	51.0 kB
legs-14.jpg	61.2 kB	legs-15.jpg	79.4 kB
legs-16.jpg	71.9 kB	legs-17.jpg	29.1 kB
legs-18.jpg	50.1 kB	legs-19.jpg	619.7 kB
legs-20.jpg	624.7 kB		

scripts/ - 16 files

File	Size	File	Size
assembly_layout.py	9.6 kB	build_bom.py	9.9 kB
build_manual.py	57.7 kB	build_video_audio.ps1	7.2 kB
deliver.py	7.5 kB	draw_robot.py	101.4 kB
electronics.py	6.6 kB	export_cad.py	7.8 kB
generate_audio.py	11.7 kB	parts.json	2.5 kB
render_video.py	35.0 kB	slice_check.py	9.2 kB
stability.py	5.6 kB	verify.py	9.5 kB
video_scene.js	42.1 kB	video_storyboard.json	6.8 kB

stl/ - 9 files

File	Size	File	Size
body_lower.stl	1.8 MB	body_upper.stl	3.3 MB
dome.stl	2.4 MB	foot_center.stl	421.8 kB
foot_outer.stl	754.6 kB	head_drive.stl	378.6 kB
leg_center.stl	351.5 kB	leg_lower.stl	357.2 kB
leg_upper.stl	528.4 kB		